Climate Change, Income Inequality, and Migration in a Spatial Economy

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Those hit hardest by climate change are often too poor to move away from it. Using Brazilian census microdata, I show that within the same origin, migration rises steeply with income, and that when the poor got richer in the 2000s, they left lagging places more. I build a dynamic spatial model in which moving costs cash, not only utility, and heat lowers farm productivity. A 500 km move costs around a year's earnings for the median least-educated worker. Under high emissions, climate change costs 2.56% of consumption, permanently, and the poorest quarter in hot regions loses 25% more than the richest. Removing the fare altogether would reduce the climate loss of the low-income workers it holds back in the hottest regions by 11.7%. A tax-financed subsidy that pays part of the fare for those with below-median income has three effects. For those who move in response, it reduces the loss from climate change, by 4.7% in the hottest regions, and it is worth 5.19% of consumption to them; and it raises national output, by 14.8% of a year's output cumulated to 2100, as workers resettle in more productive regions. Migration is often cast as a way to adapt to climate change, but it is an option only for those who can afford it; paying part of the fare opens it to the poor and raises national output along the way.

JEL codes: J61, O15, Q54, R12, R23.

Keywords: climate change, internal migration, migration costs, cash cost of moving, spatial general equilibrium, Brazil.

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1. Introduction

Trade liberalization, technological change and, increasingly, climate change shift economic opportunity across sectors and space. In principle, workers can follow these shifts by moving to where the opportunities are. In practice, many stay even when the gains from moving are large. Understanding this immobility is central to both positive and normative economics: it determines who bears the cost of local downturns, how quickly economies reallocate labour after shocks, and which policies work when the fortunes of places diverge. A growing literature suggests that the monetary cost of moving holds back low-income workers [Bryan et al., 2014, Cai, 2020, Gazeaud et al., 2023, Bastian and Black, 2024]. If moving costs cash, the workers whom climate change hits hardest are the least able to move away from it.

This paper asks how much the cash cost of moving matters for who loses from climate change, and whether the tax system can help low-income workers move away from it. Brazil is a natural setting: its North and Northeast are poorer and more agricultural than the South and Southeast, and climate change hits them hardest, so it raises the value of moving away most for those least able to pay for it. I document two facts consistent with a cash cost of moving, build a dynamic spatial general-equilibrium model in which a move costs a cash fare as well as a non-monetary cost, and estimate the fare from census migration flows. Under high emissions, climate change costs 2.56 percent of consumption, permanently, and the bottom quartile in the hottest regions loses 25 percent more than the top. A subsidy that pays part of the fare for those with below-median income, financed by a proportional tax, lets the poorest leave these places: for those who move in response, it reduces the loss from climate change by 4.7 percent in the hottest regions, and it raises national output.

My first contribution is two facts from Brazilian census microdata that add to the evidence on the monetary cost of moving. First, within an origin the low-income migrate least: within the same origin and age group, people in the top quintile of earning potential at the origin were more likely than those in the bottom quintile to have changed microregion over the preceding five years, by a margin equal to 34 percent of the mean migration rate. Second, when the incomes of the poor rose, they left lagging places more. The 2000s brought Brazil a large pro-poor income shift, as the real minimum wage rose steeply and transfers expanded, and the least educated gained most. Following groups across three censuses, those whose income grew faster left structurally lagging places more, relative to their own trend. Both facts are what a cash cost predicts: those with less money move less, and they move more when they get more of it.

Two further facts show why this matters for climate change. Third, Brazil is warming: by 2100 the median microregion's annual mean temperature is projected to rise by

5.4°C under SSP5-8.5. Fourth, heat lowers agricultural productivity: using microregion crop data, I estimate that productivity rises with temperature up to about 19.8°C and then falls. Most microregions in the hot North and Northeast are already past this optimum, so warming lowers incomes most where workers are poorest. It raises the value of leaving while making it harder to pay for: where the incentive to migrate is strongest, the means to do so are weakest.

My second contribution is a dynamic spatial general-equilibrium model with a cash cost of moving. In quantitative spatial models, migration costs are typically utility costs [e.g., Caliendo et al., 2019, Conte, 2026, Conte et al., 2021, Cruz and Rossi-Hansberg, 2024, Cruz, 2024, Desmet and Rossi-Hansberg, 2015, Jedwab et al., 2023, Morten and Oliveira, 2024], which do not depend on a worker's income; a few recent models add monetary costs together with wealth or credit [Giannone et al., 2023, Baseler and Gordeev, 2026]. In my model, the cost of moving is both non-monetary and monetary: a fare paid out of a year's income. The fare weighs more on a poorer worker, so a shock that lowers local incomes also makes leaving harder. I estimate the fare from the variation behind the second fact: a 500 km move costs the median least-educated worker about a year's earnings (101%). I use the model to measure the welfare losses from climate change across the income distribution and to study policies that pay part of the cost of moving.

My third contribution is to show what a migration subsidy financed by a tax can do for the losses from climate change. The subsidy removes the part of the burden of the fare that falls on a worker because they are poor: a worker below the median income of 2010 is paid the part of the fare that makes the move as costly, in utility, as it is at the median. A small proportional tax on all earnings balances the budget. The subsidy raises the bottom quartile's leaving rate from 3.1 to 7.8 percent over five years; removing the fare altogether would raise it to 14.9 percent. The 4.7 percent of the quartile whom the subsidy induces to leave in 2015 gain 4.51 percent of consumption, net of the tax, and 5.19 percent in the hottest regions. For them, the subsidy makes the loss from climate change 2.9 percent smaller, and 4.7 percent smaller in the hottest regions.

The subsidy also raises national output: discounted and cumulated to 2100, the output gain is 14.8 percent of national output in 2010, against outlays of 2.9 percent. This is not a gain in efficiency: in the model the fare is a private cost and moving has no externality, so the output gain measures the income that the induced moves add, not a market failure that the subsidy corrects.

The analysis abstracts from several margins. The model has no savings or credit, so the fare is paid out of current income, and the cost of moving depends on income rather than on access to liquidity as such. The dispersion of tastes is set externally, and the migration flows identify the fare given it, so the level of the monetary cost depends on

that elasticity. The model covers earners, so adults without earnings, who are poorer, are outside it. Migration networks are held fixed and do not grow with the flows a subsidy induces, and congestion works through housing rents, without agglomeration economies.

Related literature. First, the paper contributes to the literature on who loses from climate change and on migration as a way to adapt to it. Quantitative spatial models of climate change find that warming raises spatial inequality, with welfare losses of up to 20 percent in parts of Africa and Latin America and gains in northern latitudes [Cruz and Rossi-Hansberg, 2024], and that migration is a central margin of adaptation: shutting down mobility raises the welfare cost of warming more than tenfold in a stylized economy [Desmet and Rossi-Hansberg, 2015], migration costs 25 percent higher make the losses about a third larger by 2200 [Cruz and Rossi-Hansberg, 2024], and in sub-Saharan Africa warming produces millions of climate migrants [Conte, 2026]. In these models, workers within a location are identical or differ only by sector [Conte et al., 2021, Cruz, 2024], and, as in quantitative spatial models more broadly [e.g., Caliendo et al., 2019, Morten and Oliveira, 2024, Jedwab et al., 2023], the cost of moving is typically a utility cost that does not depend on a worker's income. A large empirical literature suggests otherwise: small migration subsidies raise seasonal migration in Bangladesh [Bryan et al., 2014], credit access raises rural-urban migration in China [Cai, 2020], cash transfers raise migration in the Comoros [Gazeaud et al., 2023], and expansions of the Earned Income Tax Credit raise out-migration from distressed regions of the United States [Bastian and Black, 2024]. This matters for climate change. If part of the cost of moving is paid in cash, a shock that lowers incomes also makes leaving harder, so migration responds least where losses are largest. The evidence on climate migration is indeed more mixed than these models suggest: migration responses to temperature shocks are often muted in low-income settings [Bastos et al., 2013, Brunel and Liu, 2025, Bryan et al., 2014, Bazzi, 2017], and climate change can trap the poorest by lowering the incomes from which emigration is paid [Benveniste et al., 2022]. I add a cash cost of moving, paid out of income, to a dynamic spatial general-equilibrium model of climate change in which incomes are distributed within each region and both affect and are affected by mobility. I use it to measure how the losses from climate change vary across the income distribution, within and across regions, how much of this variation is due to the cash cost holding back low-income workers, and what a tax-financed policy can do about it.

I am not the first to model a monetary cost of moving. Structural models of migration under risk and liquidity constraints [Kennan and Walker, 2011, Bryan et al., 2014, Aron-Dine, 2024, Kleemans, 2023] capture how the means to move shape individual decisions, but are partial-equilibrium, and Baseler and Gordeev [2026] and Giannone et al. [2023]

model monetary moving costs together with credit or wealth in general equilibrium, but not climate change. Lagakos et al. [2023] evaluate migration subsidies in a general-equilibrium model of seasonal migration, discussed below, also without climate change. What is new here is to bring such a cost, estimated from data, into a spatial model of climate change, and to use it to evaluate how the fiscal system can help low-income workers move away from the regions that warming hits hardest.

The engine of the model translates warming into incomes, and thereby into the incentives and the means to move. It builds on the evidence that temperatures above an agronomic optimum lower agricultural productivity and income [e.g., Schlenker and Roberts, 2009, Lobell and Gourdji, 2012, Burke et al., 2015, Feng et al., 2010], and on quantitative trade and spatial models that extend the Ricardian framework of Eaton and Kortum [2002] to agriculture under climate change [e.g., Costinot et al., 2016, Gouel and Laborde, 2021, Sotelo, 2020, Pellegrina, 2022, Conte, 2026, Farrokhi and Pellegrina, 2023, Pellegrina and Sotelo, 2025]. Warming lowers farm productivity and with it the incomes of farm workers, most in the hottest regions; this raises the value of leaving those regions while lowering the incomes from which a move is paid.

Second, the paper relates to how fiscal policy can help workers move: the evaluation of migration subsidies and the debate on whether to help people or places. Closest to this paper, Lagakos et al. [2023] evaluate migration subsidies in a general-equilibrium model of seasonal rural-urban migration, disciplined by the randomized migration subsidies of Bryan et al. [2014] in Bangladesh. They find a large non-monetary disutility of migrating. The subsidies' welfare gains therefore come mainly from better insurance against seasonal income shocks, not from relaxing credit constraints that keep workers from more productive urban jobs. My model also has a large non-monetary cost, but the cash cost is not small: a 500 km move costs the median least-educated worker about a year's earnings (101%), and paying part of the fare raises the leaving rate of the bottom quartile from 3.1 to 7.8 percent. I depart from them in three ways. First, and most importantly, I study migration subsidies as a response to climate change, which they do not consider: I ask whether a tax-financed subsidy lets low-income workers move away from the regions that warming makes worse, and how it changes their losses from climate change. Second, I study permanent migration across all of a country's regions, in a dynamic spatial equilibrium in which migration responds to and shapes local incomes. Third, I identify the cash cost from observational data, from how migration responded within origins to differences in earnings growth across education groups.

In the public-economics debate on whether to help people or places [Kline and Moretti, 2014, Austin et al., 2018], O'Connor [2025] shows that the optimal transfer to a declining region can be a tax, because taxing it encourages residents to relocate. This result rests on moving costs that are utility costs. When moving costs cash, a tax on a declining region also erodes the income out of which a move is paid, so it falls

hardest on those who can least afford to leave and can keep them in place. Where moving is a route to opportunity [Chetty et al., 2016], a cash cost makes access to that route depend on income. I study a direct remedy, a subsidy to the cost of moving for those on whom it weighs most, and compare it with targeting by place and with a cash transfer. The output gains of the subsidy add to the evidence that migration costs hinder the reallocation of labour and lower aggregate output [Morten and Oliveira, 2024, Pellegrina and Sotelo, 2025, Bryan and Morten, 2019]: when part of the cost is cash, it holds back most the low-income workers whom climate change hits hardest.

The rest of the paper is organized as follows. Section 2 describes data and measurement. Section 3 presents the four facts. Section 4 presents the model, and Section 5 estimates the fare and calibrates the model. Section 6 reports the welfare losses from climate change, Section 7 removes the fare and studies the income-targeted migration subsidy, and Section 8 discusses other policies, their implementation and open questions. Section 9 concludes.

2. Data and measurement

The analysis needs three kinds of data, each tied to a part of the argument. Census microdata on incomes, sectors and migration show who moves (Section 3), discipline the cash cost of moving, and give the model its initial incomes and migration flows. Annual crop production paired with temperature records measures how heat affects agricultural productivity, which is how climate change enters the model. Climate projections through 2100 provide the warming scenarios.

Unit of observation The unit of observation is the microregion. A microregion is a subnational statistical unit in Brazil, defined by the Brazilian Institute of Geography and Statistics (IBGE) by grouping adjacent municipalities with integrated production structures and labor markets. I use all microregions except the oceanic archipelago of Fernando de Noronha, 557 in total, which are grouped into five larger geographic regions: the North, Northeast, Centre-West, Southeast and South. While microregions are not administrative units, they offer the advantage of being stable units over time and ease inter-temporal comparisons. They are coarse enough that most routine, short-distance mobility takes place within them, so that a recorded change of microregion reflects a costly relocation, yet numerous enough to retain wide variation in climate exposure, agricultural specialization, and income. Microregions are the standard definition of local labor markets in research on Brazil [Kovak, 2013, Dix-Carneiro and Kovak, 2017, 2019, Costa et al., 2016]. Figure I, panel (a), shows the microregions of Brazil and the respective major geographical regions.

Socioeconomic data Socioeconomic data on migration flows, employment patterns, and individual incomes come from the 2000 and 2010 waves of the Brazilian Demographic Census (*Censo Demográfico*).[1] The sample consists of Brazilian nationals aged 20–65 with valid schooling and non-missing income who were resident in Brazil five years before the census and whose current and earlier municipalities can be assigned to a microregion; the top 1% of real total income is trimmed within each wave. Pooled over the two waves, this yields 22,445,097 observations; the 2010 wave alone represents about 111,847,171 individuals in the weighted population.

Migration flows are constructed from each respondent's reported microregion of residence five years before enumeration, aggregated into microregion-to-microregion flows over two five-year windows, 1995–2000 and 2005–2010. In the 2000 census, 7.4% of the sample had lived in another microregion five years earlier, and 5.9% in the 2010 census; in 2010, 2.7% had lived in another state.

I refer to the workers in the fields of crop farms as "agriculture" and to all other workers as "non-agriculture", a concept narrower than the agricultural sector of the national accounts that matches the crop-based measure of agricultural productivity below. Appendix A documents the sample, the harmonisation of the census waves, the income concepts, the definition of the farm sector and summary statistics.

Agricultural production Data on agricultural production are sourced from the *Produção Agrícola Municipal* (PAM) of IBGE, which provides annual statistics on the quantity, value, and cultivated area of the major temporary and permanent crops from 1994 to 2017, aggregated by IBGE to the microregion. This yields annual measures of crop-specific area planted, physical output, and production value.

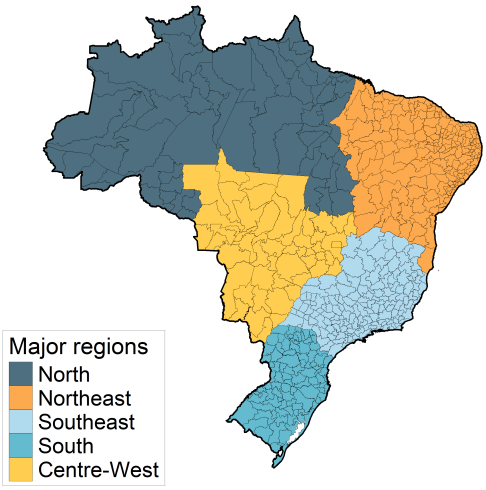
Measured agricultural productivity is defined as the value of crop output per unit of land at constant prices.[2] Fixing prices at nationwide reference levels, as is standard in the literature [e.g., Gollin et al., 2021, Restuccia et al., 2008], makes differences in measured productivity reflect variation in physical yields rather than price heterogeneity across crops, regions, or time. Fact 4 in Section 3 adjusts it for the share of land under cultivation, following Costinot et al. [2016].[3] Appendix A.3 documents the crop selection and the construction.

[1] One of the facts in Section 3 also uses the 1991 census, harmonised in the same way, for the migration window 1986–91, which provides the earlier decade against which the 2000s are compared.

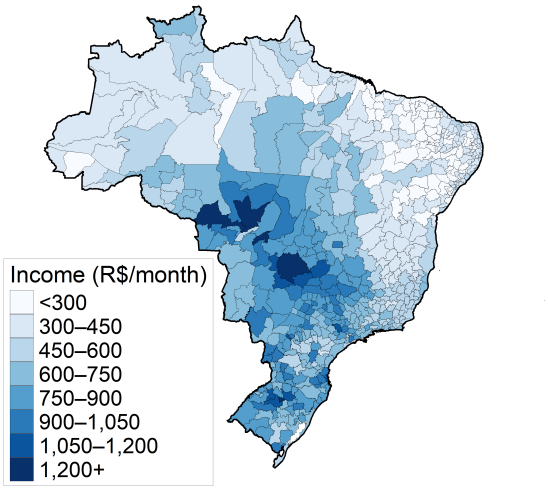
[2] For each crop c, I compute a nationwide reference price $P_c = \left(\sum_i \sum_{t \in \mathcal{T}_0} V_{c,it}\right) / \left(\sum_i \sum_{t \in \mathcal{T}_0} Q_{c,it}\right)$, where $V_{c,it}$ is the value of production in constant 2010 reais and $Q_{c,it}$ the physical quantity in region i and year t, over the base period $\mathcal{T}_0 = \{1994, \ldots, 1998\}$. Constant-price output in region i and year t is then $V_{it} = \sum_c P_c Q_{c,it}$, and measured productivity is $\tilde{A}_{it} = V_{it} / \sum_c X_{c,it}$, where $X_{c,it}$ is the land area planted with crop c.

[3] Some of the highest measured productivities are in urban microregions that farm little of their land, consistent with a model in which the most productive fields are cultivated first: output per cultivated hectare then overstates land productivity where little land is farmed.

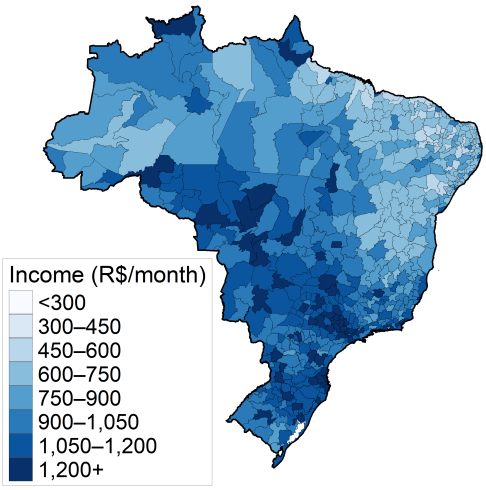
FIGURE I
Regions of Brazil, income & emigration



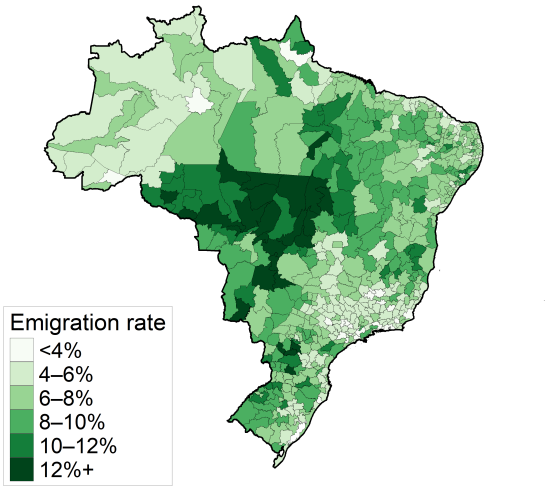
(A) Microregions and major regions of Brazil



(B) Average monthly income of agricultural workers



(C) Average monthly income of non-agricultural workers



(D) Emigration rate, 2005–2010

Notes. Panel (a) shows the five major geographic regions of Brazil, the North, Northeast, Centre-West, Southeast and South, and the 557 microregions defined by the Brazilian Institute of Geography and Statistics (IBGE). Microregion boundaries are marked with thin black lines. Panels (b)–(d) are based on microdata from the 2010 Brazilian Demographic Census (*Censo Demográfico*); the sample is the analysis sample described in Section 2, weighted by census person weights. Panels (b) and (c) show the average monthly income of workers in crop cultivation and of all other workers in each microregion in 2010. Income refers to monthly earnings from employment of workers with positive earnings, in constant 2010 Brazilian Reais (R$); both panels use the same bins. In panel (b), microregions with fewer than 25 sampled farm workers with positive earnings are shown in grey. The 2010 exchange rate was approximately R$1.76 per US dollar. Panel (d) shows the share of people who lived in the microregion in 2005 and lived in a different microregion in 2010.

Temperature data Historical temperature data come from the Brazilian Daily Weather Gridded Data of Xavier et al. [2022], a 0.1° grid interpolated from about 1,250 Brazilian weather stations, which I aggregate to annual microregion means from 1994 to 2017, matched to PAM. The temperature measure is the annual mean of daily maximum and minimum temperature, averaged over the grid cells within each microregion using area weights.

Climate projections Projections through 2100 come from the CMCC-ESM2 Earth System Model of CMIP6 under two scenarios: SSP5-8.5, high emissions, which the main analysis uses, and SSP2-4.5, intermediate emissions, used for comparison. I use only the warming the climate model projects, not its temperature levels. For each microregion, I take the model's annual mean near-surface temperature, averaged with area weights over the grid cells the microregion overlaps,[4] and compute its change relative to the model's own 2006–2015 mean. This change is added to the microregion's observed 2010 temperature, so the projections inherit the observed level and spatial pattern of temperature.

3. Income, migration and climate: four facts

In many low- and middle-income countries, households live close to subsistence, with limited savings and limited access to credit. In Brazil in 2010, 18% of the population lived in households with less than R$ 140 a month per person, the poverty line of the Bolsa Família programme, and 32% in the North and Northeast.[5] For these households, financing a large one-time expense such as a move is hard. This section documents four facts. The first two are patterns consistent with money playing a role in the decision to migrate: within an origin, the low-income migrate least (Fact 1), and when the incomes of the least educated rose, they left lagging places more (Fact 2). The last two concern where climate change hits: Brazil is warming (Fact 3), and heat lowers agricultural productivity, most in the hot and poor North and Northeast (Fact 4). Fact 2 also introduces the variation from which Section 5.3 estimates the fare.

Fact 1: Within an origin, the low-income migrate least

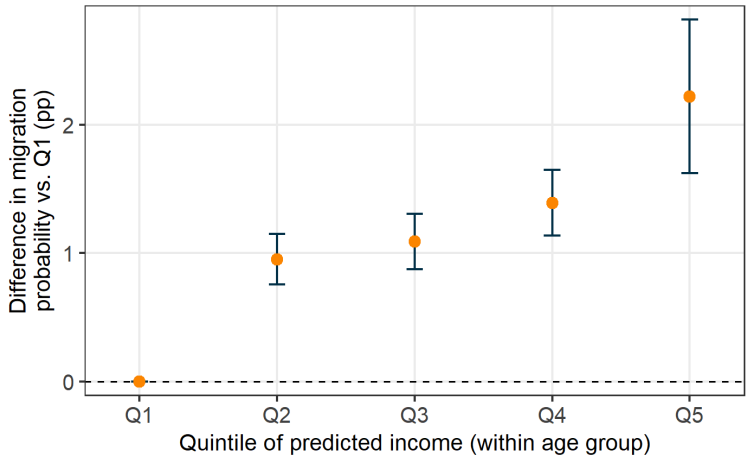
If moving costs cash, then, ceteris paribus, those with fewer resources should migrate less. In Brazil, I find that within the same origin and age group, the probability of having changed microregion over five years rises with every quintile of the income a

[4] The climate model's grid, about 1°, is coarser than most microregions, so neighbouring microregions often share grid cells.

[5] Household total income from all sources divided by the number of residents, 2010 Census, all ages, weighted by census person weights. The line is the one set by Decree 6.917 of 2009.

person would earn at origin, and the top quintile migrates more than the bottom by 34 percent of the mean migration rate (Figure II). The low-income migrate least, consistent with a cash cost of moving.

FIGURE II
Migration by quintile of predicted income at origin, within origin and age group



Notes. Coefficients β_q of equation (1): the difference in the probability of having changed microregion over the five years before the census relative to the lowest quintile of predicted income, within origin microregion × wave and five-year age group × wave, with 95% confidence intervals; standard errors clustered by origin microregion. Predicted income is the individual total income a person would earn at the origin, from regressions on stayers by origin and wave, ranked within age group and wave. Mean predicted income is R$ 236 a month in Q1 and R$ 1,845 in Q5. Brazilian nationals aged 20–65, 2000 and 2010 Censuses pooled (22,445,097 observations), census person weights.

Method and results. The census records income after the move, so for movers it measures what they earn at their destination. Following the literature on the selection of migrants [Borjas et al., 1992, Chiquiar and Hanson, 2005], I compare people through the income their characteristics command at origin: for each origin microregion and census wave, I regress the total income of stayers on schooling, age, age squared, gender and race, and predict, for everyone who lived in the microregion five years earlier, what they would earn there. With predicted incomes grouped into quintiles within each five-year age group and wave, I estimate

$$Y_{ijt} = \sum_{q=2}^{5} \beta_q \, \mathbb{1}\{\text{Pred. Inc.}_{ijt} \in Q_q\} + \mu_{jt} + \kappa_{a(i)t} + \varepsilon_{ijt}, \tag{1}$$

where Y_{ijt} is one if person i, who lived in microregion j five years before census t, lives in another microregion at the census, and μ_{jt} and $\kappa_{a(i)t}$ are origin × wave and age group × wave fixed effects. Against a mean five-year migration rate of 6.6%, the gradient resembles those documented for emigration from Mexico and poor countries [Chiquiar and Hanson, 2005, McKenzie and Rapoport, 2007, Dustmann and Okatenko,

2014, Clemens and Mendola, 2024]. Those with less income also go less far when they move (Appendix B.5).

Robustness. The gradient holds when income is predicted with other regressors or at another level of geography, when it is predicted from the census before the move, and in each census wave, for men and women, in poor and rich regions, and for moves across states; it blurs only with very fine income bins, which tie many people at the same predicted income. It is steepest where moving pays: splitting the sample by whether a person's origin has lower mean income than the places near it, for the person's education level, the top quintile migrates more than the bottom by 43% of the mean rate where the origin lags and by 21% elsewhere. Origin fixed effects and ranking within age group matter, because the low-income live in origins from which many leave and the young are both mobile and poor (Appendix B.4).

Two other readings of the gradient remain. First, it may reflect schooling rather than income: 79% of it runs through schooling (Table B.I), and the least educated may face barriers to moving other than cash, such as less information or weaker networks. Second, the poor may have less to gain from moving. On average, the gains run the other way: in 2010, the mean income of adults with less than primary schooling was 83% higher in the Southeast than in the average microregion of the North and Northeast, against 16% for university graduates (Table B.II). If anything, the least educated have stronger incentives to leave. Fact 2 addresses both readings: it follows the same groups over time, allowing each group to be less mobile for reasons unrelated to income, and shows that their emigration responds to changes in their income, given an explicit measure of the gain from moving.

Fact 2: When the least educated got richer, they left lagging places more

If the poor move less because they cannot pay for the move, they should leave disadvantaged regions when their incomes rise. When the 2000s brought Brazil a large pro-poor income shift, the least educated, whose incomes grew fastest, indeed became more likely to leave places that lag behind their neighbours.

Method and results. In the 2000s the real minimum wage rose steeply and Bolsa Família and non-contributory pensions expanded, national policies that did not respond to conditions in any one place. The mean real income of adults with less than primary schooling grew by 24% between 2000 and 2010, while that of university graduates changed by -18%; the 1990s had run the other way. As in studies that use income changes to test for a cost of moving the poor cannot finance [Cattaneo and Peri, 2016,

Bazzi, 2017, Angelucci, 2015, Bastian and Black, 2024], I relate these shifts to emigration of groups g (education level $\times$ five-year age group) from microregions j, over three five-year migration windows observed in the 1991, 2000 and 2010 censuses (12,646 cells). The *return to migration* measures how far j lags behind the places near it for the group's education level $e(g)$, from 1991 incomes:

$$R_{jg} = \underbrace{\ln \frac{\sum_{i \neq j} e^{-d_{ji}/d_0}\, \bar{y}_{i,e(g),1991}}{\sum_{i \neq j} e^{-d_{ji}/d_0}}}_{\text{income in the places near } j} - \underbrace{\ln \bar{y}_{j,e(g),1991}}_{\text{income in } j}, \tag{2}$$

with d_{ji} the distance between microregions and d_0 a decay distance that sets how fast the weight of other places falls with distance, 300 km in the baseline (robustness below varies it). The number of movers M_{jgt} out of the P_{jgt} people who lived in j five years earlier is

$$\mathbb{E}[M_{jgt}] = P_{jgt} \exp\Big(\underbrace{\alpha_{jg} + \psi_{jt} + \omega_{gt}}_{\text{fixed effects}} + \underbrace{\sum_t b_t R_{jg}}_{\text{return by census}} + \underbrace{\rho\, L_{gt} R_{jg}}_{\text{income} \times \text{return}} \Big), \tag{3}$$

estimated by Poisson pseudo-maximum likelihood, where L_{gt} is the log of the group's national mean real income relative to 1991. The coefficient of interest is ρ: how much more strongly a group's emigration responds to the return to migration when the group's income is higher. With a cash cost of moving, a richer group can better afford to act on the gain from leaving a lagging place, so $\rho > 0$; if income did not matter for moving, ρ would be zero. Formally, a rise in group income by one log point raises the semi-elasticity of emigration with respect to the return by ρ. With cell, microregion $\times$ census and group $\times$ census effects, and a linear trend in each group's response to the return, ρ is identified from the change in a group's income growth between the two decades.

Table I, column (1), gives $\rho = 0.76$ (standard error 0.30): groups whose income rose more left lagging places more. Column (2) estimates ρ separately for the 7 poorest groups, those whose national mean income in 1991 was at or below the median across groups, and for the others: 0.89 (standard error 0.23) against 0.54 (0.49). The difference has the sign a cash cost predicts but is not statistically significant (p-value 0.32); in levels of the emigration rate it is (column 4, p-value 0.02).

Robustness. The pattern is the same in levels of the emigration rate (columns 3 and 4), and with d_0 at half or twice its value, or inverse-distance weights, ρ is 0.75, 0.82 and 0.92. In the 2000–2010 comparison of Appendix C, the estimate without the least educated is -2.16 (standard error 2.47), so the estimate comes from their income shift. Bolsa Família, which appears to keep beneficiaries in place [Fontes et al., 2019], works

TABLE I
Income growth and emigration from lagging places, three censuses

	Movers, Poisson		Emigration rate (pp), OLS	
	Pooled (1)	Separate by group (2)	Pooled (3)	Separate by group (4)
Group income × return, ρ	$+0.76^{**}$		$+9.06^{***}$	
	(0.30)		(2.88)	
	[0.04]		[0.04]	
poorest groups		$+0.89^{***}$		$+10.92^{***}$
		(0.23)		(2.64)
		[0.21]		[0.21]
other groups		$+0.54$		$+5.87$
		(0.49)		(3.67)
		[0.25]		[0.08]
Test of equal ρ: poorest − other		$+0.35$		$+5.05^{**}$
		(0.35)		(2.15)
		[0.58]		[0.54]
Group time trend	✓	✓	✓	✓
Cells × censuses	37,938	37,938	37,938	37,938

Notes. Equation (3), migration windows 1986–91, 1995–2000 and 2005–10. (1)–(2): Poisson on movers, population as exposure; (3)–(4): OLS on the emigration rate in percentage points, population-weighted. All columns include cell, microregion × census and group × census effects, the return interacted with census dummies, and a linear trend in each group's response to the return. (2) and (4): ρ differs between the 7 poorest groups (national mean income in 1991 at or below the median across groups) and the others, estimated in one regression; the difference is tested with the clustered covariance of the two. Standard errors clustered by microregion in parentheses; permutation p-values over the 24 assignments of income paths across education levels in brackets. *** 1%, ** 5%, * 10%.

against ρ.

Inference. The income variation comes from only four national income paths, one per education level, so clustered standard errors may overstate the precision. As a placebo test, I re-estimate ρ after giving each education level the income path of another level, in each of the 24 possible assignments, one of which is the actual one. If income growth did not affect emigration, the actual assignment would be no more likely than any other to produce the largest estimate. It produces the largest, so the permutation p-value is $1/24 \approx 0.04$, the smallest attainable with four paths. For the difference between the poorest groups and the others, the permutation p-value is 0.58 (0.54 in levels). Appendix C reports the income shifts, the estimate as a binned scatter and in maps, the response by initial income, and the 2000–2010 comparison.

Fact 3: Brazil is warming

Brazil is warming. Under SSP5-8.5, the median microregion warms by 5.4°C by 2100, and microregions holding 32 percent of the 2010 population will have an annual mean above 30°C, a level none reached in 2010. Most of them are in the North and Northeast, where in some microregions up to 60 percent of workers are in agriculture.

Method and results. The projections are taken as given, from a CMIP6 climate model under two emission scenarios (Section 2): SSP2-4.5, intermediate emissions, and SSP5-8.5, high emissions. Figure III, Panel A, shows the 2010 temperatures (left) and the projected path of the median microregion with the spread across microregions (right). Under SSP2-4.5 the median microregion warms by 2.7°C by 2100. Under SSP5-8.5 the North, already the hottest region, reaches an average of 33.1°C and the hottest microregion 34.3°C, and 52.6 percent of microregions end the century hotter than the hottest microregion-year observed between 1994 and 2017 (29.7°C).

Fact 4: Heat lowers agricultural productivity

How warming affects incomes, and with them the incentive to migrate, depends on how agricultural productivity responds to temperature. Productivity rises with the annual mean temperature up to about 19.8°C and then falls at an increasing rate (Figure III, Panel B). Most microregions, above all in the hot North and Northeast, are already past this optimum, so warming lowers farm productivity most where people are poorest.

Method and results. I estimate the response from year-to-year variation in temperature within microregions. Because farmers cultivate the best fields first, output per cultivated hectare overstates a microregion's land productivity where little land is farmed. Following Costinot et al. [2016], I correct measured productivity $\tilde{A}_{it}$ for the share of land under cultivation s_{it}, $A_{it} = \tilde{A}_{it}\, s_{it}^{1/\theta}$, with $\theta = 2$, the midpoint of available estimates [Sotelo, 2020, Costinot et al., 2016], and estimate

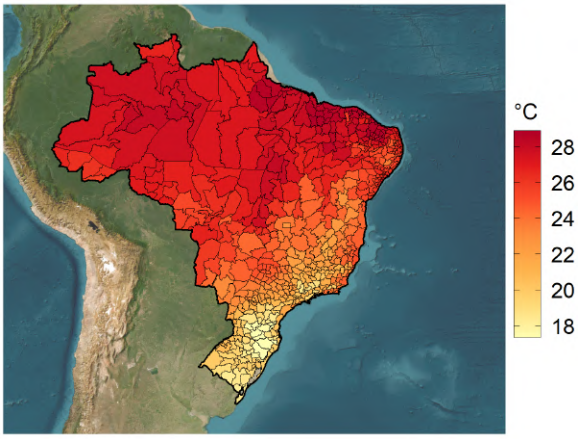
$$\ln A_{it} = \beta_1 T_{it} + \beta_2 T_{it}^2 + \mu_i + \sum_{p=1}^{3} \lambda_{r(i)p}\, t^p + \varepsilon_{it}, \tag{4}$$

with microregion fixed effects μ_i and macro-region-specific cubic time trends. The relationship is concave, with its maximum at 19.8°C (95% confidence interval [19.0, 20.6]°C).

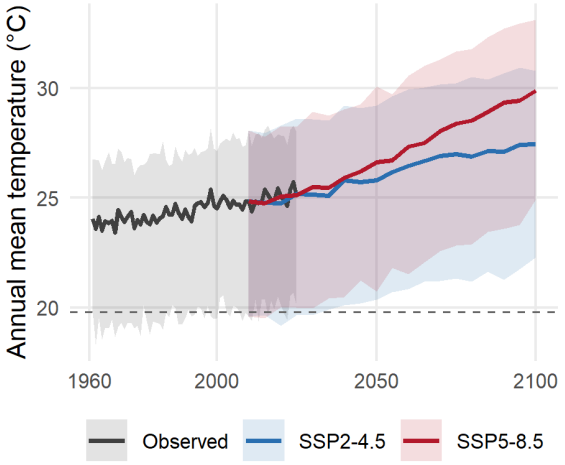
Robustness. Three choices could drive the estimate. *The land-use correction:* with half the correction ($\theta = 4$ instead of 2) or none, the curvature and the optimum barely

FIGURE III
Temperature in Brazil and agricultural productivity

Panel A: Fact 3, Brazil is warming

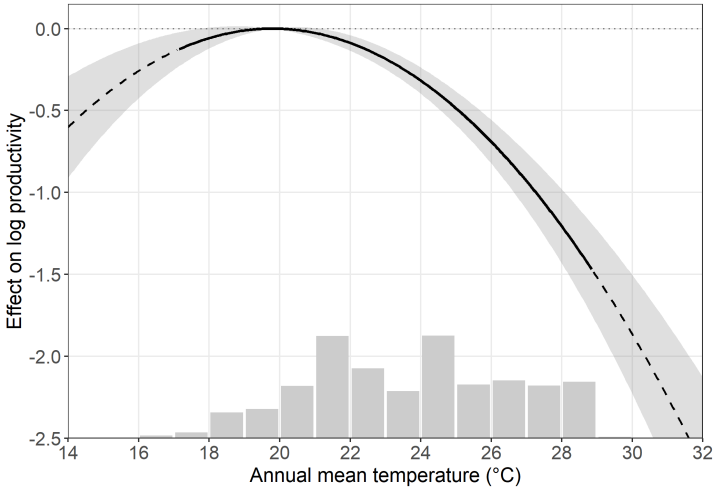


Temperature in 2010



Projected temperature under SSP2-4.5 and SSP5-8.5

Panel B: Fact 4, heat lowers agricultural productivity



Notes. Temperature is the annual mean temperature. Panel A, left: 2010, from the station-based gridded data of Xavier et al. [2022], averaged at the microregion level using area weights; the colour scale runs from the 1st to the 99th percentile across microregions, and values beyond are drawn at the end colours. Panel A, right: the median microregion's temperature, observed from 1961 to 2025 in the same gridded data (grey) and projected to 2100 under SSP2-4.5 and SSP5-8.5 (Section 2); projected values are five-year averages, shaded areas the 10th and 90th percentiles across microregions, and the dashed line the productivity-maximizing temperature of Panel B. Panel B: equation (4), log fundamental agricultural productivity (constant-price output per planted hectare of the 44 crops reported consistently, adjusted for land-use intensity) on annual mean temperature, microregions 1994–2017 (13,254 microregion-years), normalised to zero at the maximum. Shaded: 95% confidence interval, standard errors clustered by microregion. Solid over the observed range of temperatures (0.5th to 99.5th percentile), dashed where extrapolated. Grey bars: the 2010 population across 1°C bins of temperature, not read on the vertical axis.

change (Table B.IV, lower panel). *The crop mix:* farmers can respond to heat by switching to crops that fare better. Holding each microregion's crop mix fixed at its 1994–98 shares steepens the curvature of measured productivity from -0.0154 to -0.0218, so switching absorbs about 29% of it; the estimate includes this margin of adaptation (Table B.V). *Common weather shocks:* with state × year effects, which absorb most of the common variation in weather, the curvature is smaller and the optimum lower and imprecise (Table B.IV). Table B.IV also reports other fixed effects and trends. Estimated crop by crop, the optima of Brazil's main crops lie between 19.4 and 22.1°C, close to the aggregate optimum, and most places where these crops are grown are already warmer (Figure B.IV).

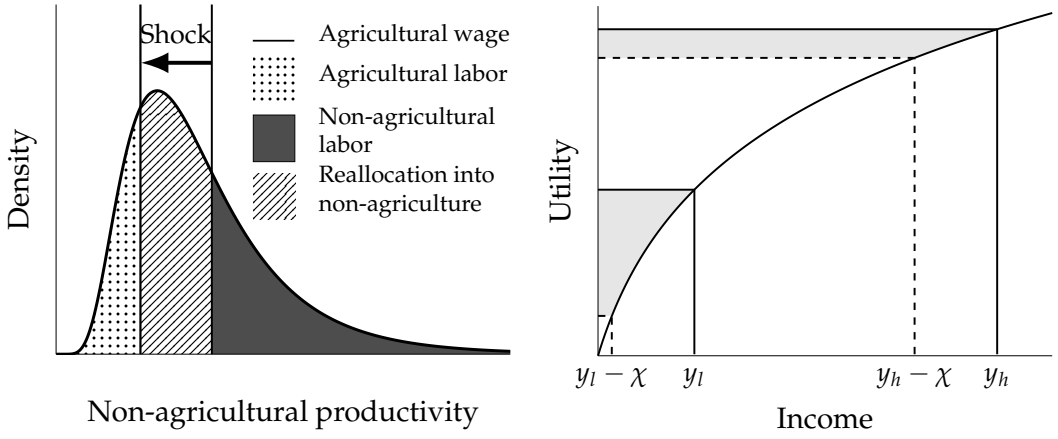
Limits. The estimate captures adaptation through the crop mix but not slower adaptation, such as new varieties or irrigation [Mendelsohn et al., 1994, Deschênes and Greenstone, 2007]; to the extent that it takes place, the projections overstate long-run damages, although in US agriculture long-run adaptation has offset little of the effect of extreme heat [Burke and Emerick, 2016]. And where projected temperatures leave the observed range (Fact 3), the damage function extrapolates.

What the facts imply

Facts 1 and 2 are consistent with a cash cost of moving. A fare of the same size weighs more on a poor household than on a rich one, so the poor leave less (Fact 1), and as their incomes rise the fare weighs less on them, so they start leaving lagging places (Fact 2). Facts 3 and 4 say where the pressure to move will come from: warming lowers farm productivity most in the hot, poor North and Northeast, where many of those least able to pay a fare live.

Figure IV sketches how the model links the two. When heat lowers a region's farm wage, workers whose non-agricultural productivity was below it move into low-productivity non-agricultural jobs, so mean income falls and more workers end up with low incomes (panel a). Evidence from Brazil and India is in line with this: persistent dryness in Brazil lowers agricultural employment, with some workers moving into local manufacturing and others migrating out [Albert et al., 2021]; heat moves Indian workers from agriculture into non-agricultural jobs [Colmer, 2021]; and farm wages fall more after productivity shocks where workers are poorer and less able to borrow or migrate [Jayachandran, 2006]. If the shock hits regions unevenly, these workers gain most from leaving, but a fare of the same size costs them more utility than it costs a richer worker (panel b), so the shock that raises the value of leaving also makes leaving harder. Where the incentive to migrate is strongest, the means to do so are weakest. Section 4 quantifies what this means for who loses from climate change.

FIGURE IV
Economic adjustment under productivity shocks.



(A) Distribution of productivity and sectoral labor allocation.

(B) Utility-burden of a monetary migration cost χ.

Notes. Panel (a) illustrates how a negative productivity shock reduces the agricultural reservation wage, reallocating marginal workers into low-productivity non-agricultural jobs. Panel (b) illustrates how a monetary migration cost χ translates into a higher utility-equivalent burden for low-income individuals than for high-income individuals, muting migration even when wage gaps across regions widen. Utility is logarithmic, as in the model.

4. Model

Section 3 suggests that part of the migration cost is paid in cash, and that warming will lower farm productivity most where people are poorest. To turn these facts into numbers, I build a dynamic spatial general-equilibrium model with a monetary migration cost, for three purposes. First, to quantify the welfare losses from projected climate change, which has not yet happened at this scale: the model traces the temperature–productivity link of Fact 4 through wages, prices, rents and migration, in the regions that warm and in those people move to. Second, to switch parts of the economy off, such as the monetary migration cost, and measure how much of the losses, and of their distribution across incomes, they account for. Third, to evaluate policies that have not been tried, such as a tax-financed subsidy to the migration cost.

The model builds on the dynamic spatial model of Caliendo et al. [2019] and departs from it in five ways: (i) migration costs have a monetary part, a fare paid out of income, alongside the non-monetary part; (ii) incomes differ within each region and skill group, through the choice between farming and non-agricultural work; (iii) agriculture is Ricardian, with heterogeneous land, and climate enters through the productivity of that land; (iv) food demand is non-homothetic; and (v) a budget-balanced fiscal system lets taxes and transfers change both consumption and the burden of the fare. Appendix D derives the expressions of this section and proves its results, and Appendix E describes how the equilibrium is computed.

4.1. Overview and timing

The model has two interacting blocks: (i) a *dynamic* block governs location choices: forward-looking workers, heterogeneous in skill and income, choose where to live subject to monetary and non-monetary migration costs; (ii) a *static* block determines, within each period, regional prices and wages given the spatial distribution of workers, through agricultural and non-agricultural production, occupational choice, and trade. The agent of the model is a worker: an adult who works, earns an income, and chooses where to live.

The economy has regions $j, i \in \mathcal{S}$ and discrete time $t \in \{1, \ldots, T\}$. Within period t:

1. The period starts with the masses $\{L_t^{ik}\}$ of workers across regions and skill groups and within-cell income distributions $\{F_t^{ik}\}$.
2. *Static equilibrium.* Given the regional skill composition, prices $\{p_t^i\}$ and agricultural wages $\{w_{A,t}^i\}$ are determined; each worker makes an occupational choice and realizes income y in its current region.
3. *Location choice.* Workers draw idiosyncratic location preference shocks and choose where to reside next period, subject to migration costs.
4. Workers relocate; the period closes with updated masses $\{L_{t+1}^{ik}\}$.

Workers thus earn and consume in their current region and relocate for the next period.

4.2. Dynamic component

Setup. A worker is characterized by its current region j, a permanent skill group $k \in \{1, \ldots, K\}$, and its current income y. Income is realized each period in the region of residence and comes from the static block (Section 4.3). The location choice takes as given, for each region i and skill group k, the distribution F_t^{ik} of income among the workers who live there.

Dynamic discrete-choice problem. A worker in region j with skill k and current income y chooses next period's region i to maximize expected lifetime utility:

$$\underbrace{U_t^j(y)}_{\text{utility today}} + \max_{i \in \mathcal{S}} \Big\{ \underbrace{\beta\, \overline{V}_{t+1}^{i,k}}_{\text{value of destination}} - \underbrace{\tau^{jk,i}(y)}_{\text{migration cost}} + \underbrace{\nu\, \xi_t^i}_{\text{taste shock}} \Big\}.$$

Here $U_t^j(y)$ is the utility of spending income y at the prices of food and housing in the current region j, logarithmic in income. A period lasts five years, and income and prices are constant within it, so $U_t^j(y)$ is five times the annual utility of spending the

year's income, and the taste shock below is drawn once per period. The discount factor $\beta \in (0,1)$ is per period, and $\overline{V}^{i,k}_{t+1}$ is the expected value of *entering* region i next period with skill k (the expectation over next period's income draw and taste shocks, defined in (9)). The shocks ξ^i_t are i.i.d. across destinations, workers, and time, drawn from a Type-I extreme value (Gumbel) distribution with *unit* scale, centered to have mean zero, and enter utility premultiplied by the dispersion parameter $\nu > 0$, so the composite taste shock $\nu\,\xi^i_t$ has scale ν and mean zero. The consumption term $U^j_t(y)$ does not depend on the destination: workers consume in their current region, and i enters only through the continuation value and the migration cost.

As in Caliendo et al. [2019], the cost of migrating is stated as an assumption.

Assumption 1 (Migration costs)**.** *The cost of migrating from j to i is additive and has two parts. The non-monetary part $\check{\tau}^{jk,i}$ depends on the origin j, the destination i and the skill group k, is time invariant, and is measured in terms of utility. The monetary part is a fare $\chi^{j,i}$ in reais, paid out of the income of the year of the move up to a share $1-\underline{s}$ of that year's income.*

Under Assumption 1, the utility cost of migrating from j to i is

$$\tau^{jk,i}(y) = \underbrace{\check{\tau}^{jk,i}}_{\text{non-monetary}} + \underbrace{\tilde{\tau}^{j,i}(y)}_{\text{monetary}}, \tag{5}$$

$$\tilde{\tau}^{j,i}(y) = -\ln \max\Big(1 - \frac{\chi^{j,i}}{Y},\, \underline{s}\Big), \qquad Y = 12\,y, \tag{6}$$

$$\chi^{j,i} = c\,\ln \mathrm{dist}^{j,i}, \qquad i \neq j. \tag{7}$$

Staying is costless: $\chi^{j,j} = \check{\tau}^{jk,j} = 0$, so every worker, however poor, can always stay.

(1) Monetary component of migration costs. The fare $\chi^{j,i}$ is a cash cost in reais, the same for every worker on the route, which rises with the logarithm of the distance between region centroids, with $c > 0$. It is a money-metric cost of relocating—search, deposits, the move itself, and earnings lost in the transition—rather than a ticket price: a subsidy of one real lowers it by one real. The fare is paid out of the income of the year of the move, Y, twelve times the monthly income y with which the rest of the model works. Paying it leaves the mover the share $1-\chi/Y$ of that year's income. Four of the period's five years are consumed out of Y and one out of $Y-\chi$, so, measured in log income, the move costs $-\ln(1-\chi/Y)$ of period utility. For a small fare this is about χ/Y, the fare's share of the year's income. It rises more than in proportion as the fare approaches a year's income, because each real is then taken from less. A mover, however, always keeps at least the share $\underline{s}$ of the year's income, as family, credit or savings bridge the rest. A fare above $1-\underline{s}$ of a year's income therefore costs $-\ln \underline{s}$, as much as living one year on the share $\underline{s}$ of one's income, whatever its size. The cost is largest for the poorest, and no worker is ever unable to move. Section 5.3 takes this cost

to the census data. Because the fare is paid in money, its utility cost falls with income under any strictly concave utility (Lemma 7).

(2) Non-monetary component of migration costs. Migration also entails costs that do not pass through the budget—psychic costs of leaving home, cultural and linguistic barriers, disrupted local networks—collected in $\check{\tau}^{jk,i}$. It is the same at every income and is otherwise unrestricted: it varies across origin–destination pairs and with distance, and it depends on the worker's skill group k, because workers with different education from the same origin rely on different networks and look for different jobs.

Expected lifetime utility. Integrating over the taste shocks gives the value of a worker with income y, and integrating over the income distribution of the cell the value of entering it,

$$V_t^{j,k}(y) = U_t^j(y) + \nu \log \sum_{i \in \mathcal{S}} \exp\left[\frac{\beta\, \overline{V}_{t+1}^{i,k} - \tau^{jk,i}(y)}{\nu}\right], \tag{8}$$

$$\overline{V}_t^{j,k} = \int V_t^{j,k}(y)\, dF_t^{j,k}(y), \tag{9}$$

which closes the recursion.

4.3. Static trade model

Taking the spatial distribution of workers and their skills as given, the static block determines regional prices and wages: production depends on local fundamentals (including climate), goods are traded subject to frictions, and labor is allocated across sectors through occupational choice. The non-agricultural good is the numeraire. This subsection states the primitives; the equilibrium objects they imply are assembled in Section 4.4.

4.3.1. Production

Agriculture. Region i is endowed with a fixed measure X^i of land, partitioned into a continuum of infinitesimal plots ω; all plot-level objects are per unit of land. A plot operated with labor $L_t^i(\omega)$ produces with Cobb–Douglas technology

$$q_t^i(\omega) \;=\; \left[L_t^i(\omega)\right]^{\alpha} \left[z_t^i(\omega)\right]^{1-\alpha}, \qquad \alpha \in (0,1),$$

where $z_t^i(\omega)$ is plot productivity. Operating a plot additionally requires a fixed per-period cost $z^{i0}(\omega)$, paid in the numeraire. Both are drawn from a joint Fréchet specifica-

tion: with scale parameters $A_t^i > 0$ and $F^{i0} > 0$ and common shape $\theta > 1$,

$$\Pr\Big(z_t^i(\omega) \le z,\ z^{i0}(\omega) \le z^0\Big) \;=\; \exp\left\{-\kappa_\theta\left[\left(\frac{z}{A_t^i}\right)^{-\theta} + \left(\frac{z^0}{F^{i0}}\right)^{-\theta}\right]\right\}, \quad \kappa_\theta \equiv \Gamma\Big(1-\tfrac{1}{\theta}\Big)^{-\theta},$$

so the marginals are independent and the normalization κ_θ delivers $A_t^i = \mathbb{E}[z_t^i(\omega)]$ and $F^{i0} = \mathbb{E}[z^{i0}(\omega)]$: A_t^i is the region's *fundamental* (unconditional mean) productivity, the margin through which climate enters production. The common shape θ governs how much plots within a region differ: the higher θ, the more alike they are, so the cultivated share responds more to the return on farming (18), and the less measured productivity exceeds fundamental productivity (19).

Non-agriculture. The non-agricultural good is produced from labor alone under constant returns, $Q_{n,t}^i = B^i N_t^i$, where N_t^i is the efficiency units of labor employed in the sector and B^i is regional non-agricultural productivity. Because the good is the freely traded numeraire, its price is one everywhere, and constant returns pin the wage per efficiency unit to productivity, $w_N^i = B^i$, however much labor is supplied locally.

4.3.2. Skills and occupational choice

Earnings and occupational choice. Farming is always available, and every farmer supplies one efficiency unit and earns the agricultural wage $w_{A,t}^i$. Non-agricultural labor is heterogeneous: a worker in skill group k who draws $\eta \sim \mathcal{N}(0, \sigma^2)$ supplies $e^{\mu_k+\eta}$ efficiency units, where μ_k is the group's skill premium. A worker compares the two sectors up to a wedge κ^{ik} and works outside agriculture iff its non-agricultural earnings exceed the agricultural wage by the factor $e^{\kappa^{ik}}$:

$$\check{y}_N^i(k,\eta) = w_N^i\, e^{\mu_k+\eta}, \tag{10}$$

$$\check{y}_N^i(k,\eta) \ge e^{\kappa^{ik}}\, w_{A,t}^i \quad\Longleftrightarrow\quad \eta \;\ge\; \bar{\eta}_t^{ik} \equiv \ln w_{A,t}^i + \kappa^{ik} - \ln w_N^i - \mu_k. \tag{11}$$

The wedge is specific to each region and skill group and constant over time. It captures what makes crop work more or less attractive than its pay, a non-monetary cost of farm work or rationed access to it, and enters the choice of occupation only, not utility; with $\kappa^{ik} = 0$ the worker chooses the sector that pays more. Realized nominal market income is

$$\check{y} \;=\; \begin{cases} w_{A,t}^i, & \text{if it farms } (\eta < \bar{\eta}_t^{ik}), \\ w_N^i\, e^{\mu_k+\eta}, & \text{if it works in non-agriculture } (\eta \ge \bar{\eta}_t^{ik}). \end{cases} \tag{12}$$

Disposable income (taxes and transfers). A generic, budget-balanced fiscal system maps market income into the disposable income that workers actually spend and carry to their migration decision,

$$y \;=\; \breve{y} - \mathcal{T}_t^i(\breve{y}) + S_t^i, \tag{13}$$

with tax schedule $\mathcal{T}_t^i(\cdot)$ and (possibly region-specific) lump-sum transfer S_t^i, balanced so that it redistributes without creating resources. Applied to (12), it delivers the within-cell distribution of *disposable* income F_t^{ik} used in the dynamic block: a mass point at the post-fiscal agricultural wage (the farmers) and a lognormal tail of non-agricultural earnings. Setting $\mathcal{T} = S = 0$ recovers $y = \breve{y}$.

4.3.3. Demand and trade

Preferences. Workers and landowners spend their income y on three goods: food, an agricultural composite $C_t^{i;a}$ at price $P_{a,t}^i$; the non-agricultural good $C_t^{i;n}$, the numeraire; and housing services H_t^i at rent r_t^i. Preferences are non-homothetic, so that poorer consumers spend a larger share of their income on food. They are of the PIGLOG class of Deaton and Muellbauer [1980], which has no closed-form utility over the goods and is defined by the indirect utility of income,

$$U_t^i(y) \;=\; 5\,\frac{\log y - \log \bar{a} - \log P_t^i}{\mathcal{B}_t^i}, \tag{14}$$

with $P_t^i = (P_{a,t}^i)^{\gamma_a}(r_t^i)^{\gamma_h}$, $\mathcal{B}_t^i = (P_{a,t}^i)^{\gamma_y}$, $\gamma_a, \gamma_h \in (0,1)$ and $\gamma_y \leq 0$; it is the period utility of the dynamic block, five years of the annual indirect utility. Housing is a non-traded service yielded by a fixed residential-land endowment $\bar{H}^i$ in each region, distinct from the farmland X^i; its local rent r_t^i is determined in (24). The agricultural composite is a CES aggregate of regional varieties,

$$C_t^{i;a} = \left(\sum_{j \in \mathcal{S}} (\psi^j)^{1/\epsilon} (c_t^{j,i;a})^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}},$$

with Armington elasticity ϵ and preference weights ψ^j (normalized to sum to one), where $c_t^{j,i;a}$ is consumption in i of the variety from origin j.

Prices and trade. Shipping the agricultural variety from j to i incurs iceberg trade costs $d^{j,i} \geq 1$, with $d^{i,i} = 1$; the non-agricultural good is freely traded. Given farmgate prices $\{p_t^j\}$, the CES price index of the agricultural composite and the regional cost-of-living index are

$$P_{a,t}^i = \left(\sum_{j \in \mathcal{S}} \psi^j \left(p_t^j d^{j,i} \right)^{1-\epsilon} \right)^{\frac{1}{1-\epsilon}}, \qquad P_t^i = \left(P_{a,t}^i \right)^{\gamma_a} \left(r_t^i \right)^{\gamma_h}, \tag{15}$$

the second equality using that the numeraire has price one everywhere and that r_t^i is the local housing rent (24); a denser, higher-income region carries a higher rent and hence a higher cost of living.

4.4. Closing the model

The two blocks interact through a fixed point. Given the masses $\{L_t^{ik}\}$, the static block determines $\{p_t^i, w_{A,t}^i\}$, and hence the income distributions $\{F_t^{ik}\}$ via (12) and the price indices via (15); the dynamic block maps these into value functions (8)–(9), migration probabilities, and the masses $\{L_{t+1}^{ik}\}$ that open the next period.

Migration probabilities and labor dynamics. The choice probabilities are multinomial logit,

$$m_t^{jk,i}(y) = \frac{\exp\left[\left(\beta\, \overline{V}_{t+1}^{i,k} - \tau^{jk,i}(y)\right)/\nu\right]}{\sum_{l\in\mathcal{S}} \exp\left[\left(\beta\, \overline{V}_{t+1}^{l,k} - \tau^{jk,l}(y)\right)/\nu\right]}. \tag{16}$$

The fare makes migration depend on income:

Proposition 1 (Income and migration)**.** *Within an origin and skill group, the probability of leaving rises with income, and, conditional on leaving, so does the expected distance of the move wherever the fares stay clear of the floor.*

This is the pattern of Fact 1. The proof is in Appendix D.4. Weighting each origin's moving propensities by its mass, the stock of skill-k workers evolves as

$$L_{t+1}^{ik} = \sum_{j\in\mathcal{S}} L_t^{jk} \int m_t^{jk,i}(y)\, dF_t^{jk}(y), \tag{17}$$

and the income distributions of period $t+1$ follow from occupational sorting at that period's wages, so only the masses carry history.

Land use and agricultural output. Taking p_t^i and $w_{A,t}^i$ as given, the grower on plot ω hires labor until its marginal product equals the wage, which leaves a variable profit $v_t^i z_t^i(\omega)$ linear in plot productivity, with $v_t^i = (1-\alpha)\alpha^{\alpha/(1-\alpha)}(p_t^i)^{1/(1-\alpha)}(w_{A,t}^i)^{-\alpha/(1-\alpha)}$. The plot is cultivated iff this covers its operating cost, $v_t^i z_t^i(\omega) \geq z^{i0}(\omega)$, so the cultivated share and measured productivity, the average productivity of cultivated land, are

$$s_t^i = \frac{\left(v_t^i A_t^i\right)^\theta}{\left(v_t^i A_t^i\right)^\theta + \left(F^{i0}\right)^\theta}, \tag{18}$$

$$\tilde{A}_t^i = A_t^i \left(s_t^i\right)^{-1/\theta} \geq A_t^i. \tag{19}$$

Aggregate output and agricultural labor demand are

$$Q^i_{a,t} = X^i \left(\frac{\alpha p^i_t}{w^i_{A,t}} \right)^{\frac{\alpha}{1-\alpha}} A^i_t \, (s^i_t)^{1-\frac{1}{\theta}}, \qquad L^{D,i}_{a,t} = X^i \left(\frac{\alpha p^i_t}{w^i_{A,t}} \right)^{\frac{1}{1-\alpha}} A^i_t \, (s^i_t)^{1-\frac{1}{\theta}}. \tag{20}$$

Operating outlays, paid in the numeraire, are $\mathcal{F}^i_t \equiv X^i F^{i0} \left[1 - (1 - s^i_t)^{1-\frac{1}{\theta}} \right]$, and land rents, the residual of revenue after wages and these outlays, accrue to landowners.

Occupational sorting and labor supply. From the cutoff (11), the share of skill group k in region i that works outside agriculture, and the region's labor supplies to the two sectors, in bodies to agriculture and in efficiency units to non-agriculture, are

$$\pi^{ik}_{N,t} = \Phi \left(\frac{\ln w^i_N + \mu_k - \ln w^i_{A,t} - \kappa^{ik}}{\sigma} \right), \tag{21}$$

$$L^{S,i}_{a,t} = \sum_k L^{ik}_t (1 - \pi^{ik}_{N,t}), \qquad N^i_t = \sum_k L^{ik}_t \, \pi^{ik}_{N,t} \, \mathbb{E} \left[e^{\mu_k + \eta} \mid \eta \geq \bar{\eta}^{ik}_t \right], \tag{22}$$

where Φ is the standard normal CDF.

Demand, housing, and goods markets. Farmland and residential land in region i are owned by immobile landowners resident there. Regional income Y^i_t, which residents spend, is labor income plus agricultural and residential land rents. By Roy's identity, a consumer with income y spends the share

$$\vartheta^i_t(y) \;=\; \gamma_a + \gamma_y \log \frac{y}{\bar{a} \, P^i_t} \tag{23}$$

of it on food, which falls with income when $\gamma_y < 0$, the constant share γ_h on housing, and the rest on the numeraire. The region spends on food the income-weighted average $\bar{\vartheta}^i_t$ of these shares, which depends on its whole income distribution and is lower the richer the region. Housing is non-traded and supplied by the fixed residential-land endowment $\bar{H}^i$, so its market clears within the region at the rent

$$r^i_t \;=\; \frac{\gamma_h \, Y^i_t}{\bar{H}^i}, \tag{24}$$

which rises with local expenditure and hence with population density, the dispersion force of the model. With CES demand, the market for each region's crop variety clears when the value of its output equals expenditure on it across all regions,

$$p^j_t \, Q^j_{a,t} \;=\; \sum_{i \in \mathcal{S}} \psi^j \, (p^j_t d^{j,i})^{1-\epsilon} \, (P^i_{a,t})^{\epsilon - 1} \, \bar{\vartheta}^i_t Y^i_t, \tag{25}$$

which, given (20), determines the farmgate prices $\{p_t^i\}$. The market for the numeraire, which residents consume and farms use for their operating outlays, then clears by Walras' law.

Labor market clearing. The agricultural labor market clears region by region,

$$L_{a,t}^{D,i}(p_t^i, w_{A,t}^i) \;=\; L_{a,t}^{S,i}(w_{A,t}^i; \{L_t^{ik}\}), \tag{26}$$

which determines $w_{A,t}^i$: demand (20) is decreasing in $w_{A,t}^i$ (through v_t^i and s_t^i), supply (22) increasing (a higher agricultural wage draws workers out of non-agriculture). The non-agricultural market requires no separate condition: labor demand is perfectly elastic at $w_N^i = B^i$.

Equilibrium. Given initial masses $\{L_0^{ik}\}$, the paths of fundamentals Θ_t (productivities, land, migration and trade costs, preference weights and the fiscal system), a competitive equilibrium is a path of prices, agricultural wages and rents $\{p_t^i, w_{A,t}^i, r_t^i\}$, values $\{V_t^{jk}, \overline{V}_t^{jk}\}$, migration probabilities, income distributions and masses such that workers and growers optimize, (11)–(13) and (18)–(22); workers solve the dynamic problem, (8)–(9) and (16); housing, goods and labor markets clear, (24), (25) and (26); the masses evolve by (17); and a terminal condition pins down $\{\overline{V}_T^{ik}\}$.

5. Estimation and calibration

Four groups of parameters have to be set: the elasticities, the parameters of the 2010 economy, the fare, and the path of agricultural productivity under climate change. How many of them have to be calibrated depends on how the model is solved. I solve it in changes relative to 2010, with the exact hat algebra of Dekle et al. [2008] within each period and the dynamic hat algebra of Caliendo et al. [2019] over time (Appendix E). In Caliendo et al. [2019], this leaves few parameters to calibrate: the levels of productivities and migration costs are summarised by the observed shares of the initial period, namely trade shares, value added, employment and the matrix of migration flows, so only the trade and migration elasticities and the discount factor have to be set from outside. The same holds here for the levels of agricultural productivity, but two features of the model require calibrating more. First, the fare is paid in reais and the food share falls with income, so the parameters of occupational choice have to be calibrated to the level and the distribution of earnings in each region and skill group, and those of food demand to how food spending varies with income. Second, because the fare makes migration depend on income, the observed flows do not summarise migration costs as they do in

Caliendo et al. [2019]. The non-monetary costs have to be calibrated to the flows, given an estimate of the fare.

Section 5.1 sets the elasticities from outside, Section 5.2 calibrates the 2010 economy, Section 5.3 estimates the fare from the change in migration between the 2000 and 2010 censuses, and Section 5.4 constructs the productivity changes from projected temperatures. The calibration of the 2010 economy and the estimation of the fare each close with moments they do not target. Appendix Table G.I lists every parameter, its value and its source.

5.1. Parameters set from outside

Six parameters are set from outside: the Armington elasticity, the dispersion of land quality, the dispersion of tastes, the discount factor, the agricultural labor share and the distance elasticity of trade costs. The model has 557 microregions and 4 education groups, and a period lasts 5 years. The discount factor is $\beta = 0.859$, an annual factor of 0.97. The Armington elasticity is $\epsilon = 5$, so that the trade elasticity $\epsilon - 1$ lies within the range estimated by Simonovska and Waugh [2014] and Eaton and Kortum [2002]. The dispersion of land quality is $\theta = 2$, between the estimates of Costinot et al. [2016] and Sotelo [2020]. The dispersion of tastes is $\nu = 1.75$, close to the values of Artuç et al. [2010], used by Caliendo et al. [2019], and of Dix-Carneiro [2014]. The agricultural labor share is $\alpha = 0.45$, and trade costs rise with distance as $\ln d^{j,i} = \delta_d \ln \text{dist}^{j,i}$, with $\delta_d = 0.15$. The fiscal wedge is zero at the baseline, $\mathcal{T} = S = 0$.

5.2. The 2010 economy

The parameters of the 2010 economy are either measured directly in the data or calibrated so that the model's 2010 equilibrium matches the census and the agricultural data. The model's population is earners: adults who work and report positive earnings; the text calls them workers. Amounts are in 2010 reais; earnings are monthly unless stated otherwise.

Migration. Given the fare of Section 5.3, the non-monetary costs $\check{\tau}^{jk,i}$ are chosen so that the 2010 economy reproduces the origin-conditional flow shares of each education group between 2005 and 2010, from the census question on the place of residence five years earlier.[6]

[6]The targeted flow shares are a weighted average of the observed shares and of shares that decline with distance and keep each cell's observed share of stayers, with weight 10^{-3} on the latter, so that every route has a positive share; the largest gap between a targeted and an observed share is 1.6×10^{-4}.

Occupations. Because the utility cost of the fare depends on each worker's earnings, the parameters of occupational choice are calibrated to the distribution of earnings within each region–skill cell, not only to its mean. The dispersion $\sigma = 0.739$ is set to the dispersion of non-agricultural log earnings, corrected for truncation at the cutoff, and the agricultural wage of each region is set to the earnings of its crop workers. Each cell then has two free parameters, the location $\ln B^i + \mu_k$ of the non-agricultural efficiency distribution and the wedge κ^{ik} of the occupational cutoff (11), and two targets, the non-agricultural participation share and the non-agricultural wage bill. The participation share (21) identifies the distance between the cutoff and the location in units of σ, the wage bill identifies the location, and the wedge is the log ratio of the earnings at the cutoff to the agricultural wage.[7] The wedge averages -0.77 (standard deviation 0.47): without it the cutoff would sit at the agricultural wage, while in the census 25% of non-agricultural workers earn less than the agricultural wage of their region.

Land and trade. Land endowments X^i and the cultivated share s^i (crop area over total land) are taken from the data. Given trade costs from distance, the demand shifter $\psi^j (p^j)^{1-\epsilon}$ of each region, which combines its preference weight and its crop price, is calibrated so that goods-market clearing (25) holds at each region's observed crop revenue; together with trade costs it gives the bilateral expenditure shares $\pi^{j,i}$. In the calibrated economy, earnings are 0.78 of regional income; the rest is housing and land rents, which accrue to resident landowners.

Demand. Because the food share falls with income, crop demand (23) has two parameters, a level and a slope. The slope, $\gamma_y = -0.0493$, is estimated as the slope of the food share in log consumption across the income classes of Brazil's household budget survey (POF 2008–2009), scaled from retail to farm prices. The level, $\gamma_a = 0.169$, is set so that national crop spending equals national crop revenue, 13.3 percent of national income. The housing share $\gamma_h = 0.139$ is set to the ratio of rent to household income among renters in the 2010 census, and 2010 rents are set to one, so that differences in the cost of housing across regions are absorbed by the calibrated non-monetary costs.

Fit. Mean non-agricultural log earnings across region–skill cells, which the calibration does not target, fit with a root-mean-squared gap of 0.07 log points, and their standard deviation is 0.68 in the model against 0.67 in the data. Appendix G.1 lists the matched moments, these fits and the food-spending data.

[7] Cells in which every worker is outside agriculture have their participation share capped at $1 - 10^{-4}$ (97 cells, 4.3% of workers).

5.3. The fare

The fare cannot be calibrated to the 2010 flows: the non-monetary costs are calibrated separately for every origin, destination and skill group, so for any value of the fare some combination reproduces the flows exactly. I estimate it instead from the variation documented in Fact 2: earnings that changed differently across skill groups of the same origin between the 2000 and 2010 censuses.

The cost in the data. Two choices take the cost (6) to the data. The census reports one month's earnings, which measure the resources available in the year of a move with error: I let these resources be $Ye^{\sigma_Y \varepsilon}$, with ε standard normal, and use the cost averaged over ε. And I replace the maximum in (6) by a smooth version, so that the cost changes smoothly with earnings near the floor.[8] The cost $\tilde{\tau} = \mathcal{K}(\chi/Y)$ of a fare that takes the share x of measured annual earnings is then

$$\mathcal{K}(x) = \mathbb{E}_\varepsilon\Big[-\ln \widetilde{\max}\big(1 - x\, e^{-\sigma_Y \varepsilon},\ \underline{s}\big)\Big]. \tag{27}$$

The fare cost then depends on three parameters: the fare c, which sets x, the floor $\underline{s}$ and the spread σ_Y. All quantitative results use this cost. At the estimates it is close to x for small fares, rises more than in proportion to the fare up to about three quarters of a year's earnings, and approaches $-\ln \underline{s}$ as the fare exceeds a year's earnings.

Estimating equation. Let N^{jkit} be the number of workers from skill group k in origin j who live in region i at the end of the five-year window of census wave t, stayers included, and $G^{jkit} \equiv E[\mathcal{K}(\chi^{j,i}/Y) \mid j,k,t]$ the mean over the group's earnings of the utility cost (27) of the fare to i. Aggregating the location choice (16) over the group's earnings and approximating the cost by its group mean gives

$$\begin{aligned} \ln \mathbb{E}\big[N^{jkit}\big] = {} & \underbrace{\mu^{jkt}}_{\text{group size}} + \underbrace{\delta^{ikt}}_{\text{value of } i} - \mathbb{1}[i \neq j]\ \underbrace{\tfrac{1}{\nu}\, G^{jkit}(c, \underline{s}, \sigma_Y)}_{\text{fare}} \\ & - \mathbb{1}[i \neq j]\Big(\underbrace{A_\ell^{jk}}_{\text{attachment}} + \underbrace{\omega^{kt}}_{\text{national trend}} + \underbrace{\zeta_\ell^{jt}}_{\text{origin shock}} \\ & + \underbrace{\eta^{\sigma(j)\sigma(i)}}_{\text{state pair}} + \underbrace{\lambda^{jk} \ln d^{ji} + \varphi_t(d^{ji})}_{\text{distance}} \Big), \end{aligned} \tag{28}$$

where d^{ji} is the distance between region centroids, $\delta^{ikt} \equiv \beta\, \overline{V}^{ikt}/\nu$, and the term in brackets is the non-monetary cost $\check{\tau}^{jkit}/\nu$, which the estimation allows to vary by skill group and wave. The attachment A_ℓ^{jk} of a skill group to its origin, for example through

[8] $\widetilde{\max}(u, \underline{s}) = \underline{s}\big[1 + \ln\big(1 + e^{8(u/\underline{s} - 1)}\big)/8\big]$, which differs from $\max(u, \underline{s})$ by at most $\underline{s} \ln 2/8$, at $u = \underline{s}$.

networks, is constant over time and may differ between short ($\ell = S$, under 100 km) and long moves. The national trend ω^{kt} captures the cost of leaving common to a skill group nationwide in each decade. The origin shock ζ_{ℓ}^{jt}, common to the skill groups of an origin, absorbs new roads, local booms or droughts. State pairs and a flexible distance profile absorb the rest of the geography. Because δ^{ikt} is common to workers who move to i and to those who stay there, $\delta^{ikt} - \delta^{jkt}$ measures the gain from moving from j to i, and the equation controls for it. I estimate (28) by Poisson pseudo-maximum likelihood on the census sample of workers in 2000 and 2010.[9]

Identification. With these effects, the fare is identified by a triple difference. Differencing between the censuses removes the attachment; comparing two skill groups of the same origin removes the origin shock; comparing such pairs across origins removes the national trend. What is left is whether the group whose earnings rose by more, relative to another group of the same origin, so that the fare weighs less on its earnings, became more likely to leave, beyond what the change in the gain from moving predicts. The identifying assumption is that, within an origin, no change in the non-monetary cost of leaving specific to one skill group moves with that group's earnings beyond the group's national trend. This is the comparison of Fact 2, with each group's own change in the utility cost of the fare in place of the national income shift and the full choice of destinations in place of the return to migration.

Because the loss is not proportional to the fare, the fare, the floor and the spread are identified separately given ν: the fare by the average response of moving to the fare's share of earnings, the floor by how the deterrence levels off among the poorest, and the spread by how gradually the deterrence rises as the fare approaches a year's earnings.

The estimate. I estimate $(c, \underline{s}, \sigma_Y)$ by minimizing the Poisson deviance of (28) over a grid, at $\nu = 1.75$. The fare is $c = 1{,}000$ reais per log kilometre, the floor $\underline{s} = 0.035$ and the spread $\sigma_Y = 0.25$.[10] A move of 500 km costs 6,215 reais, 12.2 months of the 2010 minimum wage (Figure V, Panel A).[11] Because it grows with the logarithm of distance, the fare of a short move is a large part of the fare of a long one; what differs across workers is the income it is paid from (Panel B). A 500 km move costs a worker at the median of the bottom quintile of 2010 earnings 20.1 months of earnings, and one at the median of the top quintile 2.6: for the bottom quintile the fare exceeds a year's earnings,

[9] Annual earnings are twelve times reported monthly earnings; G^{jkit} is computed from up to forty earnings quantiles per origin, skill group and wave, with the expectation over ε taken at nine Gauss–Hermite nodes.

[10] The spread is chosen on a coarse grid, $\sigma_Y \in \{0.25, 0.4\}$, on which the deviance is lower at 0.25. The standard errors, clustered by origin and conditional on ν, are evaluated at the reported estimates (Figure V, Panel A).

[11] Appendix G.4 reports the sample of the estimation.

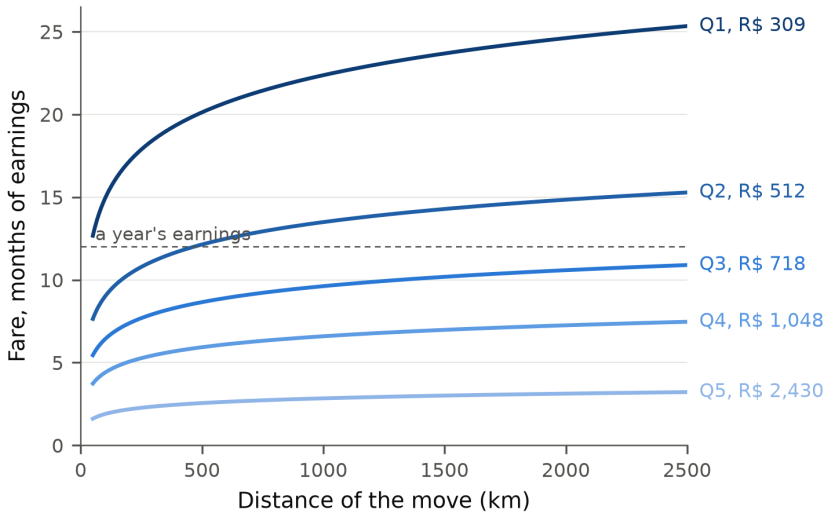
so that the floor $\underline{s}$ bounds its utility cost. Among the least educated, the median worker pays 101 percent of a year's earnings for such a move.

FIGURE V
The fare

Panel A. Estimate

	Estimate	Standard error
Fare c (reais per log km)	1,000	(43)
Floor $\underline{s}$	0.035	(0.019)
Spread σ_Y	0.25	(0.056)
Fare of a 500 km move (reais)	6,215	

Panel B. The fare in months of earnings, by distance and earnings quintile



Notes. Panel A: estimates of equation (28) by Poisson pseudo-maximum likelihood on counts of workers by origin × education × region of residence × wave, 2000 and 2010; the fare is $\chi(d) = c \ln d$ and its utility cost (27), with $(c, \underline{s}, \sigma_Y)$ chosen on a grid to minimize the Poisson deviance. Fixed effects: group size (μ^{jkt}), the value of each destination by education and wave (δ^{ikt}), state pairs, a distance profile in $\ln d$, $(\ln d)^2$ and $\ln d \times 2010$, a distance slope by origin × education, attachment by origin × education and short or long move, origin × decade effects by short or long move, and the education × decade cost of leaving. 2010 reais at $\nu = 1.75$. Standard errors, in parentheses, are clustered by origin microregion and conditional on ν. Panel B: the estimated fare divided by the median monthly earnings of each national quintile of workers in 2010 (shown next to each line, 2010 reais), by the distance of the move; census earnings of the estimation sample, pooled over all cells. Dashed line: a year's earnings.

The fare and consumption are in the same units: as in the estimation, the fare is paid out of the earnings of the year of the move, and its utility cost is a loss of log consumption in the one year of the period in which it is paid (Appendix G.4).

Fit. The estimation and the calibration of Section 5.2 target flows by origin and education group, not by earnings within a group. How emigration varies with earnings within a group is therefore not targeted, and it tests how the deterrent effect of the

fare varies across the earnings distribution. In the calibrated economy, the top quintile of workers emigrates more than the bottom quintile by 7.9 percentage points within origin and education level, against 3.9 in the 2000 and 2010 data, and by 7.5 within origin, against 4.4 in the 2010 data; the latter comparison also contains the differences between education groups, which the calibration targets. In the model and in the data, the lowest quintile is more deterred by distance than the highest: the distance elasticity of moves is -1.99 against -1.54 in the model and -1.77 against -1.59 in the data. The estimated location choice gives a Fact 2 coefficient of $\rho = 4.52$ against 6.11 in the data; this is a consistency check rather than an independent test, because the fare is estimated from the same comparison across skill groups within an origin. Appendix Table G.IV reports these moments.

5.4. Climate shocks

Climate change enters the model as a path of fundamental agricultural productivity. I construct it from projected temperatures through the quadratic temperature–productivity relationship $g(T)$ of Fact 4 (Figure III), evaluated at $\theta = 2$: region i's productivity in period t changes by $\ln(A_t^i / A_0^i) = g(T_t^i) - g(T_0^i)$, where T_0^i is its annual mean temperature in 2010. Each region moves along its own segment of the quadratic, so a common warming has different effects across regions. Projected temperatures come from CMCC-ESM2 under SSP5-8.5, with SSP2-4.5 as the alternative. Under SSP5-8.5, the median microregion loses 29.3 percent of its fundamental productivity by 2050 and 75.0 percent by 2100, and 98.9 percent of microregions lose (Appendix Table G.III). The losses are large because the estimated optimum, 19.8°C, lies below the temperatures of most microregions; late in the century many microregions leave the range of temperatures observed in the estimation, so the late-century losses extrapolate the quadratic. Appendix G.3 documents the paths.

6. Welfare losses from climate change

Under high emissions, climate change costs the workers living in 2015 2.56 percent of their consumption, permanently, and more for those living later, because the damages are back-loaded: 6.08 percent in 2050 and 10.82 percent in 2100 (Figure VI, Panel A; under SSP2-4.5 the loss in 2015 is 1.37 percent). In the hottest regions the bottom quartile loses 25 percent more than the top. This section derives these losses, shows where they fall, and explains why low-income workers lose more in most of the country but not in the South. The baseline is the transition of the calibrated economy from 2010 with agricultural productivity held at its 2010 level. In the climate counterfactual, productivity follows the paths of Section 5.4, under SSP5-8.5 unless stated otherwise

and under SSP2-4.5 as the alternative.[12] Welfare changes are consumption-equivalent variations (CEV): the permanent proportional change in consumption, from the date of measurement on, that gives a group of workers the same average change in lifetime utility as the counterfactual. The loss in 2015 is that of the workers living in 2015, from 2015 on, and likewise for later years. Workers are compared across the two worlds by their microregion, skill group and rank in the efficiency distribution of their cell, and grouped by national quartile of their earnings in the baseline. The *hottest regions* are the microregions with a mean temperature above 24.6°C in 2010; together they hold a third of workers. Appendix E.3 defines the measure.

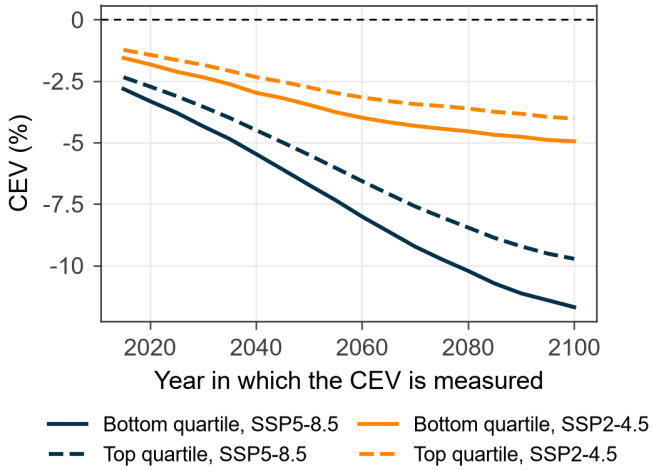
Who loses. Climate change falls hardest on those least able to move away from it. In 2015 the bottom quartile's loss is 20 percent larger than the top quartile's nationally, and 25 percent larger in the hottest regions. In every later year the bottom quartile nationally also loses more than the top quartile (Figure VI, Panel A). Panel B of Figure VI maps the losses in 2015. The bottom quartile loses most in the North and in the interior of the Northeast (map a), and that is also where its loss exceeds the top quartile's by most (map b). In the South, the coolest part of the country, the bottom quartile loses less than the top. Bottom-quartile farmers lose 15 percent more than the quartile as a whole, and the least educated 42 percent more than university graduates. Appendix H.1 reports the losses by region, temperature, skill group and sector.

Why. Two channels set the gap between low- and high-income workers, and they work in opposite directions across the country. The first is food: low-income workers spend a larger share of their income on food, so the rise in crop prices costs them more everywhere. Nationally, it alone more than accounts for the gap between the bottom and the top quartile: the other channels, rents and the option to move, work the other way and narrow it. The second is earnings: low-income workers are farmers, or non-agricultural workers close to the occupational cutoff, whose incomes follow the agricultural wage, while high-income workers work outside agriculture, at a regional wage that climate does not move, and lose through prices and rents. In the hottest regions, where warming lowers farm productivity most, the earnings channel adds to the food channel: the agricultural wage falls and the marginal farmers are pushed into low-paid non-agricultural work, and earnings account for 44 percent of the gap there. In the South, the earnings channel works the other way. Workers there lose too, because crop prices rise everywhere, by about 14 percent by 2050 in the South. But farm productivity there falls least or rises, so the higher crop prices raise the farm wage:

[12] The shock is unexpected in 2010 and learned in 2015. The location choices of 2010 are those observed in the census, which the model reproduces by construction, so news about the climate path can first change choices at the next decision date; from 2015 on, workers foresee the whole path. Both paths are solved for 40 five-year periods, to 2210, with productivity held at its 2100 level after 2100.

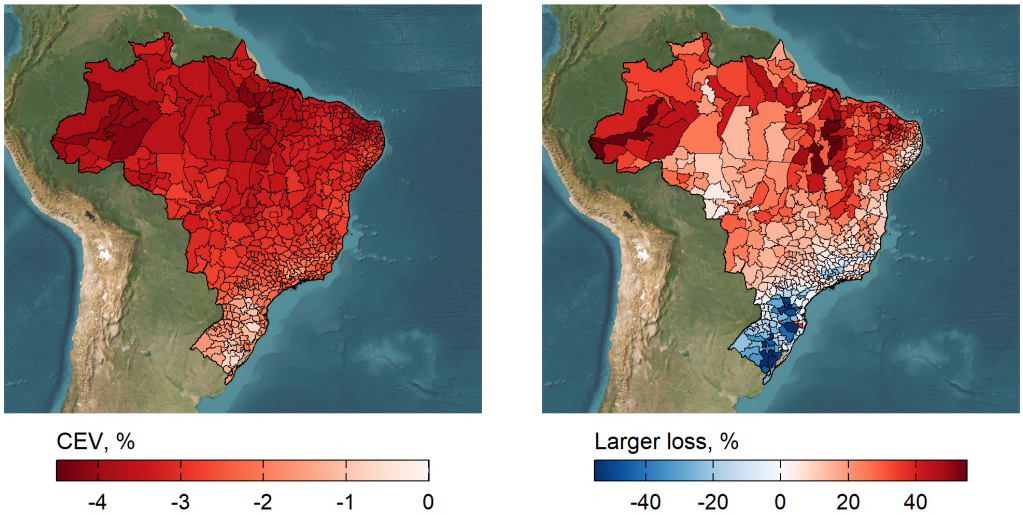
FIGURE VI
Welfare change from climate change

Panel A. By the year in which it is measured



Panel B. By microregion, 2015

(a) Bottom quartile's welfare change *(b) Bottom quartile's loss over the top quartile's*



Notes. CEV (percent) of climate change relative to the baseline; negative values are losses. The CEV in a year is that of the workers living in that year, from then on; workers are grouped by national quartile of baseline earnings in that year and weighted by the baseline population. Panel A: SSP5-8.5 and SSP2-4.5, for each year between 2015 and 2100, for the bottom and top quartiles. Panel B: SSP5-8.5, 2015, by microregion. (a) CEV of the bottom quartile. (b) The percent by which the bottom quartile's loss exceeds the top quartile's, $100\,(\text{CEV}_{Q1}/\text{CEV}_{Q4} - 1)$; negative where the bottom quartile loses less; grey where the top quartile's loss is below one percent or the two change in opposite directions. In Panel B, each colour scale runs from the 1st to the 99th percentile of its values (rounded outward); values beyond are drawn at the end colours.

by 2050 it is 5.9 percent higher than without climate change in the South, against 8.9 percent lower in the North. The incomes of low-income workers, tied to the farm wage, rise with it and offset part of their larger food bill, while high-income workers bear the higher prices without an income gain, so the bottom quartile loses less than the top (Figure VI, map (b)).

Migration. Under SSP5-8.5, by 2100, 2.55 percent of workers live in a different microregion from the one they would live in in the baseline, and the number of workers in the hottest regions is 4.26 percent lower, with the largest fall among the least educated. Migration is part of the adjustment, but the workers with the most to gain from leaving are those on whom the fare weighs most. Section 7 asks how much this matters, and what the tax system can do about it.

7. Removing the fare and an income-targeted subsidy

How much does the cash cost of moving matter for who loses from climate change, and can the tax system help low-income workers move away from it? This section answers in two steps. The first removes the fare altogether and keeps the non-monetary costs of leaving: no policy that acts on the cash cost of moving can do more, so this bounds what to expect. The second studies a policy the tax system can deliver: a subsidy that pays part of the fare for those with below-median income, financed by a proportional tax on earnings. It moves the workers it is meant to move, raises their welfare and makes their loss from climate change smaller; it also raises national output, an additional motive that has nothing to do with its target.

7.1. Without the fare

The fare holds back the workers whom climate change hits hardest. Without it, the bottom quartile's leaving rate in 2015 would be 14.9 percent over five years, against 3.1 percent with it (Figure VII, Panel B(a)): the poorest would leave their microregion more often than the richest, not less. The fare holds back 11.8 percent of the bottom quartile, and 15.0 percent in the hottest regions: workers who would leave without it and stay with it. For them, removing the fare makes the loss from climate change 7.5 percent smaller, and 11.7 percent smaller in the hottest regions (Table II). Cash is not the only barrier: even without the fare, 85 percent of the bottom quartile, and 90 percent of the top quartile, would stay in their microregion over five years, because the non-monetary cost of leaving, which cash cannot remove, keeps most workers in place.

7.2. An income-targeted subsidy

7.2.1. Design

The payment. The subsidy removes the part of the burden of the fare that falls on a worker because its income is low. When a worker whose income lies below a reference income moves, it is paid the part of the fare that makes the move as costly to it, in utility, as the same move is to a worker at the reference income. In panel (b) of Figure IV, the two shaded areas are the utility cost of the same fare χ for a low-income worker, at y_l, and for a high-income worker, at y_h: the same fare costs the poorer worker more utility. The subsidy is designed to compensate for this difference: the payment shrinks the utility cost of the low-income worker to that of a worker at the reference income. With the fare $\chi^{j,i}$ of a move from j to i valued out of a year's income, the payment to a worker with monthly disposable income y is

$$\varsigma^{j,i}(y) = \chi^{j,i}\left(1 - \frac{y}{y_{\text{ref}}}\right) \quad \text{if } y < y_{\text{ref}}, \qquad \varsigma^{j,i}(y) = 0 \quad \text{otherwise.} \tag{29}$$

The subsidy pays the share of the fare by which the worker's income falls short of the reference income. Staying is never paid, and every move is paid, including return moves.

Eligibility. The reference income is the median of the 2010 distribution of model earnings, $y_{\text{ref}} = \text{R\$}\,737$ a month (about US\$ 419), fixed at this value in 2010 reais. Every worker whose disposable income is below it is eligible, in every region and skill group. The subsidy is in place from 2010 to the end of the horizon and, like the climate shock, comes unexpectedly. The rule is a benchmark that removes the part of the fare's burden due to low income, not an optimum (Section 8).

Financing. The subsidy is financed inside the model by a proportional tax on the labour earnings of every worker, constant over time, set so that the present value of revenue equals the present value of outlays. The tax rate that balances the budget is 0.109 percent of earnings under SSP5-8.5, about R\$ 15 a year for the average worker in 2015. The rate is low relative to the payment per move because the subsidy is paid once per move, on the moves of low-income workers, while the tax is levied on all earnings in every year: in 2015, 7.4 percent of workers leave their microregion within five years, 47 percent of these moves are subsidised, and a subsidised move is paid R\$ 2,387 on average. All welfare gains below are net of the tax. Panel A of Figure VII, (a) and (b), shows who pays and who receives. Payments are concentrated in the bottom half of the earnings distribution and fall with earnings, while the tax rises in proportion to them. Net of the tax, the lowest decile receives the largest transfer relative to its earnings, the

transfer turns negative around the median, and above it every decile pays the same share of its earnings.

7.2.2. Whom the subsidy moves

The subsidy moves the workers it is meant to move. In 2015, the bottom quartile leaves its microregion within five years at a rate of 7.8 percent with the subsidy, against 3.1 percent without it, 154 percent more; the top quartile, not eligible, leaves at 8.3 percent with and without the subsidy (Figure VII, Panel B(a)). The subsidy thus closes part of the distance to the economy without the fare, where the bottom quartile would leave at 14.9 percent. Nationally the leaving rate rises by 28 percent, from 5.8 to 7.4 percent.

The subsidy induces 4.7 percent of the bottom quartile to leave its microregion in 2015, and 5.9 percent in the hottest regions: workers who, with the same tastes, leave with the subsidy and stay without it (Figure VII, Panel B(b)). They are the workers the policy is for: 46 percent of them are in the first earnings decile and 61 percent live in the North and Northeast, and farm workers are more common among them than in the quartile as a whole (Appendix Table H.VII). The subsidy moves only those who were close to leaving anyway; for most low-income workers, the non-monetary cost of leaving is too large. Workers who leave anyway are paid as well: 39 percent of the bottom quartile's subsidised moves in 2015 would have been made without the subsidy.

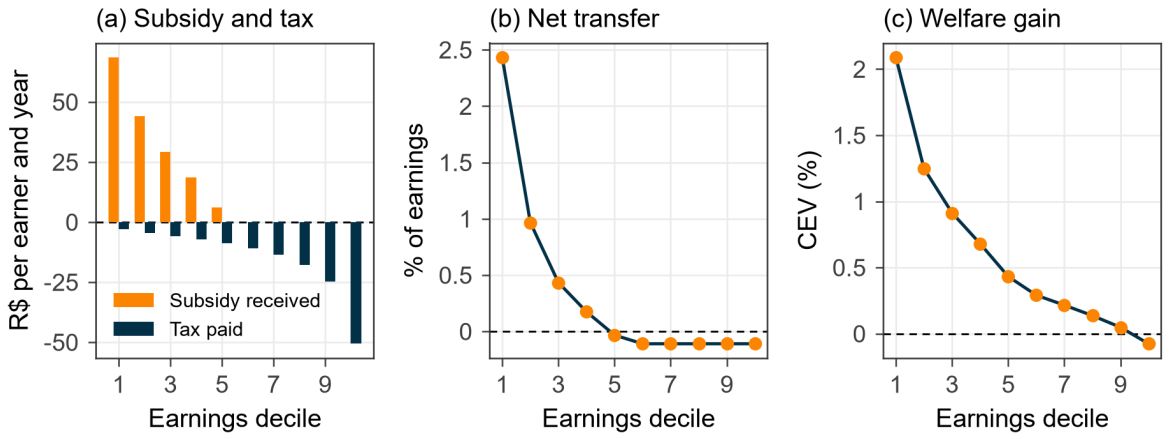
7.2.3. What it does for those it moves

For the workers it moves, the subsidy is a large gain and makes their loss from climate change smaller. I measure both on these workers: an average over a whole group also counts the many whose choice the subsidy does not change, most of whom would stay even if moving were free, and so does not show what the subsidy does for those who respond to it. Figure VIII maps both effects across the country, and Table II reports them by group, next to the economy without the fare.

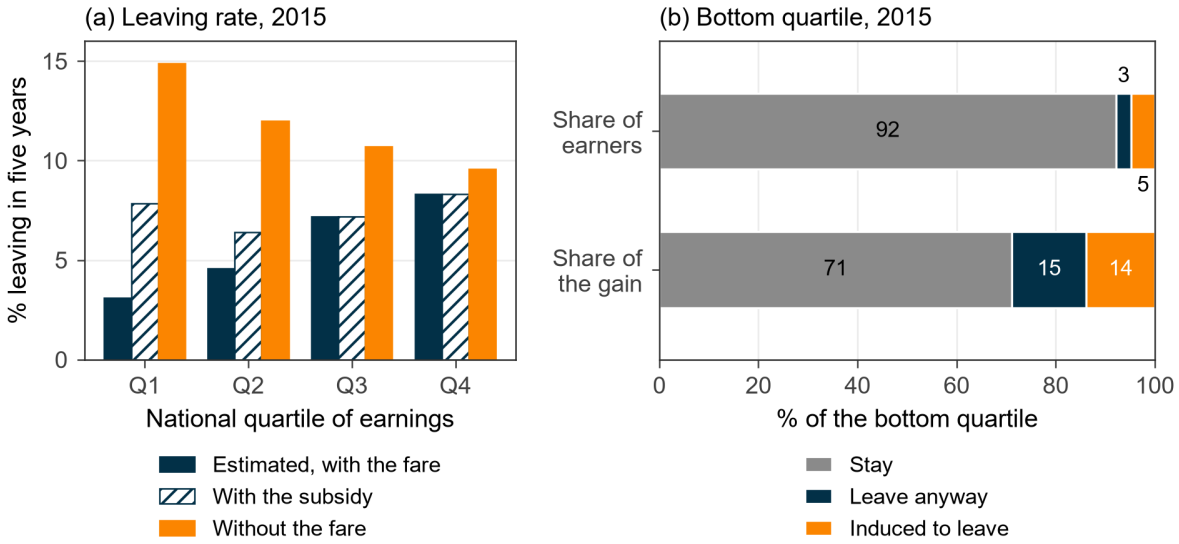
Welfare. Net of the tax, the workers whom the subsidy induces to leave in 2015 gain the equivalent of 4.51 percent of their consumption, permanently. Their gains are largest where climate change hits hardest: 5.47 percent in the North and Northeast and 5.19 percent in the hottest regions, against 3.03 percent in the South, Southeast and Centre-West (Figure VIII, map (a)). Most of the gain would be there without climate change, 4.45 percent, and 5.07 percent in the hottest regions: the regions they leave lag behind even without warming.

FIGURE VII
The subsidy: who pays, who gains and whom it moves, 2015

Panel A. The subsidy, the tax and the welfare gain, by earnings decile



Panel B. Whom the subsidy moves



Notes. The subsidy financed by a proportional tax, under SSP5-8.5, in 2015. Panel A, by national decile of baseline earnings, weighted by the population of the period: (a) the expected subsidy payment (one fifth of the expected payment of the five-year period, at disposable income) and the tax paid (drawn as negative), in 2010 reais per worker and year; (b) the subsidy minus the tax, in percent of the decile's earnings; (c) CEV (percent) of the subsidy, net of the tax, against the same economy without it. The budget is balanced in present value over the horizon, not in each period. Workers above the reference income, R$ 737 a month in 2010 reais, are not eligible in 2015. Panel B: (a) the share of workers in each national quartile of baseline earnings who leave their microregion within five years, in percent, in the calibrated economy, in the same economy with the subsidy, and in the economy without the fare with the same non-monetary costs; (b) the workers of the bottom quartile, by what the subsidy does to their choice in 2015, comparing each worker with the same tastes in the economies with and without the subsidy. *Stay*: they stay with and without the subsidy; *leave anyway*: they leave with and without it; *induced to leave*: they leave with the subsidy and stay without it. Upper bar: share of the quartile's workers, in percent; lower bar: share of the quartile's gain from the subsidy, in percent of the sum of the gains in lifetime utility.

The option to leave. Workers whom the subsidy does not move gain as well. The bottom quartile's workers who stay with and without the subsidy gain 1.18 percent of consumption, and they receive 71 percent of the quartile's gain (Figure VII, Panel B(b)). They do not move now, but they may want to later: their incomes are redrawn every period, and climate change keeps making their region worse. The subsidy makes that future move cheaper, so it raises the value of the option to leave, and with it their expected lifetime welfare today.

The loss from climate change. By letting them leave the places that climate change hits, the subsidy also makes the loss from climate change itself smaller. For the workers it induces to leave in 2015, the loss is 2.9 percent smaller with the subsidy than without it, each against its own path without climate change; 4.6 percent in the North and Northeast and 4.7 percent in the hottest regions (Table II; Figure VIII, map (b)). For comparison, removing the fare altogether makes the loss of the workers it holds back 7.5 percent smaller (Section 7.1).

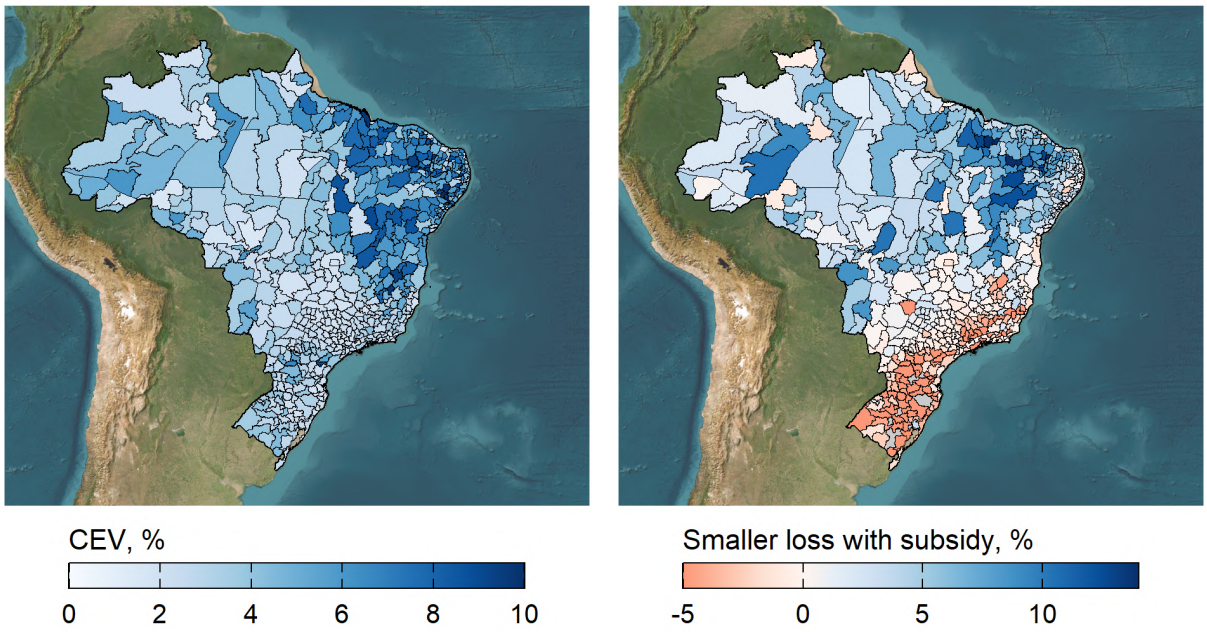
Where the subsidy helps. The subsidy is worth more where staying becomes worse relative to the places a worker could move to. Climate change lowers the value of the North and Northeast more than that of most destinations, so there the option to leave cheaply gains value; in the cooler South, it lowers the value of the destinations more than that of the origin, and the option to leave loses value. The subsidy moves the induced movers of the South mostly to other places in the South, which climate change hurts more than their origin, so their loss is larger with it. In percent this increase is large in some southern microregions (Figure VIII, map (b)) because there the loss without the subsidy is small, so a small change in lifetime utility is a large share of it. The difference shows up in where workers live: by 2100, climate change lowers the number of workers in the North and Northeast by 6.07 percent with the subsidy, against 5.33 percent without it. Appendix H.2 reports the gains and loss reductions averaged over all workers of each group.

7.2.4. An additional motive: national output

The subsidy is designed to move low-income workers away from the places that climate change hits. It also raises national output, a by-product that has nothing to do with this target. The induced moves take workers from low-wage farm regions to destinations where their labour earns more, and the gain accumulates as the population shifts: output is 0.232 percent higher than without the subsidy in 2015 and 0.642 percent in 2050, and then levels off. By 2050, 1.6 percent of workers live in another microregion than they would without it, and each of them raises national output by about 39 percent

FIGURE VIII
The subsidy and the bottom quartile's induced movers, by microregion, 2015

(a) Their welfare gain from the subsidy *(b) How the subsidy changes their climate loss*



Notes. SSP5-8.5, 2015, for the workers of the bottom national quartile of baseline earnings whom the subsidy induces to leave their microregion in 2015 (they leave with the subsidy and stay without it, with the same tastes; Appendix E.5), by microregion of origin. Map (a): CEV (percent) of the financed subsidy; the colour scale runs from zero to the 99th percentile of the values (rounded up), larger values are drawn at the darkest colour. Map (b): the percent by which the loss from climate change is smaller in the economy with the subsidy than in the economy without it, each against its own path without climate change (100 times one minus the ratio of the change in lifetime utility from climate change with the subsidy to the same change without it, on group means); positive (blue): the loss is smaller with the subsidy; grey: microregions where the loss without the subsidy is below 1 percent, where the ratio is unstable. In map (b), values below -5 percent are drawn at the end of the colour scale.

of the output per worker. In each period, the outlays are 0.097 percent of national output, the sum of regional incomes, in 2015 and 0.077 percent in 2050, and the tax revenue that finances them is about the same. Discounted and cumulated (Figure IX), the output gain reaches 10.1 percent of national output in 2010 by 2050 and 14.8 percent by 2100, against outlays of 2.4 and 2.9 percent; the cumulated gain exceeds the cumulated outlays from 2015 on. Over the whole horizon of the model, the present value of the output gain is 527 percent of the present value of the outlays, and 370 percent when each region's income is deflated by its cost of living, which rises with rents where the movers arrive.

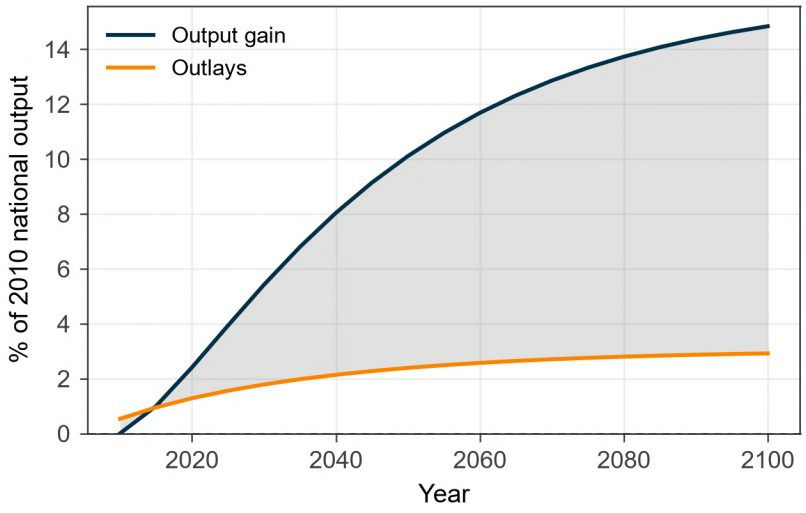
The output gain exceeds the outlays because the two accrue differently. A subsidised move is paid once, a share of a fare that for a 500 km move is about a year's earnings of the median least-educated worker, while an induced mover earns more in every year it stays at its destination. The induced movers had not moved before because the move is costly to them, in cash and in the non-monetary cost of leaving their origin, not because their income would not rise, so output can rise much more than the outlays while the

TABLE II
The subsidy, the fare and the loss from climate change, 2015

	All workers of the group				Workers induced to move				
	Climate change (%)	Loss smaller by (%)	Without fare: smaller by (%)	Gain (%)	Held back by fare (%)	Share of group (%)	Climate change (%)	Loss smaller by (%)	Gain (%)
Bottom quartile	−2.82	1.0	3.5	+1.53	11.8	4.7	−2.97	2.9	+4.51
North and Northeast	−3.20	2.5	8.4	+2.71	16.0	6.2	−3.27	4.6	+5.47
Hottest regions	−3.21	2.4	8.2	+2.36	15.0	5.9	−3.31	4.7	+5.19
South, Southeast, Centre-West	−2.48	−0.7	−2.1	+0.49	8.1	3.5	−2.51	−0.6	+3.03
Top quartile	−2.34	−0.1	−0.3	+0.01	1.3	0.0	–	–	–

Notes. SSP5-8.5, 2015. *Climate change*: CEV (percent; Section 6) of climate change against the baseline, in the economy without the subsidy, for the workers of the block; negative values are losses. *Loss smaller by*: the percent by which the loss from climate change is smaller in the economy with the subsidy than in the economy without it, each against its own path without climate change: 100 times one minus the ratio of the change in lifetime utility from climate change with the subsidy to the same change without it, on group means; negative where the loss is larger with the subsidy. *Without fare: smaller by*: the same, for the economy without the fare, with the same non-monetary costs. *Held back by fare*: the share of the group that leaves in 2015 without the fare and stays with it, for each worker with the same tastes. *Gain*: CEV (percent) of the subsidy, net of the tax, under SSP5-8.5 against the same economy without it. *Workers induced to move*: workers who, under SSP5-8.5, leave their microregion in 2015 with the subsidy and stay without it, comparing each worker with the same tastes in the two economies (Appendix E.5); the top quartile is not eligible in 2015, and the subsidy induces none of it to move. Workers compared at the same microregion, skill group and efficiency rank, weighted by the baseline population, and grouped by national quartile of baseline earnings in 2015. The North and Northeast and the South, Southeast and Centre-West partition the country; the hottest regions, as in Section 6, are a separate cut, mostly in the North and Northeast.

FIGURE IX
The migration subsidy: cumulative output gain and outlays



Notes. The subsidy financed by a proportional tax, under SSP5-8.5. The change in national output (the sum of regional incomes) against the path without the subsidy and the outlays of the subsidy, both discounted at 3 percent a year and summed from 2010 to the year shown, in percent of national output in 2010 (one year); the shaded area is the difference. The tax balances revenue and outlays in present value over the horizon of the model, 2210. Appendix Figure H.II shows the output gain, the outlays and the tax revenue in each period.

movers' own welfare gain is smaller than their income gain, since they also bear the non-monetary cost of leaving. The ratio of output gain to outlays is not a fiscal return: the government collects the tax share of the extra output and no more. Nor is it a gain in efficiency in the welfare sense. In the model the fare is a private cost, and moving has no externality, so workers who did not move without the subsidy judged the move not worth its cost; the output gain is the extra income of those whom the subsidy moves, not a market failure that it corrects.

8. Discussion

Migration is often cast as a way to adapt to climate change, but the results of Section 7 say that it is an option only for those who can afford it, and that a policy designed for the move can open it to the poor. This section places the subsidy among the alternatives discussed in the literature, asks why workers do not borrow to pay the fare themselves, and sets out what the results leave open.

People or places. A long debate asks whether policy should help people or places: whether to support the regions that a shock hits, or the people in them, including by helping them leave [Kline and Moretti, 2014, Austin et al., 2018]. Place-based policies keep workers where their productivity is falling; helping people move lets them follow opportunity, but leaves behind those who cannot afford to go. The income-targeted subsidy is a policy for people, aimed at the ones the fare holds back. Appendix H.4 compares it with two alternatives, each financed by a proportional tax on earnings: the same payment paid only on moves out of the hottest regions, a policy targeted by place, and a cash transfer of the same present value to the same eligible workers, paid whether or not they move. The place-targeted subsidy costs less and yields more output per real of outlays, but low-income workers outside the hottest regions pay the tax for it and are rarely paid. For the question of this paper, the income-targeted subsidy is preferable: it reaches low-income workers wherever they live, and it moves more of them. The bottom quartile's leaving rate in 2015 rises from 3.1 percent without a policy to 7.8 percent with it, against 4.2 percent with the place-targeted subsidy. Targeting by place buys output more cheaply, at the expense of the low-income workers outside the targeted regions.

Why not a cash transfer? A transfer of the same present value spread over every eligible worker in every month is small: at most R$ 6.37 a month, and R$ 2.54 for the average recipient, against R$ 2,387 paid on an average subsidised move. A move needs a large sum at one moment, and a transfer spread over time and over all eligible workers rarely provides it: the transfer does not change who moves, and the bottom

quartile's leaving rate in 2015 is 3.1 percent with it, against 7.8 percent with the subsidy. To move people, redistribution has to deliver a critical amount at the moment of the move, and only a policy designed for the move does so.

Why do workers not borrow? If moving raises their lifetime welfare, why do the workers held back by the fare not borrow to pay for it? A large literature finds that they cannot. A move is an investment whose return is uncertain and comes later, at the destination, and the poor have little to borrow against: they have little collateral, often work informally, and leave the place where a local lender could enforce repayment. Where cash or credit for the move is provided, migration rises: a small grant or a zero-interest loan for the trip raises seasonal migration in Bangladesh by similar amounts [Bryan et al., 2014], randomized access to credit raises rural–urban migration in China [Cai, 2020], and cash transfers and pensions that relax liquidity raise migration in Mexico and South Africa [Angelucci, 2015, Ardington et al., 2009]. In Brazil, many low-income households in the 2000s had no bank account and no access to formal credit, and the informal workers and subsistence farmers among them could not borrow against future earnings. A subsidy is one way to relax this constraint; a loan for the move, repaid out of earnings at the destination, would be another.

Implementation. The income-targeted subsidy could be delivered through the social-protection systems that already reach low-income households and record their incomes, as a relocation grant paid on registration in the new municipality, or as a tax credit, which would exclude informal workers and subsistence farmers outside the formal tax system. Two features of the subsidy as modelled would need care in practice. It pays every move, including return moves; in the model the payment never exceeds the fare, so no one gains from moving only to collect it. And it is paid on the fare, which in the model is a money-metric cost of relocating, including search and earnings lost in the transition, rather than a ticket price; a scheme that paid only for transport would cover part of it.

Open questions. The subsidy studied here is a benchmark rule that removes the part of the fare's burden due to low income, not an optimum. Appendix H.5 gives, at given prices, the first-order welfare effect of changing the payment at each income in terms of three statistics: the value of a real paid to movers at that income, the response of moves to the payment, and the fiscal externality. Characterising the optimal schedule, and how climate change shifts it, is left for future work.

9. Conclusion

This paper studies how income shapes migration responses to climate change and their welfare consequences. Using Brazilian census data, I document that within an origin the low-income migrate least, and that when the poor got richer in the 2000s, they left structurally lagging places more (Section 3). I build a dynamic spatial general equilibrium model in which a move costs a cash fare, valued with log utility, and estimate the fare from income changes within origins: a 500 km move costs the median least-educated worker about a year's earnings.

The model application yields two findings. First, climate change imposes large losses on every group of workers, and they grow over the century: under SSP5-8.5, workers living in 2015 lose the equivalent of 2.56 percent of their consumption, permanently, and the loss from 2100 on is 10.82 percent. The losses are regressive: among workers living in 2015, the bottom quartile loses 20 percent more than the top quartile nationally, and 25 percent more in the hottest regions (3.21 against 2.57 percent), and the gap persists over the century.

Second, because the fare weighs most on low-income workers, paying part of it for them raises their welfare and national output. A subsidy that pays low-income workers the part of the fare that makes its utility cost equal to that of a worker at the median income, financed by a proportional tax, induces 4.7 percent of the bottom quartile of workers living in 2015 to leave their microregion. These workers gain 4.51 percent of consumption, and 5.19 percent in the hottest regions, and the subsidy raises national output. Because it lets them leave the places climate change hits, it also reduces their loss from climate change, by 2.9 percent, and by 4.7 percent in the hottest regions. When moving costs cash, a subsidy to leave reaches the workers whom climate change hits hardest and who are least able to leave. The subsidy studied here is a benchmark rule, not an optimum. Characterising the optimal schedule, and how climate change shifts it, is the subject of ongoing work.

References

C. Albert, P. Bustos, and J. Ponticelli. The effects of climate change on labor and capital reallocation. NBER Working Paper 28995, National Bureau of Economic Research, 2021.

Manuela Angelucci. Migration and financial constraints: Evidence from Mexico. *Review of Economics and Statistics*, 97(1):224–228, 2015.

Cally Ardington, Anne Case, and Victoria Hosegood. Labor supply responses to large social transfers: Longitudinal evidence from South Africa. *American Economic Journal: Applied Economics*, 1(1):22–48, 2009.

S. Aron-Dine. Rebuild or relocate? Recovery after natural disasters. Job market paper, Stanford University, 2024.

Erhan Artuç, Shubham Chaudhuri, and John McLaren. Trade shocks and labor adjustment: A structural empirical approach. *American Economic Review*, 100(3):1008–1045, 2010.

Benjamin Austin, Edward Glaeser, and Lawrence H. Summers. Jobs for the heartland: Place-based policies in 21st-century America. *Brookings Papers on Economic Activity*, 2018(Spring):151–255, 2018.

T. Baseler and S. Gordeev. The macroeconomic effects of lifting barriers to migration. Working paper, University of Rochester and Texas Christian University, 2026.

J. E. Bastian and D. A. Black. Relaxing financial constraints with tax credits and migrating out of rural and distressed America. *Journal of Public Economics*, 234:105111, 2024.

P. Bastos, M. Busso, and S. Miller. Adapting to climate change: Long-term effects of drought on local labor markets. IDB Working Paper Series IDB-WP-466, Inter-American Development Bank, 2013.

S. Bazzi. Wealth heterogeneity and the income elasticity of migration. *American Economic Journal: Applied Economics*, 9(2):219–255, April 2017.

H. Benveniste, M. Oppenheimer, and M. Fleurbaey. Climate change increases resource-constrained international immobility. *Nature Climate Change*, 12(7):634–641, 2022.

George J. Borjas, Stephen G. Bronars, and Stephen J. Trejo. Self-selection and internal migration in the United States. *Journal of Urban Economics*, 32(2):159–185, 1992.

C. Brunel and M. Y. Liu. Out of the frying pan: Weather shocks and internal migration in Brazil. *World Development*, 188:106859, 2025.

G. Bryan, S. Chowdhury, and A. M. Mobarak. Underinvestment in a profitable technology: The case of seasonal migration in Bangladesh. *Econometrica*, 82(5):1671–1748, 2014.

Gharad Bryan and Melanie Morten. The aggregate productivity effects of internal

migration: Evidence from Indonesia. *Journal of Political Economy*, 127(5):2229–2268, 2019.

M. Burke, S. M. Hsiang, and E. Miguel. Global non-linear effect of temperature on economic production. *Nature*, 527(7577):235–239, 2015.

Marshall Burke and Kyle Emerick. Adaptation to climate change: Evidence from US agriculture. *American Economic Journal: Economic Policy*, 8(3):106–140, 2016.

S. Cai. Migration under liquidity constraints: Evidence from randomized credit access in China. *Journal of Development Economics*, 142:102247, 2020.

Lorenzo Caliendo, Maximiliano Dvorkin, and Fernando Parro. Trade and labor market dynamics: General equilibrium analysis of the China trade shock. *Econometrica*, 87(3): 741–835, 2019.

C. Cattaneo and G. Peri. The migration response to increasing temperatures. *Journal of Development Economics*, 122:127–146, 2016.

Raj Chetty, Nathaniel Hendren, and Lawrence F. Katz. The effects of exposure to better neighborhoods on children: New evidence from the Moving to Opportunity experiment. *American Economic Review*, 106(4):855–902, 2016.

Daniel Chiquiar and Gordon H. Hanson. International migration, self-selection, and the distribution of wages: Evidence from Mexico and the United States. *Journal of Political Economy*, 113(2):239–281, 2005.

Michael A. Clemens and Mariapia Mendola. Migration from developing countries: Selection, income elasticity, and Simpson's paradox. *Journal of Development Economics*, 171:103359, 2024.

J. Colmer. Temperature, labor reallocation, and industrial production: Evidence from India. *American Economic Journal: Applied Economics*, 13(4):101–124, 2021.

B. Conte. Climate change and migration: The case of Africa. *Journal of the European Economic Association*, 2026. doi: 10.1093/jeea/jvag035. Advance article, published online 29 May 2026.

B. Conte, K. Desmet, D. K. Nagy, and E. Rossi-Hansberg. Local sectoral specialization in a warming world. *Journal of Economic Geography*, 21(4):493–530, 2021.

Francisco Costa, Jason Garred, and João Paulo Pessoa. Winners and losers from a commodities-for-manufactures trade boom. *Journal of International Economics*, 102: 50–69, 2016. doi: 10.1016/j.jinteco.2016.04.005.

A. Costinot, D. Donaldson, and C. Smith. Evolving comparative advantage and the impact of climate change in agricultural markets: Evidence from 1.7 million fields around the world. *Journal of Political Economy*, 124(1):205–248, 2016.

J.-L. Cruz. Global warming and labor market reallocation. SSRN Scholarly Paper 4946752, 2024.

J.-L. Cruz and E. Rossi-Hansberg. The economic geography of global warming. *Review*

of Economic Studies, 91(2):899–939, 2024.

Angus Deaton and John Muellbauer. An almost ideal demand system. *American Economic Review*, 70(3):312–326, 1980.

Robert Dekle, Jonathan Eaton, and Samuel Kortum. Global rebalancing with gravity: Measuring the burden of adjustment. *IMF Staff Papers*, 55(3):511–540, 2008.

Olivier Deschênes and Michael Greenstone. The economic impacts of climate change: Evidence from agricultural output and random fluctuations in weather. *American Economic Review*, 97(1):354–385, 2007.

K. Desmet and E. Rossi-Hansberg. On the spatial economic impact of global warming. *Journal of Urban Economics*, 88:16–37, 2015.

Rafael Dix-Carneiro. Trade liberalization and labor market dynamics. *Econometrica*, 82 (3):825–885, 2014.

Rafael Dix-Carneiro and Brian K. Kovak. Trade liberalization and regional dynamics. *American Economic Review*, 107(10):2908–2946, 2017. doi: 10.1257/aer.20161214.

Rafael Dix-Carneiro and Brian K. Kovak. Margins of labor market adjustment to trade. *Journal of International Economics*, 117:125–142, 2019. doi: 10.1016/j.jinteco.2019.01.005.

Christian Dustmann and Anna Okatenko. Out-migration, wealth constraints, and the quality of local amenities. *Journal of Development Economics*, 110:52–63, 2014.

J. Eaton and S. Kortum. Technology, geography, and trade. *Econometrica*, 70(5):1741–1779, 2002.

F. Farrokhi and H. S. Pellegrina. Trade, technology, and agricultural productivity. *Journal of Political Economy*, 131(9):2509–2555, 2023.

S. Feng, A. B. Krueger, and M. Oppenheimer. Linkages among climate change, crop yields and Mexico–US cross-border migration. *Proceedings of the National Academy of Sciences*, 107(32):14257–14262, 2010.

Jesús Fernández-Huertas Moraga. New evidence on emigrant selection. *Review of Economics and Statistics*, 93(1):72–96, 2011.

Luiz Felipe Campos Fontes, Paulo de Andrade Jacinto, and Marco Tulio França. Programas de transferência de renda e migração interna: Evidências do Programa Bolsa Família. *Análise Econômica*, 37(72), 2019.

J. Gazeaud, E. Mvukiyehe, and O. Sterck. Cash transfers and migration: Theory and evidence from a randomized controlled trial. *Review of Economics and Statistics*, 105 (1):143–157, 2023.

E. Giannone, Q. Li, N. Paixão, and X. Pang. Unpacking moving: A quantitative spatial equilibrium model with wealth. Staff Working Paper 2023-34, Bank of Canada, 2023.

D. Gollin, C. W. Hansen, and A. M. Wingender. Two blades of grass: The impact of the green revolution. *Journal of Political Economy*, 129(8):2344–2384, 2021.

C. Gouel and D. Laborde. The crucial role of domestic and international market-

mediated adaptation to climate change. *Journal of Environmental Economics and Management*, 106:102408, 2021.

S. Jayachandran. Selling labor low: Wage responses to productivity shocks in developing countries. *Journal of Political Economy*, 114(3):538–575, 2006.

R. Jedwab, F. Haslop, R. Zarate, and C. Rodriguez Castelan. The effects of climate change in the poorest countries: Evidence from the permanent shrinking of Lake Chad. IZA Discussion Paper 16396, IZA, 2023.

John Kennan and James R. Walker. The effect of expected income on individual migration decisions. *Econometrica*, 79(1):211–251, 2011.

Marieke Kleemans. Migration choice under risk and liquidity constraints. Working paper, University of Illinois, Urbana-Champaign, March 2023.

Patrick Kline and Enrico Moretti. People, places, and public policy: Some simple welfare economics of local economic development programs. *Annual Review of Economics*, 6 (1):629–662, 2014.

Brian K. Kovak. Regional effects of trade reform: What is the correct measure of liberalization? *American Economic Review*, 103(5):1960–1976, 2013. doi: 10.1257/aer.103.5.1960.

D. Lagakos, A. M. Mobarak, and M. E. Waugh. The welfare effects of encouraging rural–urban migration. *Econometrica*, 91(3):803–837, 2023.

D. B. Lobell and S. M. Gourdji. The influence of climate change on global crop productivity. *Plant Physiology*, 160(4):1686–1697, 2012.

David McKenzie and Hillel Rapoport. Network effects and the dynamics of migration and inequality: Theory and evidence from Mexico. *Journal of Development Economics*, 84(1):1–24, 2007.

Robert Mendelsohn, William D. Nordhaus, and Daigee Shaw. The impact of global warming on agriculture: A Ricardian analysis. *American Economic Review*, 84(4): 753–771, 1994.

M. Morten and J. Oliveira. The effects of roads on trade and migration: Evidence from a planned capital city. *American Economic Journal: Applied Economics*, 16(2):389–421, 2024.

D. G. O'Connor. Revitalize or relocate: Optimal place-based transfers for local recessions. Working paper, Massachusetts Institute of Technology; Chapter 2 of the author's PhD thesis, *Essays on Spatial Economics*, MIT, 2025.

H. S. Pellegrina. Trade, productivity, and the spatial organization of agriculture: Evidence from Brazil. *Journal of Development Economics*, 156:102816, 2022.

H. S. Pellegrina and S. Sotelo. Migration, specialization, and trade: Evidence from Brazil's march to the west. *Journal of Political Economy*, 133(12):3993–4049, 2025.

D. Restuccia, D. T. Yang, and X. Zhu. Agriculture and aggregate productivity: A

quantitative cross-country analysis. *Journal of Monetary Economics*, 55(2):234–250, 2008.

W. Schlenker and M. J. Roberts. Nonlinear temperature effects indicate severe damages to US crop yields under climate change. *Proceedings of the National Academy of Sciences*, 106(37):15594–15598, 2009.

Ina Simonovska and Michael E. Waugh. The elasticity of trade: Estimates and evidence. *Journal of International Economics*, 92(1):34–50, 2014.

S. Sotelo. Domestic trade frictions and agriculture. *Journal of Political Economy*, 128(7): 2690–2738, 2020.

A. C. Xavier, B. R. Scanlon, C. W. King, and A. I. Alves. New improved Brazilian daily weather gridded data (1961–2020). *International Journal of Climatology*, 42(16): 8390–8404, 2022. Data version v_3.2.4 (1961–2025) used.

Online Appendix I: Data and Empirical Evidence

A. Data

A.1. Geographic units and the municipality–microregion crosswalk

All variables enter the analysis at the level of the *microregion* (*microrregião geográfica*), the intermediate spatial unit into which the Brazilian statistical institute (IBGE) groups contiguous municipalities with integrated production structures and labor markets. The division comprises 558 microregions and was the country's official regional partition throughout the period I study: it was in force from 1989 until it was superseded by the *regiões geográficas imediatas e intermediárias* in 2017.[13] I exclude the oceanic archipelago of Fernando de Noronha, leaving 557 mainland microregions. Boundaries are fixed across all years and sources.

I rely on microregions instead of municipalities for two reasons. First, municipalities were not fixed over the window: driven by *desmembramento* (the splitting-off of new municipalities from existing ones), the number of Brazilian municipalities rose from about 4,490 in 1991 to 5,565 in 2010 and 5,570 from 2013 on. While the underlying municipalities changed, their unique assignment to microregions, and the microregion boundaries did not. Second, microregions are designed to be a combination of adjacent municipalities with integrated production structures and labor markets. They are coarse enough that most routine, short- distance mobility takes place within them, so that a recorded change of microregion reflects a costly relocation, yet numerous enough to retain wide variation in climate exposure, agricultural specialization, and income.

Constructing the municipality-to-microregion crosswalk. The Census reports the location of current and past residence at the municipal level and must be mapped to microregions. I build the crosswalk from official IBGE geometries distributed through the `geobr` package.[14] For every municipality, I take its full-resolution polygon and assign it to the microregion that contains the largest share of its area.[15]

How each source reaches the microregion. The mapping differs by source, but every source rests on the *same* fixed microregion division, which is what makes them

[13] IBGE, *Divisão Regional do Brasil em Mesorregiões e Microrregiões Geográficas* (Resolução PR n. 51, 31/07/1989). See https://www.ibge.gov.br/geociencias/organizacao-do-territorio/estrutura-territorial/.

[14] Pereira and Gonçalves, *geobr: Download Official Spatial Data Sets of Brazil*; https://ipeagit.github.io/geobr/.

[15] In later years, boundaries are precise. Earlier municipality shapefiles are only available at a much lower spatial accuracy, so that municipal and microregion boundaries may not overlap exactly.

comparable (Table A.I). The Census is aggregated to microregions through the crosswalk above. The agricultural data (PAM) are instead retrieved *already aggregated to the microregion* from IBGE's SIDRA system,[16] so the municipal aggregation is performed by IBGE on the official division rather than by me. The gridded climate variables, observed and projected, are mapped to microregions by area-weighted averaging of the grid cells intersecting each microregion polygon.

TABLE A.I
Spatial handling by data source

Source	Coverage	Native level	Aggregation to microregion
Demographic Census	2000, 2010	municipality	crosswalk (spatial, = official)
PAM	1994–2017	microregion	by IBGE (SIDRA), server-side
Climate, observed (BR-DWGD)	1961–2025	0.1° grid	area-weighted zonal averaging onto microregion polygons
Climate, projected (CMIP6 CMCC-ESM2)	1994–2100	1.25° × 0.94° grid	area-weighted zonal averaging onto microregion polygons

Centroids, areas, and distances. Finally, the geometry supplies the spatial primitives used elsewhere in the analysis. From the microregion polygons I compute, on the WGS84 sphere, each microregion's centroid and its area; the full matrix of bilateral migration distances is the great-circle distance between microregion centroids.

A.2. The Demographic Census

The individual-level data are the microdata of the Brazilian Demographic Census (*Censo Demográfico*) for 2000 and 2010, collected by IBGE. Income, migration, industry and occupation are asked only in the census's long questionnaire, which goes to a sample of households: about one in ten nationally, and a larger share in small municipalities. The microdata cover this sample and come with person weights that restore population totals.[17] Brazil's main source of individual panel data is RAIS (*Relação Anual de Informações Sociais*), the administrative register of formal employment, which follows workers across employers and places over time. It covers only formal jobs, however, and so misses informal and self-employed workers and much of farm labour, the low-income earners on whom this paper focuses; a worker who moves into informal work or out of work leaves the data. The census covers everyone, formal or not, and records where each person lived five years earlier. It does not follow individuals across waves, so the analysis compares two cross-sections, each with its own five-year migration window, rather than following the same people over time.

[16] https://sidra.ibge.gov.br/; PAM pulled at the `MicroRegion` level.

[17] Retrieved through the `censobr` package, which serves IBGE's public microdata files; see https://ipeagit.github.io/censobr/. I download every column of each wave verbatim and perform all variable selection and harmonisation myself, so the extract is reproducible from the raw files.

Harmonisation. The waves do not share a variable dictionary, so every object used in the paper is built from its wave-specific code (Table A.II). Three harmonisation choices are worth stating.

First, *who counts as working*. The employment-status variables are not comparable: the 2000 question (`V0439`) asks whether the respondent worked *for pay*, while the 2010 question (`V6920`) asks whether they were *occupied*. Using them would drop most of 2000's unpaid and own-consumption workers while keeping 2010's, which inflates the measured agricultural share in 2010 for purely definitional reasons. I instead define a worker as someone reporting a codable industry of work, which is asked identically in both waves.

Second, *which income*. Total income (all sources) and labour earnings (all jobs) are separate variables, and the census leaves the earnings variables blank for anyone who did not work. Total income is used wherever the concept is ability to pay, with missing values set to zero (the census category "sem rendimento"); labour earnings are used wherever the concept is productivity and are deliberately left missing for non-workers, since their labour income is unrevealed rather than zero. All nominal values are deflated to 2010 reais with the IPCA (*Índice Nacional de Preços ao Consumidor Amplo*), IBGE's official consumer price index.

Third, *which classification*. The industry classification changed between the waves, from CNAE-Domiciliar 1.0 in 2000 to 2.0 in 2010, and the 2010 wave codes occupation in two classifications. The paragraphs on industry and occupation below show which codes are comparable across the waves and use only those.

TABLE A.II
Census variables by wave

Object	2000	2010	Classification
Person weight	`P001`	`V0010`	—
Age, gender	`V4752`, `V0401`	`V6036`, `V0601`	—
Educational attainment	`V4300`	`V6400`	years of study vs. attainment level
Industry of work	`V4462`	`V6471`	CNAE-Domiciliar 1.0 / 2.0
Occupation	`V4452`	`V6462`	CBO-Domiciliar 2000 (both)
Position in employment	`V0447`	`V6930`	—
Total income, all sources	`V4614`	`V6527`	monthly, deflated to 2010 R$
Labour earnings, all jobs	`V4525`	`V6525`	monthly, deflated to 2010 R$
Municipality of residence	`code_muni`	`code_muni`	mapped to microregions (§A.1)
Residence five years ago	`V4250`, `V4260`	`V6264`, `V6262`	municipality / state
Urban or rural	`V1006`	`V1006`	household situation

Migration. Both waves ask where the respondent lived five years before the census, coded to the municipality. A *mover* is someone whose microregion of residence differs from the microregion of that earlier residence, so the 2000 and 2010 waves measure the

migration windows 1995–2000 and 2005–2010. Individuals who lived abroad five years earlier, or who were not yet born, are dropped, since neither has an origin inside the model. Within-microregion moves are not counted as migration.

Industry: the agricultural sector. Industry is the five-digit CNAE-Domiciliar activity code. Five-digit codes are not comparable across the two vintages in general, but the agricultural block is: *agriculture* (divisions 01–02: crops, livestock, agricultural services, hunting and forestry, excluding fishing) and *crop cultivation* (group 011, a strict subset) are defined identically in CNAE-Dom 1.0 and 2.0.

The model's farm sector is crop cultivation rather than the broader *agropecuária*, so that labour, output and area all describe the same activity (Section 5). Crop cultivation employs 11.1% of working non-movers in the analysis sample against 14.9% for broad agriculture (Table A.III); the difference is livestock, forestry and agricultural services, which the model therefore places in the non-farm sector. The table also reports the model's farm sector itself, crop cultivation in an agricultural occupation (defined below): 10.5% of workers in the pooled sample.

Occupation: who does farm work. The model's farm sector is defined by two codes, because each answers a different question. The industry code classifies the *establishment*: it says what is produced, and so ties the worker to crop cultivation, the activity whose output and area PAM records (Appendix A.3). It does not say what the worker does. An agronomist, a payroll clerk and a production director employed by a large soybean operation are all recorded in crop cultivation, alongside the field labour the model means by farm work. The occupation code says what the worker does: both waves carry a four-digit CBO-Domiciliar 2000 code, whose major group 6 is agricultural, forestry and fishery work.[18] Occupation alone would not do either, since major group 6 also covers livestock, forestry and fishery workers, whose output PAM does not record. A farm worker in the model is therefore a worker in crop cultivation *and* in an agricultural occupation; every other worker is in the non-farm sector. The crop-industry workers this rule places in the non-farm sector, managers, professionals, technicians and clerks, are few, 2.6% of crop-industry workers in 2000 and 4.5% in 2010, but in 2010 they receive 13.4% of the crop industry's income; counted as farm workers, they would make part of the measured change in the farm wage between the waves a change in who is counted.

[18] The 2010 wave also carries a CBO-Domiciliar 2010 code (`V6461`) with a different structure. I use the CBO-Dom 2000 code (`V6462`) in both waves, so that occupational shares are comparable. Workers whose occupation is uncodable (0.78% of crop-industry workers in 2000 and 0.45% in 2010) are not counted as farm workers.

Analysis sample. I restrict to individuals aged 20–65 (the working-age window over which the five-year migration decision is observed for both origin and destination), Brazilian nationals (by birth or naturalisation; foreign nationals are excluded), with a determined level of schooling, with non-missing income, resident in Brazil five years earlier, and with both the current and the earlier municipality mapping to a mainland microregion. The top 1% of the real total-income distribution is trimmed within wave, a standard guard against top-coded outliers, computed within wave because the two waves differ in composition even after deflation. Employment status is *not* a sample restriction: non-workers are retained and flagged, so that the same file serves the migration analysis, which needs everyone, and the sectoral analysis, which needs workers. The restrictions leave 22,445,097 individuals, a weighted 6.6% of whom changed microregion in the five years before their census.

Descriptive statistics for the analysis sample, split by migration status, appear in Table A.III.

Summary statistics. Table A.III shows summary statistics for the analysis sample used throughout the paper, pooled over the 2000 and 2010 waves, and splits it by migration status, so that it serves both as the overall summary and as the comparison of movers with non-movers. The pooled sample has 22,445,097 individuals. Movers are younger and more educated than non-movers and have higher incomes. These characteristics are measured *ex post*, at the census, after the move, so Section 3 ranks individuals by the income predicted at their origin instead (Appendix B.1), and Appendix B.2 shows that education carries the largest part of the resulting income gradient in migration.

Figure A.I shows the employment share in crop cultivation by microregion in 2010, and Figure I, panel (b), the average agricultural monthly income. The employment share in crop cultivation is highest in the North and Northeast regions, while the average agricultural income is highest in the South and Southeast regions. Agricultural income is the monthly labour earnings of farm workers—crop-cultivation workers in an agricultural occupation—in constant 2010 reais.

A.3. Agricultural production data

The primary data source for agricultural production is the *Produção Agrícola Municipal* (PAM), a survey conducted annually by the Brazilian Institute of Geography and Statistics (IBGE). The PAM provides detailed information on the area planted or destined for harvest, harvested area, physical production quantities, and the nominal value of production for a wide range of temporary and permanent crops.

For the empirical analysis, I use PAM data for the years 1994–2017 (Table A.I). The model's farm block uses the PAM totals over all temporary and permanent crops: the

TABLE A.III
Descriptive statistics: non-movers vs. movers

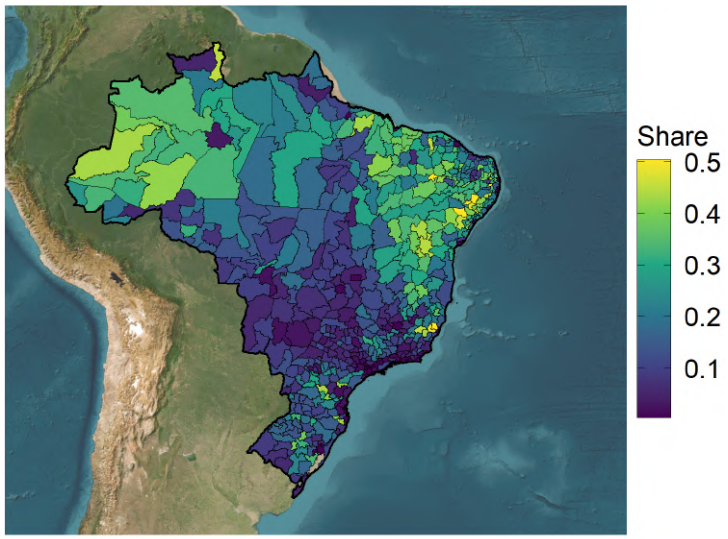
	Non-movers	Movers
Demographics		
Age	38.3 (12.4)	34.0 (11.2)
Female (%)	51.8	50.3
White (%)	50.9	52.6
Black or brown (%)	47.6	46.0
Married or in union (%)	57.9	55.5
Household size	4.1 (2.0)	3.8 (2.0)
Years in municipality	28.1 (14.7)	2.0 (1.4)
Educational attainment (%)		
Less than primary	50.4	46.2
Primary	16.1	16.9
Secondary	25.5	27.0
University	7.9	9.8
Labor market and income		
Employed (%)	62.2	61.8
In agriculture (% of workers)	14.9	11.9
In crop cultivation (% of workers)	11.1	7.9
In the farm sector (% of workers)	10.7	7.3
Monthly total income (2010 R$)	797 (1,149)	869 (1,281)
Monthly labor earnings (2010 R$)	1,084 (1,167)	1,200 (1,326)
Observations	20,943,237	1,501,860
Share of sample (%)	93.4	6.6

Notes. Pooled 2000 and 2010 Brazilian Demographic Census microdata, analysis sample (individuals aged 20–65, Brazilian nationals, with valid schooling and non-missing income, resident in Brazil five years earlier, and current and lagged municipality both mapping to a microregion; the top 1% of real total income is trimmed within year). All statistics are weighted by census person weights. A *mover* is an individual whose microregion of residence differs from that five years before the census; non-movers include within-microregion movers. Continuous variables report the weighted mean with the standard deviation in parentheses; remaining rows are weighted percentages. Income is monthly, in constant 2010 reais; labor earnings are those of workers with positive earnings. Agriculture is the share of employed workers in crop and livestock activities (CNAE-Domiciliar divisions 01–02); crop cultivation is its subset in group 011.

value of production, deflated to 2010 reais, and the area planted (temporary crops) or destined for harvest (permanent crops). For the aggregate productivity measure of Figure III, and for its crop-level counterpart in Fact 4 (Section 3) and Appendix B.6, I restrict attention to crops with strictly positive planted area in 2010 that are reported in every year, and exclude crops that display implausible jumps in yield between 2000 and 2001, ensuring a consistent set of crops across all years of the analysis.[19]

[19]The jumps reflect a change in units: from 2001, PAM reports most fruit in tonnes rather than in thousands of fruits or bunches. The rule keeps 44 crops. Output is valued at base-period (1994–1998) national prices in 2010 reais and divided by the area planted or destined for harvest, so that crops lost to adverse weather count as low productivity. The crop-specific optima of Appendix B.6 use the 8 of these

FIGURE A.I
Employment share in crop cultivation by microregion, 2010



Notes. The number of employed individuals working in crop cultivation in an agricultural occupation divided by the total number of employed individuals in each microregion, 2010. The colour scale runs from the 1st to the 99th percentile across microregions; values beyond are drawn at the end colour.

Table A.IV shows the major crops in Brazil by area planted, quantity produced, and value of production over 1994–2014. By area planted, the most important crops are soybeans, maize and sugarcane, which together account for 66 percent of the area planted. By value of production, the most important crops are soybeans, sugarcane and maize, which together account for 52 percent of the value of production.

Since the empirical analysis is conducted at the microregion level, the PAM data are retrieved already aggregated to microregions from IBGE's SIDRA system, so that the aggregation from municipalities is performed by IBGE on the official, fixed division (Appendix A.1). I merge the resulting panel with the total land area of each microregion, computed from the microregion polygons, to obtain the share of total land under crops in each microregion-year. This share is the cultivated share s_t^i of (18); because PAM sums planted area over crops and seasons, it can exceed one where land is double-cropped.

To obtain a measure of constant-price agricultural output per hectare, I revalue crop-level production using reference prices. Specifically, I construct crop-specific reference prices as the ratio of the total real value to the total quantity produced over the base period 1994–1998. Each crop's production is then revalued at these constant prices and aggregated across crops to obtain total output at the microregion level. Dividing by the area planted yields an index of agricultural land productivity at constant prices, which forms the basis of the productivity analysis presented in the main text. Figure A.II shows the share of land used in agriculture and measured agricultural productivity by

crops with the largest national revenue in 1994–1998.

TABLE A.IV
Major crops in Brazil, 1994–2014

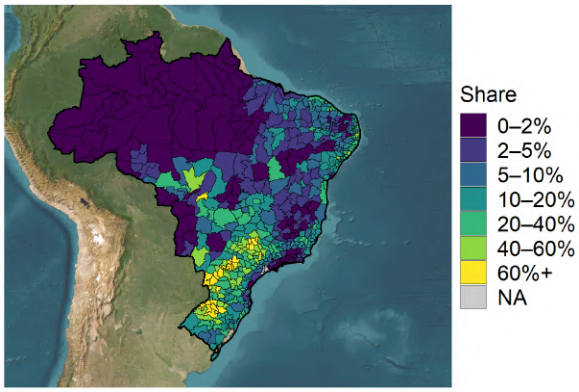
Crop	Area (%)	Quantity (%)	Value of production (%)
Soybean (grain)	31.6	6.9	23.9
Sugarcane	11.2	68.1	16.0
Maize (grain)	22.6	6.5	11.6
Coffee (bean), total	3.7	0.4	7.5
Paddy rice	5.4	1.6	4.8
Cassava	3.0	3.3	4.6
Orange	1.4	6.6	3.8
Beans (grain)	7.0	0.4	3.6
Herbaceous cotton (seed cotton)	1.6	0.4	3.0
Tobacco (leaf)	0.7	0.1	2.7
Banana (bunch)	0.9	0.7	2.4
Tomato	0.1	0.5	2.0
Potato	0.3	0.4	1.8
Wheat (grain)	3.3	0.6	1.4
Grape	0.1	0.2	1.2
Top 15 combined	93.0	96.5	90.5

Notes. IBGE Produção Agrícola Municipal (PAM), all microregions, 1994–2014. Each column is the crop's share of the total over all crops and years. Area is area planted (temporary crops) or area destined for harvest (permanent crops); quantity is tonnes; value of production is deflated to 2010 reais with the IPCA before summing. All PAM crops are included (coffee sub-series excluded to avoid double counting); crops are ordered by value share and the 15 largest are shown.

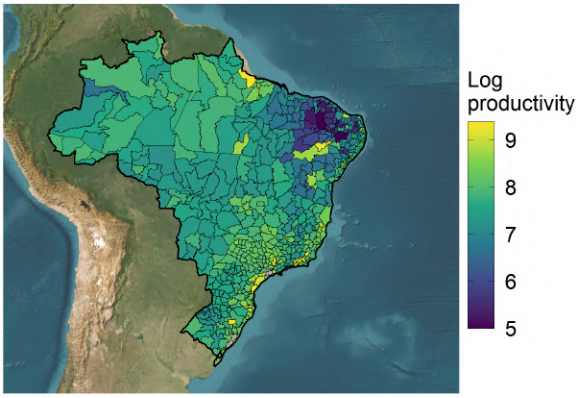
microregion in 2010.

In the "raw" measured agricultural productivity, where land-use intensity is not accounted for, some of the highest agricultural productivity levels are observed in urban microregions. This is driven by the fact that these use a small share of their land for agriculture, but the land that is used is highly productive. By (19), measured productivity $\tilde{A}^i$ exceeds fundamental productivity A^i wherever only part of the land is cultivated, so the map overstates productivity where the cultivated share is small. The model, however, never recovers A^i in levels: in (20) it enters together with the farmgate price p^i, and the counterfactuals use only its changes over time. Panel (c) is therefore the data-side ln A of Figure III, not the model's A^i, whose level would mix productivity with the price level.

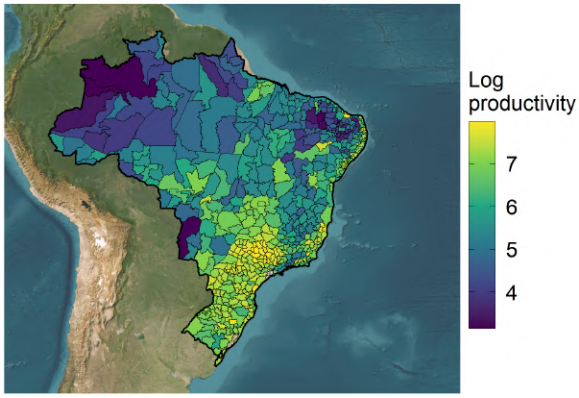
FIGURE A.II
Land use, measured productivity, and fundamental productivity by microregion, 2010



(A) Share of land used in agriculture



(B) Measured agricultural productivity



(C) Fundamental agricultural productivity

Notes. Panel (a) shows the share of land used in agriculture by microregion in 2010, computed as the area planted for all crops divided by the total land area of each microregion. Panel (b) shows measured agricultural productivity, the log of constant-price output per planted hectare of the crops selected in this appendix. Panel (c) shows fundamental agricultural productivity, the log of constant-price output per hectare adjusted for land use intensity with $\theta = 2$, as in Figure III. Microregions without crop data are grey. In panels (b) and (c), colour scales run from the 1st to the 99th percentile across microregions; values beyond are drawn at the end colours.

B. Empirical evidence

B.1. Construction of predicted counterfactual incomes

The motivating evidence in Section 3 is based on the 2000 and 2010 Brazilian Demographic Censuses, which report a respondent's current microregion of residence and the microregion of residence five years earlier. The sample is the analysis sample of Appendix A.2: Brazilian nationals (foreign nationals excluded) aged 20–65 with valid schooling and non-missing income, resident in Brazil five years earlier, yielding 22,445,097 observations across the two waves. Using the data, I estimate an income regression, for each origin microregion j and census wave t, of the form

$$\mathbb{E}\left[\, y_{ijt} \mid X_{ijt} \,\right] \;=\; \exp\!\left(X_{ijt}'\, b_{jt}\right), \tag{30}$$

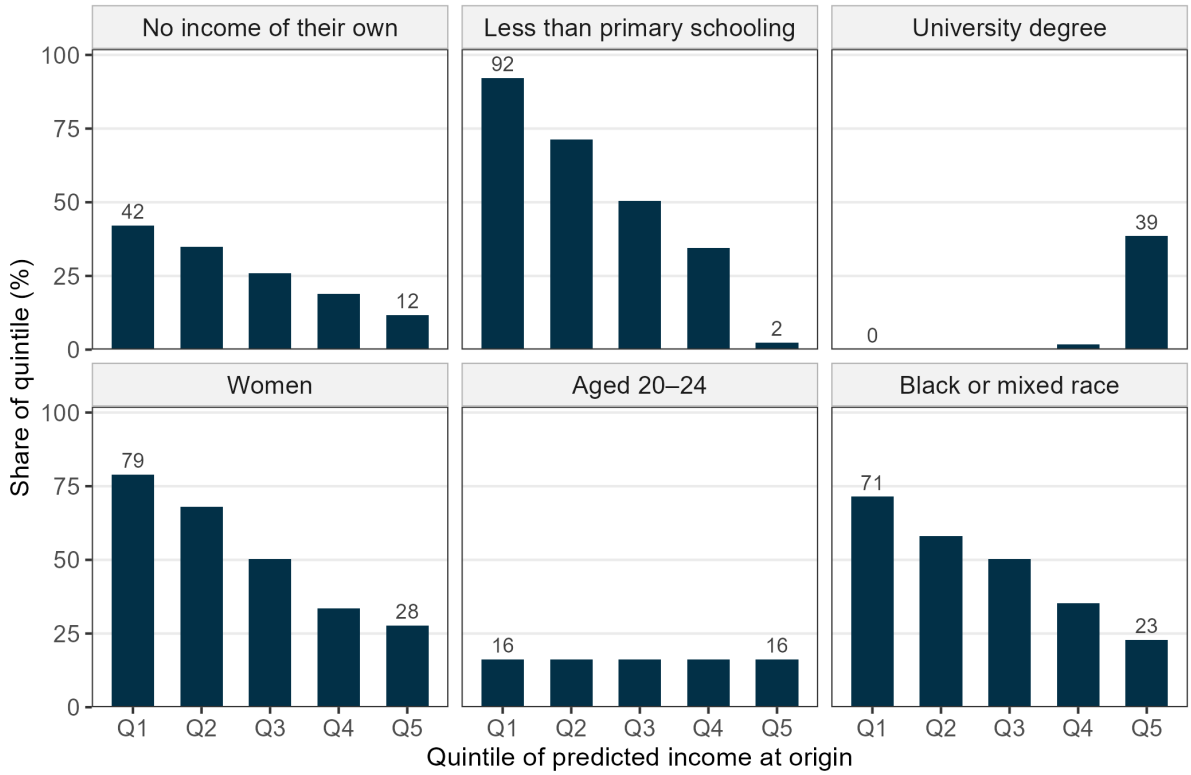
where i indexes individuals, y_{ijt} is individual total income from all sources (in 2010 R$ per month), and X_{ijt} includes primary, secondary, and university schooling, age, age squared, gender, and race. The regression is estimated by Poisson pseudo-maximum-likelihood on income levels, zeros included, on the stayers of each microregion and wave (9,766,019 in 2000 and 11,177,218 in 2010), weighted by census person weights. Where a microregion has few stayers, its slope coefficients are shrunk toward those of its state; its intercept, the level of income, is its own.

Using the estimated $\hat{b}_{jt}$, I then predict incomes for all individuals as if they had stayed in their origin microregion: each individual is evaluated at the coefficients of the microregion where they lived five years earlier, in the same wave. I interpret these fitted values as the counterfactual incomes that individuals would have had, in expectation, had they remained in their origin microregion, at their current age. Predicted values are grouped into quintiles within each five-year age group and wave, using census person weights, so that each person is ranked against others of the same age; Appendix B.4 explains why.

Figure B.I shows who is in each quintile. The lowest quintile is mostly women and people with less than primary schooling, and many of its members report no income of their own. Because quintiles are formed within age groups, every quintile has the same age composition. Almost all university graduates are in the top quintile.

What predicted income measures. The predictions are interpreted as income in the absence of migration under two assumptions. First, conditional on observables, the income structure of stayers at the census reflects the opportunities residents had five years earlier. Second, migration does not move the incomes of stayers much: net migration, which shifts local labour supply, averages 2.9% of a microregion's population over five years in absolute value. Predicted income measures what a person can expect

FIGURE B.I
Composition of the quintiles of predicted income



Notes. Share of each quintile of predicted individual income at origin (Figure II) that has the characteristic named in the panel, in percent; labels give the values for Q1 and Q5. "No income of their own" is a reported individual total income of zero at the census. Quintiles are formed within each five-year age group and census wave; shares pool the 2000 and 2010 waves, weighted by census person weights. Mean predicted income is R$ 236 a month in Q1 and R$ 1,845 in Q5.

to earn at origin year after year: the persistent part of their endowments, which a cash cost makes matter in every period. Income observed in the year before a move, where panels record it [e.g., Fernández-Huertas Moraga, 2011], would add the transitory part, which matters for a cash cost too, but it can itself reflect the decision to move: those who plan to leave may already work, save or earn differently before they go. The model has both parts: a worker's income combines the persistent position of their cell with a draw of labour efficiency that changes from period to period. Predicted income does not capture differences in ability, wealth or family support between people with the same characteristics in the same place.

B.2. Which characteristics carry the income gradient?

Because quintiles are formed within age groups, predicted income ranks people of the same age by their schooling, gender and race, so the gradient in Figure II necessarily runs through these characteristics. Table B.I asks through which of them, using the decomposition of Gelbach [2016]. Starting from the regression behind Figure II, I add

education, gender and race fixed effects, each interacted with the census wave, and attribute the change in each quintile's coefficient to the characteristic that absorbs it. The attribution is exact and does not depend on the order in which the characteristics are added.

Of the 2.22 percentage-point difference between the top and the bottom quintile, education accounts for 79%, race for 10% and gender for 13%; together they account for essentially all of it (the unexplained remainder is -2%). For the lower quintiles a larger part stays unexplained. It comes from how the characteristics combine in predicted income and from origin-specific returns to them, neither of which additive fixed effects for each characteristic absorb.

These shares do not separate the income channel from other channels that run through the same characteristics. Schooling is the main determinant of earnings capacity, so the decomposition shows which observable the income ranking runs through, not whether income or some other correlate of schooling, such as information or access to networks, makes people move.

TABLE B.I
Which characteristics carry the migration–income gradient?

	Q2	Q3	Q4	Q5
	(difference to Q1, percentage points)			
Base gradient	$+0.95^{***}$	$+1.09^{***}$	$+1.39^{***}$	$+2.22^{***}$
	(0.10)	(0.11)	(0.13)	(0.31)
Explained by:				
Education	+0.25 [26%]	+0.44 [40%]	+0.65 [47%]	+1.75 [79%]
Gender	+0.09 [10%]	+0.18 [16%]	+0.26 [19%]	+0.29 [13%]
Race	+0.06 [6%]	+0.09 [8%]	+0.16 [12%]	+0.23 [10%]
Unexplained (full model)	$+0.55^{***}$ [58%]	$+0.38^{*}$ [35%]	+0.31 [23%]	−0.05 [−2%]
	(0.15)	(0.22)	(0.33)	(0.23)

Notes. Gelbach (2016) decomposition of the difference in the five-year migration probability between each quintile of predicted income at origin and the lowest (Q1), in percentage points. The base row is Figure II: a linear probability model of migration (changing microregion) on quintile dummies with origin × wave and five-year age group × wave fixed effects. The full model adds education × wave, gender × wave and race × wave fixed effects; each "explained by" row is the part of the base coefficient that runs through that characteristic. The "unexplained" row is the quintile coefficient of the full model: the migration gap to Q1 that remains among people of the same education, gender and race. The three "explained by" rows and the unexplained row sum exactly to the base gradient, so all entries are in percentage points; brackets give each row's share of the base gradient, in percent. Standard errors of the base and full-model coefficients, clustered by origin microregion, in parentheses. Sample, weights and predicted income as in Figure II.

B.3. What each education level could earn elsewhere

Schooling carries most of the income gradient in migration, so the low-income might migrate less because they gain less from moving. Table B.II compares, for each education level, mean income in the Southeast with mean income in the origin. The gap falls steeply with schooling. For origins in the North and Northeast, the mean income of adults with less than primary schooling is about twice as high in the Southeast in 1991 and 2000 and 83% higher in 2010, while that of university graduates is 16% higher. These are differences in means, not the gains of a given mover, but on average the least educated have the most to gain from moving.

TABLE B.II
Mean income in the Southeast relative to the origin, by education (percent)

	Origins in the North and Northeast			All origins
Education	1991	2000	2010	2010
Less than primary	104	106	83	27
Primary	73	75	70	16
Secondary	50	48	47	11
University	16	17	16	4

Notes. For each origin microregion and education level, $100\,(\bar{y}_{\mathrm{SE},e}/\bar{y}_{j,e} - 1)$, where $\bar{y}_{j,e}$ is the mean real total income of adults with positive income in microregion j and $\bar{y}_{\mathrm{SE},e}$ the population-weighted mean of $\bar{y}_{j,e}$ across the microregions of the Southeast, computed from the mean log gap and averaged across origins with population weights. Cells with at least 25 observations. Census person weights.

B.4. Robustness of the income gradient

The census records income only after the move, so Figure II ranks people by income predicted at their origin. Behind that ranking are several choices: how income is predicted, at what level of geography (origin cells can be sparse), when relative to the move, and within which group people are compared. This section varies each of them, keeping the comparison of Figure II: people of the same origin and age, ranked by the income they would have had at origin at their current age. In total I estimate 400 binned specifications. Every regressor is discrete, so the census collapses exactly to cells of origin, wave, schooling, gender, race, single-year age and marital status, and each linear probability model reproduces the person-level estimate. Table B.III condenses the results.

Age: cash versus attachment. The young have little cash, but they are the most mobile group (Figure B.II): over five years, 9.2% of those aged 20–29 change microregion, against 4.0% of those aged 45–65. Their mobility reflects weak attachment to home and

a long horizon over which to recoup the cost of moving, not the absence of a cash cost. The question is therefore whether, among people of the same age, those with less cash leave less. The age group fixed effects absorb the higher mobility of the young. The bins matter as well. With national quintiles of predicted income, 34% of those under 30 fall into the lowest quintile, against 10% of those aged 50 and over (panel c). Within the young, the lowest national quintile then mixes those with little cash for their age with those who are simply young, and the comparison at the bottom is blurred. With national quintiles and age fixed effects, Q2 migrates 0.13 points more than Q1 (standard error 0.10), and several of the lowest national ventiles migrate less than the lowest (by up to 0.45 points); without age fixed effects every national quintile migrates less than the lowest (Q2: -1.14, Q5: -0.43 points), because the lowest is full of the young. Ranking each person against others of the same age resolves this (Table B.III, Panel A): Q2 migrates 0.95 points more than Q1, and Q5 2.22 points more.

Sorting across origins. Without origin fixed effects there is no clear gradient: the raw five-year migration rate is 6.9% in Q1 and 7.0% in Q5. Low-income individuals live in origins from which many people leave. The average origin of a Q1 member loses 7.7% of its residents over five years, that of a Q5 member 6.1%, and across origins emigration falls with mean income. This composition lowers the raw Q5–Q1 gap by 1.61 percentage points.

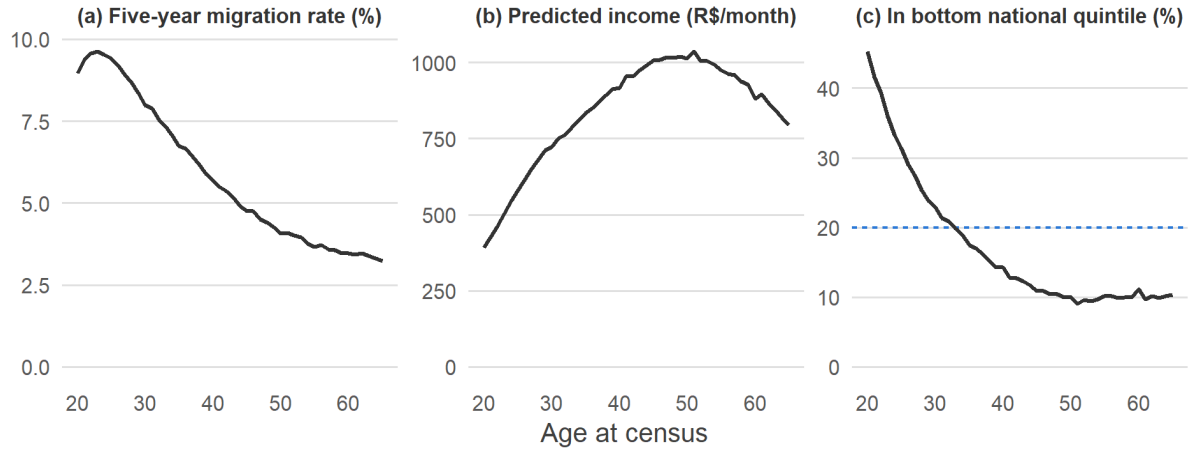
Income prediction. Panel B of Table B.III replaces the income regression by (i) k-nearest-neighbour matching, which predicts a person's income as the mean income of the $k = 30$ stayers in the same origin and wave with the same schooling, gender and race who are closest in age (if the cell has fewer than k stayers, race is dropped, then the state replaces the origin); (ii) a Mincer regression of log income on stayers with positive income; (iii) regressions at the state or national level, which avoid sparse origin cells; (iv) labour earnings instead of total income; (v) interactions of schooling and gender with age, and marital status; and (vi) a fit on all current residents of the origin rather than on stayers. In each case the lowest bin migrates least and migration rises with predicted income.

Income before the move. Migrants in the 2010 Census moved between 2005 and 2010, so their income at origin could instead be predicted from the 2000 Census, at their age at the start of that window. For 2010 migrants aged 25–65, the Q5–Q1 gap is then 2.24 points (standard error 0.18), against 2.37 with same-year income.

Samples. Panel C of Table B.III shows the same in each census wave, for men (3.40 points) and women (2.65), in poor and rich regions, for moves across states (1.03) and

with standard errors clustered by state. Across all 297 specifications that compare people of the same origin and age, the lowest bin migrates significantly less than every other bin in 98% of cases. The 5 exceptions all use very fine bins (ventiles of a prediction with national coefficients, which ties many people at the same value, and deciles formed within origin and age group); in each, the second bin migrates less than the lowest, while the top bin migrates more.

FIGURE B.II
Age, income and migration



Notes. By single year of age at the census, 2000 and 2010 pooled, weighted by census person weights. (a) Share who changed microregion in the five years before the census. (b) Mean predicted income at origin (2010 R$ per month). (c) Share in the lowest national quintile of predicted income, formed by wave; the dashed line is the 20% that quintiles formed within age groups, as in Figure II, give at every age.

B.5. Distance and income

Do low-income movers face a steeper cost of distance? The distance that movers travel is a poor guide, because it mixes the cost of distance with where each group wants to go: higher-income movers may travel farther simply because the labour markets they target are far away. I therefore estimate a gravity equation by income quintile,

$$M_{odqt} = \exp\left(\alpha_{oqt} + \delta_{dqt} - \beta_q \log \text{dist}_{od}\right) \eta_{odqt}, \tag{31}$$

where M_{odqt} is the census-weighted number of people in income quintile q (formed within age group, as in Figure II) who moved from origin microregion o to destination microregion d in the five years before census t, including all pairs with no movers, and dist_{od} is the distance between microregion centroids. The origin fixed effects α_{oqt} absorb how many people of each group leave each origin. The destination fixed effects δ_{dqt} are specific to each income group, so they absorb how attractive each destination is to

TABLE B.III
Robustness of the migration–income gradient

	Q5 – Q1 (pp)	(SE)	Q1 lowest	Mean rate (%)
A. Comparison group				
Fact 1	$+2.22^{***}$	(0.31)	✓	6.6
1-year age × wave FE	$+2.24^{***}$	(0.31)	✓	6.6
origin × age group × wave FE	$+2.13^{***}$	(0.29)	✓	6.6
+ gender and race × wave FE	$+2.33^{***}$	(0.42)	✓	6.6
no age FE	$+2.46^{***}$	(0.29)	✓	6.6
no origin FE	$+0.04$	(0.27)		6.6
National quintiles (not within age group)	$+2.08^{***}$	(0.41)		6.6
no age FE	-0.43	(0.27)		6.6
B. Income prediction				
k-nearest-neighbour matching ($k = 30$)	$+1.85^{***}$	(0.30)	✓	6.6
OLS on log income (Mincer)	$+2.60^{***}$	(0.30)	✓	6.6
State coefficients	$+2.04^{***}$	(0.27)	✓	6.6
National coefficients	$+1.74^{***}$	(0.33)	✓	6.6
+ interactions and marital status	$+2.13^{***}$	(0.30)	✓	6.6
Fit on all current residents	$+2.21^{***}$	(0.30)	✓	6.6
Labour earnings	$+2.08^{***}$	(0.28)	✓	6.6
C. Samples, outcome, inference				
2000 census	$+1.95^{***}$	(0.45)	✓	7.4
2010 census	$+2.43^{***}$	(0.19)	✓	5.9
Ages 20–29	$+1.92^{***}$	(0.50)	✓	9.2
Ages 30–44	$+2.51^{***}$	(0.31)	✓	6.5
Ages 45–65	$+1.87^{***}$	(0.11)	✓	4.0
Men	$+3.40^{***}$	(0.36)	✓	6.8
Women	$+2.65^{***}$	(0.50)	✓	6.4
Origins in the North and Northeast	$+2.90^{***}$	(0.19)	✓	7.0
Other origins	$+2.35^{***}$	(0.31)	✓	6.3
Outcome: moved state	$+1.03^{***}$	(0.22)	✓	3.0
SE clustered by origin state	$+2.22^{***}$	(0.46)	✓	6.6

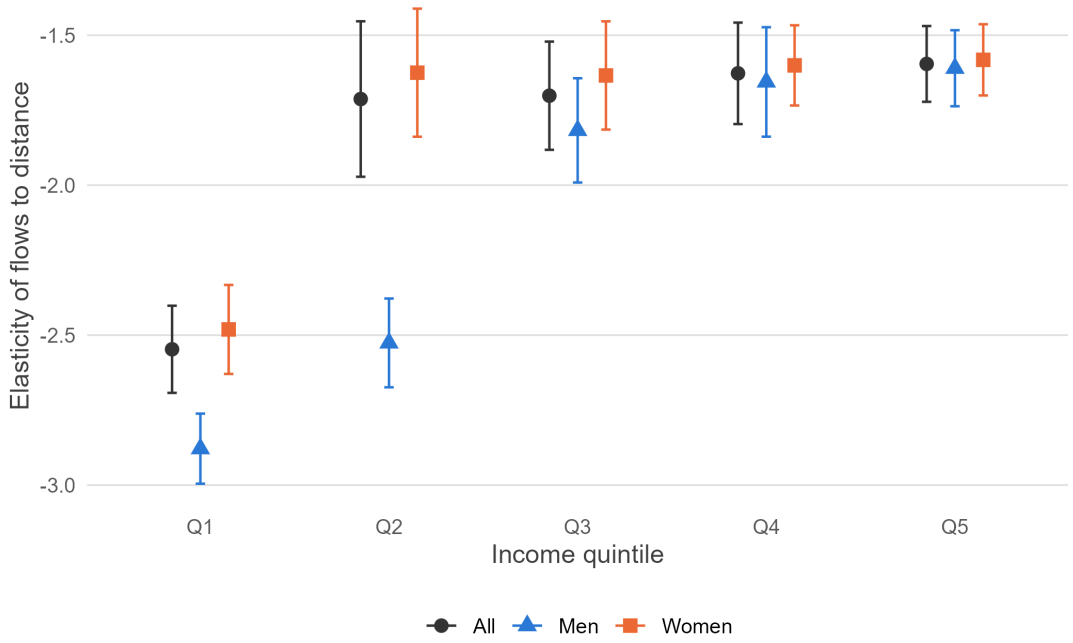
Notes. Each row is a linear probability model of migration (changing microregion over the five years before the census) on dummies for bins of predicted income at origin, with the lowest bin omitted, weighted by census person weights; standard errors clustered by origin microregion unless stated. Unless stated, bins are quintiles of predicted income formed within five-year age group × wave (Figure II), and the fixed effects are origin × wave and five-year age group × wave. "Q5 – Q1" is the coefficient of the top quintile, in percentage points. "Q1 lowest" marks rows in which every other quintile migrates significantly more than Q1 (all coefficients positive, the smallest significant at 5%). "Mean rate" is the five-year migration rate of the sample. *, **, ***: significant at 10, 5, 1%.

that group. The distance elasticity β_q is then identified only from people of the same group who reach the same destination from origins at different distances. I estimate it by Poisson pseudo-maximum likelihood, with standard errors clustered by origin and by destination.

Figure B.III shows the result. Distance deters the lowest quintile far more than the

others: the elasticity is -2.55 in Q1 and between -1.71 and -1.60 in Q2 to Q5, a difference between Q1 and Q5 of 0.95 (standard error 0.12). Allowing destination preferences to differ by income matters: with destination fixed effects common to all quintiles, the elasticities are -1.81 in Q1 and -1.47 in Q5, a difference of 0.34. The pattern is not driven by the many women in the lowest quintile: the Q1 and Q5 elasticities are -2.88 and -1.61 among men, and -2.48 and -1.58 among women. As in any gravity equation, distance also proxies for information and networks, which fade with distance and may matter more for the poor, so the result is consistent with a cost of moving that rises with distance and weighs more at low income, rather than proof of it.

FIGURE B.III
Distance elasticity of migration, by income



Notes. Coefficients $-\beta_q$ of a Poisson pseudo-maximum likelihood gravity regression of bilateral migration flows between microregions on log distance interacted with income quintile, with origin × quintile × wave and destination × quintile × wave fixed effects; 2000 and 2010 Censuses pooled; all origin–destination pairs, including those with no movers. Income quintiles of predicted income at origin, formed within five-year age group × wave (Figure II). Men and women: the same regression on flows of men and of women only. 95% confidence intervals, standard errors clustered by origin and by destination microregion.

B.6. Temperature and agricultural productivity

Robustness of the temperature–productivity relationship. Table B.IV re-estimates the quadratic of Figure III with other fixed effects and time trends, and with other land-share adjustments. The relationship is concave in every specification.

TABLE B.IV
Temperature and fundamental productivity: robustness

	Curvature $\hat{\beta}_2$		Optimum (°C)		
	estimate	s.e.	estimate	95% CI	Obs.
Fixed effects and trends ($\theta = 2$)					
Main: microregion FE + macro-region cubic trends	−0.0179	(0.0025)	19.8	[19.0, 20.6]	13,254
Microregion FE only	−0.0143	(0.0030)	21.4	[20.6, 22.2]	13,254
Microregion FE + year FE	−0.0206	(0.0029)	19.8	[19.0, 20.6]	13,254
Microregion FE + microregion cubic trends	−0.0238	(0.0026)	20.7	[20.2, 21.1]	13,254
Microregion FE + year FE + microregion linear trends	−0.0232	(0.0024)	20.4	[19.9, 20.9]	13,254
Microregion FE + state × year FE	−0.0102	(0.0035)	14.9	[8.9, 20.9]	13,230
Land-share adjustment (main fixed effects)					
$\theta = 4$	−0.0167	(0.0023)	20.1	[19.4, 20.8]	13,254
No adjustment (measured productivity)	−0.0154	(0.0022)	20.4	[19.7, 21.1]	13,254

Notes. Quadratic regressions of log fundamental agricultural productivity on annual mean temperature, microregion-years 1994–2017, as in Figure III. The first row is the specification of Figure III. The lower panel changes only the land-share adjustment θ in (19): $\theta = 4$ halves the adjustment of the first row, whose $\theta = 2$ is also the model's; without adjustment the dependent variable is measured productivity. The optimum is $-\hat{\beta}_1/(2\hat{\beta}_2)$ with a delta-method 95% confidence interval. Standard errors are clustered by microregion.

Microregion level and crop level. Fundamental productivity aggregates the output of all crops grown in a microregion. Table B.V compares it with estimates at the crop level, on the same sample: the 44 crops of Figure III, output per planted hectare at the same prices, and the same temperature data. A crop that is planted and yields little therefore counts as low productivity at both levels. A crop that is planted and yields nothing at all has no log yield and drops out of the crop-level regressions; this applies to 0.28% of planted crop-microregion-years. The crop-level regressions include crop × microregion and crop × year fixed effects.

Rows (1) and (5) show that the two levels give the same answer. The curvature of log crop yields, -0.0168, is statistically indistinguishable from that of fundamental productivity, -0.0179, and the optimum, 20.5°C, lies within the confidence interval of Figure III. Rows (2) to (4) show what the microregion-level estimate adds. Row (3) holds the crop mix of each microregion fixed at its 1994–98 revenue shares, so that only the yields of the crops it grew then can respond to temperature. This steepens the curve from -0.0154 to -0.0218: as temperatures change, farmers move towards crops that fare better, which absorbs about 29% of the curvature. Row (4), the crop-level regression weighted by the same 1994–98 revenue shares, is its crop-level counterpart

and is similarly steep (-0.0253). The microregion-level relationship used in the model therefore includes this margin of adaptation; a damage function built from crop-level yields at a fixed crop mix would omit it.

TABLE B.V
Temperature and productivity at the microregion and crop level

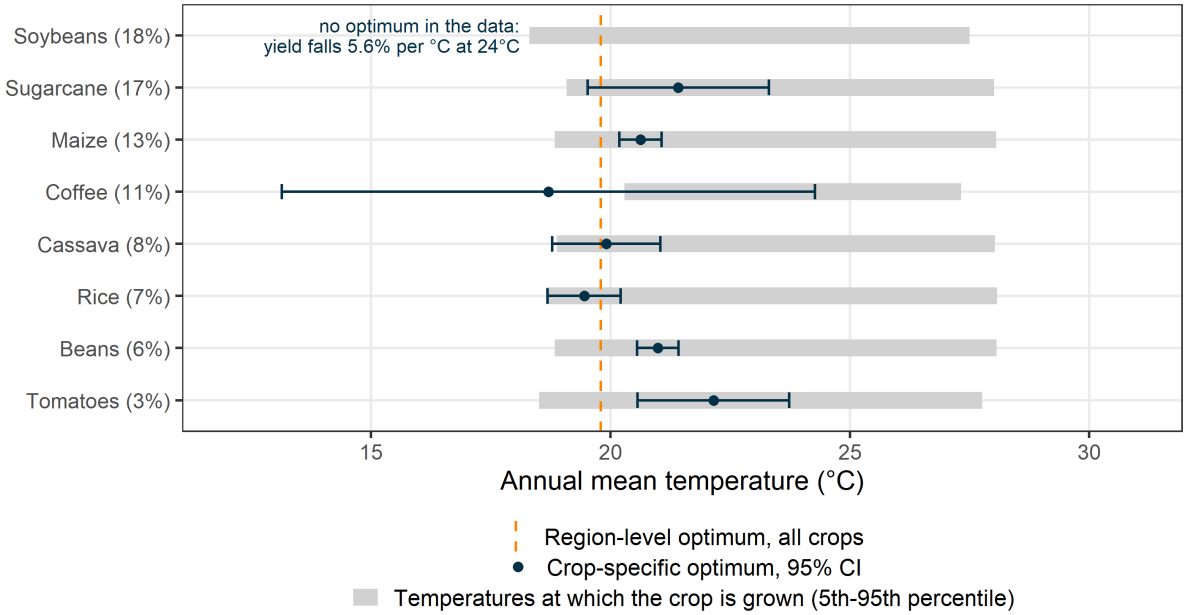
	Curvature $\hat{\beta}_2$		Optimum (°C)		
	estimate	s.e.	estimate	95% CI	Obs.
Microregion level: micro FE + macro-region cubic trends					
(1) Fundamental productivity ($\theta = 2$; Figure III)	−0.0179	(0.0025)	19.8	[19.0, 20.6]	13,254
(2) Measured productivity	−0.0154	(0.0022)	20.4	[19.7, 21.1]	13,254
(3) Crop mix fixed at 1994–98	−0.0218	(0.0020)	20.7	[20.3, 21.1]	13,231
Crop level: crop × microregion and crop × year FE					
(4) Log yield, weighted by 1994–98 revenue share	−0.0253	(0.0019)	20.7	[20.3, 21.1]	157,574
(5) Log yield, unweighted	−0.0168	(0.0013)	20.5	[20.1, 21.0]	168,535

Notes. Quadratic regressions on annual mean temperature, 1994–2017; the 44 crops selected as in Appendix A.3, output per planted hectare at 1994–98 national prices. (1) Log fundamental productivity, as in Figure III. (2) Log measured productivity, constant-price output per planted hectare. (3) Log of a fixed-basket yield index, $\sum_c \omega_{ic}\, y_{cit} / \bar{y}_{ci}$, where y_{cit} is the yield of crop c, $\bar{y}_{ci}$ its mean over the period and ω_{ic} its share of the microregion's 1994–98 crop revenue. (4) Log yield of crop c, weighted by ω_{ic}. (5) Log yield of crop c, unweighted. Rows (1)–(3) include microregion fixed effects and macro-region cubic trends; rows (4)–(5) include crop × microregion and crop × year fixed effects. Standard errors are clustered by microregion.

Individual crops. Figure B.IV estimates the relationship separately for each of the 8 crops with the largest 1994–98 revenue, with microregion and year fixed effects. The figure shows two things for each crop. The grey bar is the range of temperatures at which the crop is actually grown: the 5th to 95th percentile of annual mean temperature across all microregion-years in which it is planted. It is the part of the temperature axis on which that crop's curve is estimated from data. The point and its whiskers are the estimated optimum and its 95% confidence interval. Read together, they show whether an optimum is located by the data or extrapolated from them. For 6 crops the optimum lies inside the grey bar, between 19.4 and 22.1°C, and near the bar's left end: most of the microregion-years in which these crops are grown are warmer than their optimum, so further warming lowers their yields. Coffee's optimum lies below the range in which coffee is grown, so it is an extrapolation; the data say only that coffee yields fall across that range, which is why its confidence interval is wide. Soybeans have no optimum in the data: their yield falls almost linearly across the whole range in which they are grown, by 5.6% per degree at 24°C. The dashed line is the optimum of Figure III. It lies at or near the left end of most bars, which restates the main finding at

the crop level: most places where Brazil's main crops are grown are already warmer than the temperature at which they are most productive.

FIGURE B.IV
Crop-specific temperature optima



Notes. For each of the 8 crops with the largest national revenue in 1994–98 (revenue share in parentheses), log yield (output per planted hectare) is regressed on a quadratic in annual mean temperature with microregion and year fixed effects, 1994–2017. Points and whiskers: the optimum $-\hat{\beta}_1/(2\hat{\beta}_2)$ and its delta-method 95% confidence interval, standard errors clustered by microregion. Grey bars: the 5th to 95th percentile of annual mean temperature across the microregion-years in which the crop is planted, i.e. the temperatures from which its curve is estimated. An optimum outside its bar is an extrapolation. For soybeans the fitted curve has no maximum in this range; the label gives the slope at 24°C. Dashed line: the optimum of Figure III.

C. The Income Shifts behind Fact 2

This appendix gives the data behind Fact 2 (Section 3) and auxiliary results: the estimate shown graphically, the response by initial income, and the comparison of the 2000 and 2010 censuses on its own. As in the main text, a *group* is an education level combined with a five-year age group, and a *cell* is a group within the microregion where its members lived five years before the census.

The income shifts. Table C.I reports the growth of mean real income by education level in the two decades. The income path of a group in equation (3) is $L_{gt} = \ln(\bar{Y}_{gt}/\bar{Y}_{g,1991})$, where $\bar{Y}_{gt}$ is the mean over all adults of group g at census t, transfers included and including those without income, in 2010 reais. The outcome counts M_{jgt} and P_{jgt} are census-weighted, and the sample is the cells with at least 25 census observations in every census.

TABLE C.I
Growth of mean real income by education level

	Less than primary	Primary	Secondary	University
1991–2000	+29	+17	+21	+39
2000–2010	+24	+1	−15	−18

Notes. Percent growth of the national mean real total income of adults aged 20–65 with the education level, transfers included and including those without income, between the censuses, in 2010 reais (IPCA). Education as reported at each census. Analysis samples of the 1991, 2000 and 2010 censuses, weighted by census person weights.

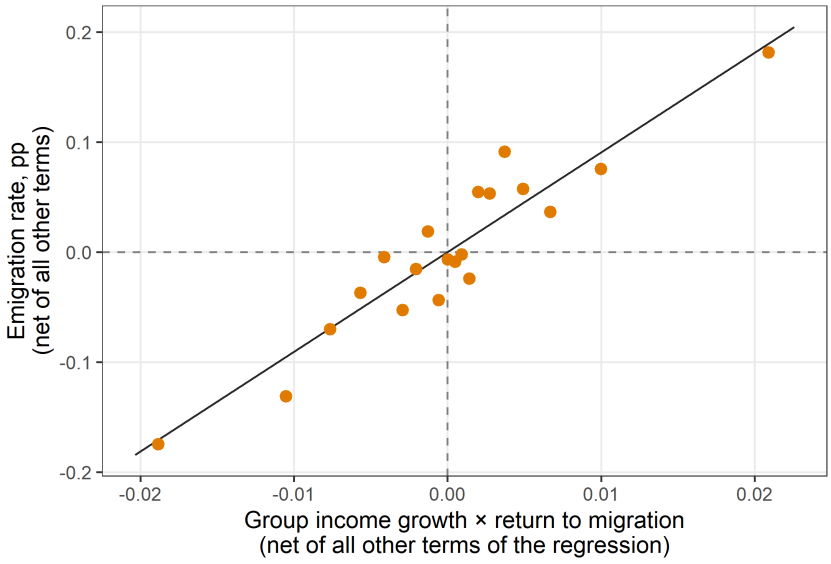
The 1991 census. The 1991 census asked every person who was not born in the municipality, or had not always lived there, for the state and municipality of residence on 1 September 1986, which gives the 1986–91 migration window. I use the person file distributed by IPEA's `censobr` project, harmonised to the 2000 and 2010 censuses: years of study are mapped to the same four education levels, municipalities to the paper's microregions, and the sample restrictions are those of Appendix A.2. Incomes are deflated to 2010 reais with the IPCA, from August 1991, the reference month of the census.

Measurement. Four features of the measurement matter for what ρ can mean. First, $\bar{Y}_{gt}$ is the mean income of whoever is in the group at the census: as schooling expanded, the least educated became a smaller, more selected group, so income growth also reflects who is in the group; group $\times$ census effects absorb this in levels. Second, incomes are measured at each census, at the end of the window over which moves are counted,

and movers' incomes are counted where they arrive. Third, the return is measured in 1991, at the end of the 1986–91 window, so earlier selective emigration shapes it; the cell effects absorb this in levels, and with the return built from 2000 incomes instead, $\rho = 1.05$ (standard error 0.39). The return is persistent: the returns built from 2000 and from 2010 incomes have a correlation of 0.90 across the 2,016 origin $\times$ education cells. Fourth, 1991 incomes were reported under high inflation; an error in the deflator common to all groups rescales the 1990s growth of every group by the same factor and leaves their ranking unchanged.

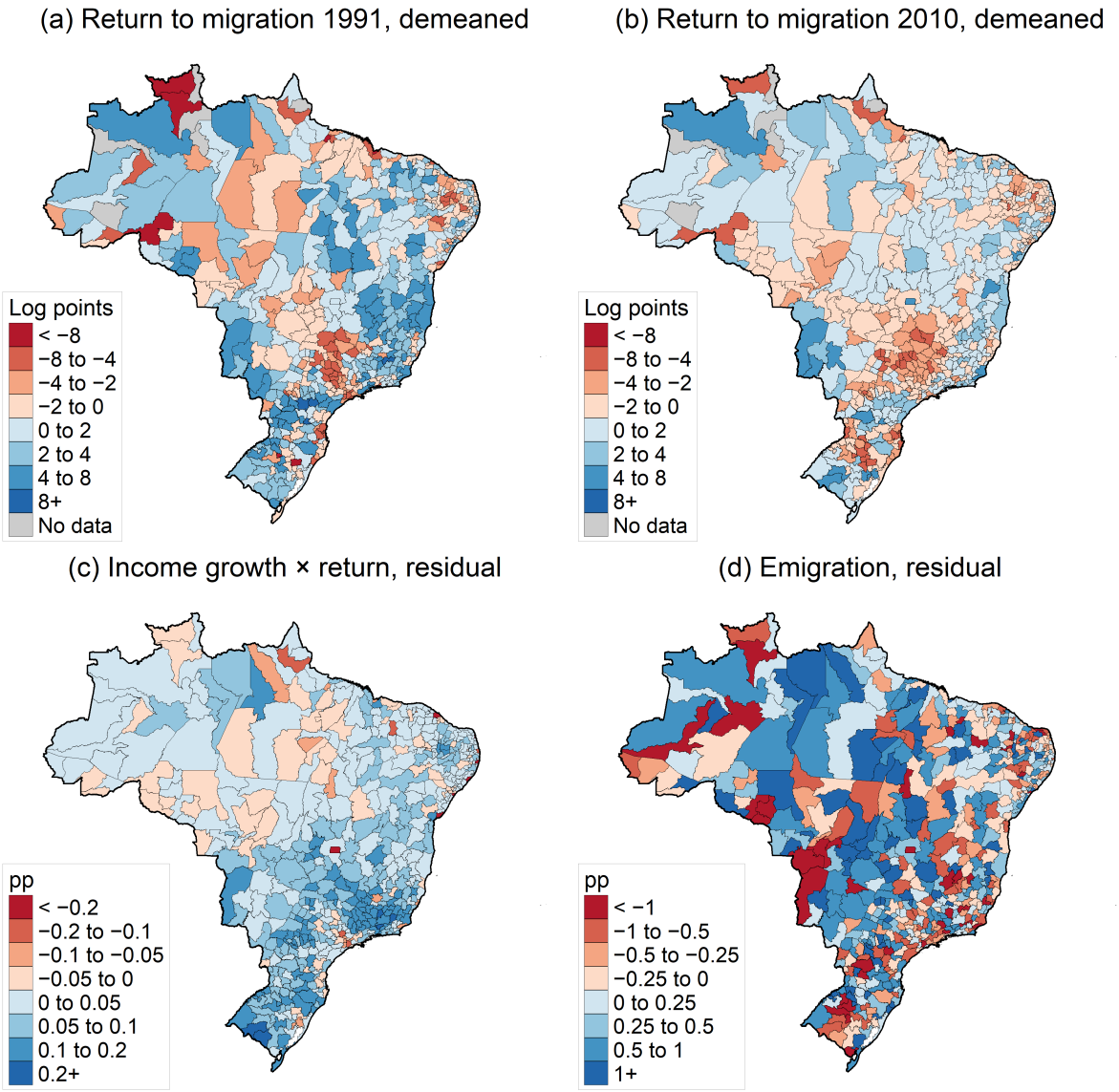
The estimate, graphically. Figure C.I shows the estimate of Table I, column (3), as a binned scatter: the emigration rate against the regressor $L_{gt}R_{jg}$, each net of all other terms of the regression. Its slope is the coefficient, 9.06 (standard error 2.88). The figure uses the OLS version because the Poisson estimate has no such decomposition. Figure C.II maps the same variation for adults with less than primary schooling: the return built from 1991 incomes, the one the regression uses (panel a), the same return built from 2010 incomes (panel b), and the two axes of the binned scatter (panels c and d), whose correlation across the 551 microregions is 0.35.

FIGURE C.I
Income growth and emigration from lagging places: binned scatter, three censuses



Notes. Frisch–Waugh–Lovell on the OLS version of equation (3) (Table I, column 3). The emigration rate (percentage points) and the regressor $L_{gt}R_{jg}$ (both centred, log points $\times$ log points) are each residualised on all other terms of the regression, weighting by population. Cells are sorted into 20 bins of the residualised regressor with equal population weight; each dot is the population-weighted mean of both residuals in a bin. The line passes through the origin with slope equal to the OLS coefficient, 9.06.

FIGURE C.II
The return to migration, the regressor and emigration, net of origin and group effects, adults with less than primary schooling, by microregion



Notes. Cells of the three-census specification (Table I): observed in 1991, 2000 and 2010, with a 1991 return; P_t is the cell's population and $W = P_{1991} + 4P_{2000} + P_{2010}$. Panels (a) and (b): the return to migration R_{jg}, equation (2), built from 1991 incomes (a) and from 2010 incomes (b), net of microregion and education effects in a regression across the four education levels, weighted by the sum of W over the age groups; log points ($\times 100$). Panels (c) and (d): the regressor $L_{gt}R_{jg}$ and the emigration rate, each residualised on all other terms of the OLS version of equation (3) (Table I, column 3): the axes of Figure C.I. Within a cell these residuals follow $(1, -2, 1)$ over the three censuses, up to a small remainder; each cell is summarised by its population-weighted projection on this pattern, shown as the change from 2000 to 2010 minus the change from 1991 to 2000. Panel (c) is multiplied by the slope 9.06, so both panels are in percentage points of emigration. The residuals sum to zero over all groups, not within the least educated, whose income accelerated most in the 2000s, so panel (c) is mostly positive. Age groups are aggregated within a microregion with weights W, so that the age composition is held fixed. Colour bins are symmetric around zero; grey: no estimate (microregions where the least educated are the only education level in the sample, or no 2010 return).

The response by initial income. Under a cash cost, the same proportional income gain lowers the utility cost of the fare more for a poor group than for a rich one: to first order the cost is χ/Y, and a gain g lowers it by $(\chi/Y)\,g/(1+g)$. The response ρ should therefore be strongest for the groups that start poorest. Table C.II reports the Poisson estimate of Table I, column (1), with ρ allowed to differ by initial income. Column (1) repeats the split of Table I, column (2); column (2) interacts ρ with the log of a group's 1991 income; column (3) lets ρ differ by education level. The response is largest, and only precisely estimated, for the poorest groups, and the interaction has the predicted sign; neither the difference between the two sets of groups nor the interaction is statistically significant, and by education level the estimates are not monotone.

TABLE C.II
Income growth and emigration from lagging places, by initial group income

	(1)	(2)	(3)
ρ, groups with low 1991 income	$+0.89^{***}$		
	(0.23)		
ρ, other groups	$+0.54$		
	(0.49)		
ρ		$+0.75^{**}$	
		(0.32)	
$\rho \times$ log 1991 group income		-0.29	
		(0.28)	
ρ, less than primary			$+0.91^{***}$
			(0.25)
ρ, primary			-0.10
			(0.27)
ρ, secondary			$+0.74$
			(0.66)
ρ, university			$+1.24$
			(0.78)
Equality across groups, p-value	0.32		
Group trends in response to return	✓	✓	✓
Cells $\times$ censuses	37,938	37,938	37,938

Notes. Equation (3) by Poisson on movers with population as exposure, three censuses, as in Table I, column (1), with ρ allowed to differ as stated. Low 1991 income: national mean real income in 1991 at or below the population-weighted median across the 36 groups. Log 1991 group income is centred. Standard errors clustered by microregion; the equality test compares the two coefficients in column (1). *** 1%, ** 5%, * 10%.

The 2000–2010 comparison. With the 2000 and 2010 censuses alone, the shock of a group is the growth of its national mean real total income between the censuses, over

all adults of the group, including those without income,

$$\text{Shock}_g \;=\; \bar{Y}_{g,2010} / \bar{Y}_{g,2000} - 1, \tag{32}$$

and the return to migration is built as in equation (2), from 2000 incomes,

$$R_{jg} \;=\; \ln \frac{\sum_{i \neq j} e^{-d_{ji}/d_0}\, \bar{y}_{i,e(g),2000}}{\sum_{i \neq j} e^{-d_{ji}/d_0}} \;-\; \ln \bar{y}_{j,e(g),2000}, \tag{33}$$

using microregions with at least 25 adults of the education level. With the emigration rate s_{jgt} (percent) and the mean real income $\bar{y}_{jgt}$ of the cell's residents with positive income, two regressions compare the 1995–2000 and 2005–10 windows:

$$\ln \bar{y}_{jgt} = \alpha_{jg} + \psi_{jt} + \pi\, \text{Post}_t\, \text{Shock}_g + u_{jgt}, \tag{34}$$

$$s_{jgt} = \alpha_{jg} + \psi_{jt} + \omega_{gt} + b\, \text{Post}_t\, R_{jg} + \rho\, \text{Post}_t\, \text{Shock}_g\, R_{jg} + u_{jgt}, \tag{35}$$

weighted by population, with shock and return centred. The emigration equation is also estimated by Poisson on the number of movers, with the cell's population as exposure. Table C.III reports both. The shift reached local incomes: $\pi = 0.76$. In levels, $\rho = 9.13$ (standard error 3.02); by Poisson, the estimate is positive and not significant. Without the groups with less than primary schooling, ρ in levels is -2.16 (standard error 2.47), so the estimate comes from their income shift.

TABLE C.III
The 2000–2010 income shift: income, and emigration from lagging places

	Log income	Emigration	
		Rate (pp), OLS	Movers, Poisson
	(1)	(2)	(3)
Post × shock	+0.756***		
	(0.013)		
Post × return		+0.007	−0.042
		(0.615)	(0.060)
Post × shock × return		+9.128***	+0.603
		(3.021)	(0.386)
Cell effects	✓	✓	✓
Microregion × census	✓	✓	✓
Group × census		✓	✓
Cells × waves	30,576	30,576	30,542

Notes. Column (1) is equation (34); column (2) is equation (35), weighted by population; column (3) estimates equation (35) by Poisson on the number of movers with the cell's population as exposure, so that its coefficients are proportional effects. Cells with at least 25 census observations in both waves; cell-waves with no movers in either window drop from column (3). Standard errors clustered by microregion, which do not account for the small number of independent shocks. *** 1%, ** 5%, * 10%.

Online Appendix II: Model

D. Derivations for the model

This appendix proves the closed forms and equilibrium statements of Section 4. Equations of the main text are cited, not restated. Static-equilibrium objects carry a time subscript (p_t^i, $w_{A,t}^i$, $\bar{\eta}_t^{ik}$, s_t^i, v_t^i, r_t^i, Y_t^i); primitives that do not change along the paths of the paper (α, θ, ϵ, γ_a, γ_y, γ_h, $\bar{a}$, ψ^j, $d^{j,i}$, B^i, F^{i0}, X^i, $\bar{H}^i$, μ_k, σ, κ^{ik}, $\chi^{j,i}$, $\check{\tau}^{jk,i}$, $\underline{s}$, σ_Y) carry none. Φ and ϕ are the standard normal CDF and density, and $\Lambda(x) \equiv \phi(x)/\Phi(-x)$ is the inverse Mills ratio. Two normalizations of Section 4 are used throughout: the taste shocks ξ_t^i are i.i.d. Type-I extreme value with unit scale and mean zero and enter utility as $\nu\xi_t^i$; staying is costless, $\tau^{jk,j}(y) = 0$ for all $y > 0$. The utility cost of the fare (6) is written $\mathcal{K}\big(\chi^{j,i}h(y)\big)$, with $h(y) \equiv 1/(12y)$ for monthly income y, and $\mathcal{K}(x)$ the loss from paying a fare that is the share x of the year's income. Section 4 writes this loss as $-\ln\max(1-x, \underline{s})$; the quantitative model uses its calibrated form (27),

$$\mathcal{K}(x) \equiv \mathbb{E}_\varepsilon\Big[-\ln \widetilde{\max}\big(1 - x\, e^{-\sigma_Y \varepsilon},\ \underline{s}\big)\Big],$$

with the expectation over $\varepsilon \sim \mathcal{N}(0,1)$. The results below are stated for the calibrated $\mathcal{K}$; for the cost of Section 4 they hold wherever the floor does not bind. The calibrated $\mathcal{K}$ is smooth and strictly increasing, bounded above by $-\ln \underline{s}$, with $\mathcal{K}(0) \leq 0$ (the smooth maximum at $u = 1$ lies slightly above one; $\mathcal{K}(0) \approx 0$ at the estimates), and $\mathcal{K}(x) = e^{\sigma_Y^2/2}x + O(x^2)$ as $x \to 0$. At the estimates $\mathcal{K}$ is convex for fares below about three quarters of the year's income and concave above, as it approaches its bound. The consumption-equivalent variation is derived in Appendix E (equation (58)), and the static equilibrium in proportional changes in Appendix F.

D.1. Agricultural technology and land selection

Fix a region i and a period t. Inside proofs write $z \equiv z_t^i(\omega)$, $z^0 \equiv z^{i0}(\omega)$, $A \equiv A_t^i$, $F^0 \equiv F^{i0}$, $v \equiv v_t^i$, $s \equiv s_t^i$, $p \equiv p_t^i$ and $w_A \equiv w_{A,t}^i$.

Lemma 1 (Fréchet normalization). Under the joint distribution of (z, z^0) in Section 4.3, with $\theta > 1$ and $\kappa_\theta = \Gamma(1 - \frac{1}{\theta})^{-\theta}$: (a) z and z^0 are independent; (b) each is Fréchet with shape θ, with scales $A/\Gamma(1-\frac{1}{\theta})$ and $F^0/\Gamma(1-\frac{1}{\theta})$; (c) $\mathbb{E}[z] = A$ and $\mathbb{E}[z^0] = F^0$.

Proof. (a) The joint CDF is the product $\exp\{-\kappa_\theta(x/A)^{-\theta}\}\exp\{-\kappa_\theta(x^0/F^0)^{-\theta}\}$, and letting $x^0 \to \infty$ (resp. $x \to \infty$) shows that each factor is a marginal. (b) $\kappa_\theta(x/A)^{-\theta} = \big(x/(A\kappa_\theta^{1/\theta})\big)^{-\theta}$ with $\kappa_\theta^{1/\theta} = \Gamma(1-\frac{1}{\theta})^{-1}$. (c) For a Fréchet variable with CDF $e^{-(x/\varsigma)^{-\theta}}$, the substitution $u = (x/\varsigma)^{-\theta}$ gives $\mathbb{E}[X] = \varsigma\int_0^\infty u^{-1/\theta}e^{-u}\,du = \varsigma\,\Gamma(1-\frac{1}{\theta})$, finite because $\theta > 1$; with the scales of (b) the means are A and F^0. □

Lemma 2 (Plot optimum). On an operated plot with productivity z, the profit-maximizing labor input is $L_t^i(\omega) = (\alpha p/w_A)^{\frac{1}{1-\alpha}} z$, output is $(\alpha p/w_A)^{\frac{\alpha}{1-\alpha}} z$, the wage bill is the share α of revenue, and variable profit (revenue net of the wage bill) is $v_t^i z$, the share $1-\alpha$ of revenue, with v_t^i as defined in Section 4.4.

Proof. The problem $\max_{L\geq 0} pL^\alpha z^{1-\alpha} - w_A L$ is strictly concave, and its first-order condition $\alpha p L^{\alpha-1} z^{1-\alpha} = w_A$ gives the labor input. Substituting, output is $(\alpha p/w_A)^{\frac{\alpha}{1-\alpha}} z$, the wage bill $w_A L = \alpha\, p\, (\alpha p/w_A)^{\frac{\alpha}{1-\alpha}} z$ is α times revenue, and the residual is

$$(1-\alpha)\, p\Big(\frac{\alpha p}{w_A}\Big)^{\frac{\alpha}{1-\alpha}} z = (1-\alpha)\,\alpha^{\frac{\alpha}{1-\alpha}} p^{\frac{1}{1-\alpha}} w_A^{-\frac{\alpha}{1-\alpha}} z = v_t^i z.$$

□

Proposition 2 (Cultivated share and measured productivity)**.** *A plot is cultivated iff $vz \geq z^0$. Then (a) the cultivated share is* (18)*; (b) conditional on cultivation, z is Fréchet with shape θ and mean $As^{-1/\theta}$, which is* (19)*, with $\tilde{A}_t^i \geq A_t^i$ and equality iff $s = 1$; (c) $\mathbb{E}\big[z^0\,\mathbf{1}\{\text{cult}\}\big] = F^0\big[1-(1-s)^{1-\frac{1}{\theta}}\big]$.*

Proof. (a) By independence, with the density $f_z(x) = \kappa_\theta \theta A^\theta x^{-\theta-1} e^{-\kappa_\theta A^\theta x^{-\theta}}$,

$$\Pr(vz \geq z^0) = \int_0^\infty F_{z^0}(vx) f_z(x)\, dx = \int_0^\infty \kappa_\theta \theta A^\theta x^{-\theta-1} e^{-Dx^{-\theta}}\, dx = \frac{\kappa_\theta A^\theta}{D},$$

where $D \equiv \kappa_\theta[A^\theta + (F^0/v)^\theta]$, since $D\theta x^{-\theta-1} e^{-Dx^{-\theta}}$ is the derivative of $e^{-Dx^{-\theta}}$. Cancelling κ_θ and multiplying through by v^θ gives (18).

(b) Integrating only up to x, $\Pr(z \leq x, \text{cult}) = s\, e^{-Dx^{-\theta}}$, so the conditional CDF is $e^{-Dx^{-\theta}} = \exp\{-\kappa_\theta (x/(As^{-1/\theta}))^{-\theta}\}$, using $D = \kappa_\theta A^\theta / s$. This is the marginal of z with A replaced by $As^{-1/\theta}$, whose mean is $As^{-1/\theta}$ by Lemma 1(c). Since $s \in (0,1]$ and $\theta > 1$, $s^{-1/\theta} \geq 1$ with equality iff $s = 1$.

(c) Conditioning on $z^0 = u$, $\Pr(z^0 \leq x, \text{cult}) = \int_0^x [1 - F_z(u/v)] f_{z^0}(u)\, du = F_{z^0}(x) - (1-s)e^{-Ex^{-\theta}}$ with $E \equiv \kappa_\theta[(vA)^\theta + (F^0)^\theta]$, by the same antiderivative and $\kappa_\theta (F^0)^\theta / E = 1 - s$. The first term is a Fréchet CDF with mean F^0; the second is $1-s$ times a Fréchet CDF with mean $E^{1/\theta}\Gamma(1-\frac{1}{\theta}) = [(vA)^\theta + (F^0)^\theta]^{1/\theta} = F^0(1-s)^{-1/\theta}$. Hence $\mathbb{E}[z^0 \mathbf{1}\{\text{cult}\}] = F^0 - (1-s)F^0(1-s)^{-1/\theta}$. □

Proposition 3 (Aggregation, operating outlays and land rents)**.** *Plots form an i.i.d. continuum of measure X^i, so aggregates are X^i times plot-level expectations. Let $Z_t^i \equiv [(v_t^i A_t^i)^\theta + (F^{i0})^\theta]^{1/\theta}$.*

(a) Output and agricultural labor demand are (20)*.*

(b) Operating outlays are $\mathcal{F}_t^i = X^i F^{i0}\big[1-(1-s_t^i)^{1-\frac{1}{\theta}}\big] \in (0, X^i F^{i0})$ for $s_t^i \in (0,1)$.

(c) *Variable profit is* $\rho_t^i \equiv (1-\alpha)p_t^i Q_{a,t}^i = X^i Z_t^i s_t^i$, *and land rents are*

$$R_t^i \equiv \rho_t^i - \mathcal{F}_t^i = X^i Z_t^i \left[1 - (1-s_t^i)^{1/\theta}\right] \;\geq\; 0. \tag{36}$$

(d) *The ratio of operating outlays to variable profit depends on the cultivated share and* θ *alone,*

$$\frac{\mathcal{F}_t^i}{\rho_t^i} = \frac{(1-s_t^i)^{1/\theta}\left[1-(1-s_t^i)^{1-\frac{1}{\theta}}\right]}{s_t^i}, \tag{37}$$

so outlays and rents follow from farm revenue and s_t^i, *with no level of* X^i, A_t^i *or* F^{i0}.

(e) *The elasticity of agricultural labor demand with respect to the agricultural wage is*

$$\frac{\partial \ln L_{a,t}^{D,i}}{\partial \ln w_{A,t}^i} = -\,\frac{1+\alpha(\theta-1)(1-s_t^i)}{1-\alpha} \;<\; 0. \tag{38}$$

Proof. (a) By Proposition 2(b), $\mathbb{E}[z\mathbf{1}\{\text{cult}\}] = s \cdot As^{-1/\theta} = As^{1-\frac{1}{\theta}}$; plot output and labor are linear in z (Lemma 2) and idle plots contribute nothing. (b) Proposition 2(c) times X^i; the bounds hold because $(1-s)^{1-\frac{1}{\theta}} \in (0,1)$. (c) Variable profit is $X^i v\,\mathbb{E}[z\mathbf{1}\{\text{cult}\}] = X^i vAs^{1-\frac{1}{\theta}}$, which is $(1-\alpha)pQ_{a,t}^i$ by Lemma 2. By (18), $vA = Zs^{1/\theta}$ and $F^0 = Z(1-s)^{1/\theta}$, so $\rho = X^i Zs$ and $\mathcal{F} = X^i Z(1-s)^{1/\theta}[1-(1-s)^{1-\frac{1}{\theta}}]$; subtracting gives (36). Rents are non-negative plot by plot, since a plot is cultivated only if $vz - z^0 \geq 0$. (d) Divide the two expressions in (c). (e) From (20), $\ln L_{a,t}^{D,i}$ is, up to a constant, $-\frac{1}{1-\alpha}\ln w_A + (1-\frac{1}{\theta})\ln s$; by (18), $\partial \ln s/\partial \ln v = \theta(1-s)$, and $\partial \ln v/\partial \ln w_A = -\alpha/(1-\alpha)$. □

D.2. Occupational choice and the income distribution

Fix a cell (i,k) and a period t, and write $\ell^{ik} \equiv \ln w_N^i + \mu_k$ for the location of log non-agricultural earnings, so that $\ln \breve{y}_N^i(k,\eta) = \ell^{ik} + \eta$ by (10). The calibration treats ℓ^{ik} as a free cell-level location (Section 5.2); nothing below uses its additive form beyond $\hat{w}_N^i = \hat{B}^i$, which shifts every ℓ^{ik} of region i by $\ln \hat{B}^i$ (Appendix F). Write $\tilde{\eta}_t^{ik} \equiv \bar{\eta}_t^{ik}/\sigma$ for the standardized cutoff; inside proofs $\tilde{\eta} \equiv \tilde{\eta}_t^{ik}$.

Lemma 3 (Cutoff, participation, and truncated moments). The worker works outside agriculture iff $\eta \geq \bar{\eta}_t^{ik}$, with the cutoff (11), $\bar{\eta}_t^{ik} = \ln w_{A,t}^i + \kappa^{ik} - \ell^{ik}$. Then

(a) $\pi_{N,t}^{ik} = \Phi(-\tilde{\eta}_t^{ik})$, which is (21);

(b) the mean efficiency of entrants is

$$\mathbb{E}\left[e^{\mu_k+\eta} \mid \eta \geq \bar{\eta}_t^{ik}\right] = e^{\mu_k}\, e^{\sigma^2/2}\, \frac{\Phi\left(\sigma - \bar{\eta}_t^{ik}/\sigma\right)}{\Phi\left(-\bar{\eta}_t^{ik}/\sigma\right)}. \tag{39}$$

which exceeds the unconditional mean $e^{\mu_k+\sigma^2/2}$;

(c) $\mathbb{E}[\eta \mid \eta \geq \bar{\eta}_t^{ik}] = \sigma\Lambda(\tilde{\eta}_t^{ik})$ and $\mathrm{Var}(\eta \mid \eta \geq \bar{\eta}_t^{ik}) = \sigma^2\big[1 + \tilde{\eta}_t^{ik}\Lambda(\tilde{\eta}_t^{ik}) - \Lambda(\tilde{\eta}_t^{ik})^2\big] < \sigma^2$.

Proof. Taking logs of $\check{y}_N^i(k,\eta) \geq e^{\kappa^{ik}} w_{A,t}^i$ gives the cutoff. (a) $\Pr(\eta \geq \bar{\eta}) = 1 - \Phi(\tilde{\eta}) = \Phi(-\tilde{\eta})$. (b) Completing the square, $\eta - \eta^2/(2\sigma^2) = \sigma^2/2 - (\eta - \sigma^2)^2/(2\sigma^2)$, so $\mathbb{E}[e^{\eta}\mathbf{1}\{\eta \geq \bar{\eta}\}] = e^{\sigma^2/2}\Phi(\sigma - \tilde{\eta})$; divide by $\Phi(-\tilde{\eta})$ and multiply by e^{μ_k}. The ratio exceeds $e^{\mu_k+\sigma^2/2}$ because $\Phi(\sigma - \tilde{\eta}) > \Phi(-\tilde{\eta})$ for $\sigma > 0$. (c) These are the mean and variance of a normal truncated from below at $\tilde{\eta}$ standard deviations; the bracket is below one because $\Lambda(\tilde{\eta}) > \tilde{\eta}$. □

Part (c) is what the calibration uses to recover σ: in each region–skill cell the observed standard deviation of non-agricultural log earnings is divided by the square root of the bracket at $\tilde{\eta}^{ik} = -\Phi^{-1}(\pi_N^{ik})$, a quantity that does not involve σ, and the cell values are averaged with weights equal to non-agricultural employment (Section 5.2).

Proposition 4 (The within-cell income distribution)**.** *Market income* (12) *has the CDF*

$$\check{F}_t^{ik}(\check{y}) = \Phi(\tilde{\eta}_t^{ik})\,\mathbf{1}\{\check{y} \geq w_{A,t}^i\} + \Big[\Phi\Big(\frac{\ln\check{y} - \ell^{ik}}{\sigma}\Big) - \Phi(\tilde{\eta}_t^{ik})\Big]^+, \tag{40}$$

with $[x]^+ \equiv \max\{x, 0\}$*: an atom of mass* $1 - \pi_{N,t}^{ik} = \Phi(\tilde{\eta}_t^{ik})$ *at* $w_{A,t}^i$ *and a lognormal tail, of mass* $\pi_{N,t}^{ik}$*, that starts at* $w_{A,t}^i e^{\kappa^{ik}}$*. Hence:*

(a) the lowest income is $w_{A,t}^i \min\{1, e^{\kappa^{ik}}\}$*; the support is* $[w_{A,t}^i e^{\kappa^{ik}}, \infty)$ *if* $\kappa^{ik} \leq 0$ *and* $\{w_{A,t}^i\} \cup [w_{A,t}^i e^{\kappa^{ik}}, \infty)$ *if* $\kappa^{ik} > 0$*;*

(b) the share of non-agricultural workers who earn less than the agricultural wage is

$$\frac{\big[\Phi\big(\tilde{\eta}_t^{ik} - \kappa^{ik}/\sigma\big) - \Phi(\tilde{\eta}_t^{ik})\big]^+}{\Phi(-\tilde{\eta}_t^{ik})}, \tag{41}$$

positive iff $\kappa^{ik} < 0$*; with* $\kappa^{ik} > 0$ *no income falls in* $(w_{A,t}^i, w_{A,t}^i e^{\kappa^{ik}})$*; with* $\kappa^{ik} = 0$ *market income is* $\max\{w_{A,t}^i, \check{y}_N^i(k,\eta)\}$*;*

(c) expected log market income is, in closed form,

$$\mathbb{E}\big[\ln\check{y}\big] = \big(1 - \pi_{N,t}^{ik}\big)\ln w_{A,t}^i + \pi_{N,t}^{ik}\big[\ell^{ik} + \sigma\Lambda(\tilde{\eta}_t^{ik})\big]; \tag{42}$$

(d) the fiscal wedge (13) *maps market into disposable income through* $g_t^i(\check{y}) \equiv \check{y} - \mathcal{T}_t^i(\check{y}) + S_t^i$*; if* g_t^i *is strictly increasing, the distribution of disposable income is* $F_t^{ik}(y) = \check{F}_t^{ik}\big((g_t^i)^{-1}(y)\big)$*, with the atom at* $g_t^i(w_{A,t}^i)$ *and the tail transported by* g_t^i*.*

Under the proportional tax of Section 7, $g(\breve{y}) = (1 - t_{\text{tax}})\breve{y}$, the tax adds $\ln(1 - t_{\text{tax}})$ to (42) and leaves $\pi^{ik}_{N,t}$ unchanged.

Proof. A worker earns at most $\breve{y}$ if it farms and $w^i_{A,t} \leq \breve{y}$, an event of probability $\Phi(\tilde{\eta})\mathbf{1}\{\breve{y} \geq w_A\}$, or if it works outside agriculture with $\ell + \eta \leq \ln \breve{y}$, the event $\bar{\eta} \leq \eta \leq \ln \breve{y} - \ell$, of probability $[\Phi((\ln \breve{y} - \ell)/\sigma) - \Phi(\tilde{\eta})]^+$. Summing gives (40). The lowest non-agricultural income is $e^{\ell+\bar{\eta}} = w_A e^{\kappa}$, which gives (a). (b) Evaluate the tail term at $\breve{y} \uparrow w_A$: $(\ln w_A - \ell)/\sigma = \tilde{\eta} - \kappa/\sigma$, and divide by the tail mass $\Phi(-\tilde{\eta})$; the bracket is positive iff $\tilde{\eta} - \kappa/\sigma > \tilde{\eta}$. With $\kappa = 0$ the tail starts at w_A and the two pieces do not overlap. (c) The atom contributes $(1 - \pi_N) \ln w_A$ and the tail $\pi_N(\ell + \mathbb{E}[\eta \mid \eta \geq \bar{\eta}])$; use Lemma 3(c). (d) A strictly increasing map transports the law point by point; under the proportional tax $\ln y = \ln(1 - t_{\text{tax}}) + \ln \breve{y}$, and occupational choice depends on market income only. □

With $\kappa^{ik} < 0$, as calibrated, efficiency rank and income rank differ: the lowest-ranked non-agricultural workers earn less than the farmers, who rank below them in efficiency. This is why incidence in Appendix E is computed by efficiency rank. Equation (42) is the expected log income that enters the change in the cell's mean period utility in Appendix E.

Corollary 1 (Inversion of the occupational block)**.** *Given σ, the participation share $\pi^{ik}_N \in (0,1)$, the mass L^{ik}, the agricultural wage w^i_A and the non-agricultural wage bill $W^{ik}_N > 0$ of a cell, the location and the wedge are, in closed form,*

$$\begin{aligned} \tilde{\eta}^{ik} &= -\Phi^{-1}(\pi^{ik}_N), \qquad \kappa^{ik} = \ell^{ik} + \sigma\tilde{\eta}^{ik} - \ln w^i_A, \\ \ell^{ik} &= \ln \frac{W^{ik}_N}{L^{ik}\pi^{ik}_N} - \frac{\sigma^2}{2} - \ln \Phi(\sigma - \tilde{\eta}^{ik}) + \ln \Phi(-\tilde{\eta}^{ik}). \end{aligned} \tag{43}$$

Proof. The cutoff inverts Lemma 3(a). By Lemma 3(b), mean non-agricultural earnings are $e^{\ell+\sigma^2/2}\,\Phi(\sigma - \tilde{\eta})/\Phi(-\tilde{\eta})$; equating them to $W_N/(L\pi_N)$ gives ℓ. The wedge solves $\sigma\tilde{\eta} = \ln w_A + \kappa - \ell$. □

Proposition 5 (Labor supplies and the agricultural labor market)**.** *By the law of large numbers within cells, the labor supplies are (22). Total labor income of the region's workers is $\sum_k L^{ik}_t \mathbb{E}[\breve{y}] = w^i_{A,t} L^{S,i}_{a,t} + B^i N^i_t$. Given $p^i_t > 0$ and $\sum_k L^{ik}_t > 0$, the agricultural labor market (26) has exactly one clearing wage: the supply elasticity*

$$\frac{\partial \ln L^{S,i}_{a,t}}{\partial \ln w^i_{A,t}} = \frac{\sum_k L^{ik}_t \phi(\tilde{\eta}^{ik}_t)/\sigma}{L^{S,i}_{a,t}} \;>\; 0 \tag{44}$$

is positive and the demand elasticity (38) negative.

Proof. Each farmer supplies one efficiency unit, so $L^{S,i}_{a,t} = \sum_k L^{ik}_t (1 - \pi^{ik}_{N,t})$, and $N^i_t = \sum_k L^{ik}_t \mathbb{E}[e^{\mu_k + \eta} \mathbf{1}\{\eta \geq \tilde{\eta}^{ik}_t\}]$. Labor income is w_A per farmer plus $w^i_N e^{\mu_k + \eta}$ per non-agricultural worker, and $w^i_N = B^i$. The wedge κ^{ik} is not a payment, so it does not enter income. For the supply elasticity, $L^{S,i}_{a,t} = \sum_k L^{ik}_t \Phi(\tilde{\eta}^{ik}_t)$ and $\partial \tilde{\eta}^{ik}_t / \partial \ln w^i_{A,t} = 1/\sigma$. Excess demand $\ln L^{D,i}_{a,t} - \ln L^{S,i}_{a,t}$ is therefore continuous and strictly decreasing in $\ln w^i_{A,t}$; it tends to $+\infty$ as $w^i_{A,t} \to 0$, where demand diverges and supply vanishes, and to $-\infty$ as $w^i_{A,t} \to \infty$, where demand vanishes and supply tends to $\sum_k L^{ik}_t$. □

D.3. Demand, price indices, and market clearing

The regional accounts require an assumption on who owns land and how the fiscal system enters regional spending. Section 4 states local ownership by immobile landowners who are not among the workers, and Section 5.2 that the population whose location choice is modelled is workers. How the fiscal system enters regional spending, part (ii) below, is a modelling choice of the quantitative exercises.

Assumption 2 (Ownership and the regional accounts)**.** *(i) Farmland X^i and residential land $\bar{H}^i$ are owned by landowners resident in region i. They are immobile, are not among the workers whose location choice is modelled, receive the agricultural rents R^i_t and the residential rents $r^i_t \bar{H}^i$, and spend them in region i with the housing share γ_h and the crop share $\bar{\vartheta}^i_t$ of the region's workers (46). Regional expenditure is then $Y^i_t = w^i_{A,t} L^{S,i}_{a,t} + B^i N^i_t + R^i_t + r^i_t \bar{H}^i$. (ii) Regional expenditure is evaluated at market income. This is exact when $\mathcal{T} = S = 0$ or when the fiscal system (13) is balanced region by region in every period. For the subsidy of Section 7, whose tax balances the budget in present value and whose payments are made per move, it is a modelling choice: neither the tax nor the payments enter regional expenditure (Remark 1). Accordingly, the crop share (46) is evaluated at market income.*

Lemma 4 (PIGLOG demands). A resident of region i with income $y > 0$ and indirect utility $U^i_t(y)$ spends the share $\vartheta^i_t(y)$ of (23) on the agricultural composite, γ_h on housing and $1 - \gamma_h - \vartheta^i_t(y)$ on the numeraire. Its marginal utility of income is $1/(\mathcal{B}^i_t y)$.

Proof. Two-stage budgeting applies because utility depends on the varieties only through $C^{i;a}_t$, available at unit cost $P^i_{a,t}$ (Lemma 5). Write $U = (\ln y - \ln \bar{a} - \gamma_a \ln P^i_{a,t} - \gamma_h \ln r^i_t)/\mathcal{B}$ with $\ln \mathcal{B} = \gamma_y \ln P^i_{a,t}$. Then $\partial U / \partial \ln y = 1/\mathcal{B}$, $\partial U / \partial \ln r^i_t = -\gamma_h/\mathcal{B}$ and $\partial U / \partial \ln P^i_{a,t} = -\left[\gamma_a + \gamma_y (\ln y - \ln \bar{a} - \ln P^i_t)\right]/\mathcal{B}$. Roy's identity, the budget share of good g being $-(\partial U / \partial \ln p_g)/(\partial U / \partial \ln y)$, gives the stated shares, and the numeraire takes the rest by the budget constraint. The function is a valid indirect utility where the shares lie in $[0, 1]$ [Deaton and Muellbauer, 1980]; Appendix G.2 reports how the crop share is bounded in the computations where (23) would leave that range. □

Lemma 5 (CES demands and bilateral expenditure). The unit cost of the agricultural composite is the first part of (15), and region i's expenditure on origin j's variety is

$$X_t^{j,i} = \pi_t^{j,i}\,\bar{\vartheta}_t^i Y_t^i, \qquad \pi_t^{j,i} \equiv \frac{\psi^j (p_t^j d^{j,i})^{1-\epsilon}}{(P_{a,t}^i)^{1-\epsilon}}, \qquad \sum_j \pi_t^{j,i} = 1. \tag{45}$$

Crop spending is not linear in income, so the region's crop share

$$\bar{\vartheta}_t^i \;=\; \frac{\sum_k L_t^{ik} \int \vartheta_t^i(y)\, y\, dF_t^{ik}(y)}{\sum_k L_t^{ik} \int y\, dF_t^{ik}(y)} \tag{46}$$

depends on its whole income distribution; given that share, aggregation over workers and landowners is exact by Assumption 2. Market clearing for each variety is (25).

Proof. Minimizing $\sum_j p_t^j d^{j,i} c^j$ subject to the CES aggregate being at least $\bar{C}$ gives $c^j = \psi^j (p_t^j d^{j,i}/\lambda)^{-\epsilon}\bar{C}$; the constraint then implies $\lambda^{1-\epsilon} = \sum_j \psi^j (p_t^j d^{j,i})^{1-\epsilon}$, so the multiplier λ is $P_{a,t}^i$ and minimized expenditure is $P_{a,t}^i \bar{C}$. Spending on variety j is $\pi_t^{j,i}$ times spending on the composite, which is $\bar{\vartheta}_t^i Y_t^i$ summed over residents (Lemma 4 and Assumption 2). Iceberg costs destroy quantity, not value: multiplying $Q_{a,t}^j = \sum_i d^{j,i} c_t^{j,i;a}$ by p_t^j gives (25). □

Proposition 6 (Walras' law and regional balance)**.** *Under Assumption 2, suppose that the agricultural labor market (26) and every goods market (25) clear. Then:*

(i) housing clears locally at the rent (24), and residential rents $r_t^i \bar{H}^i = \gamma_h Y_t^i$ are a within-region transfer;

(ii) tradable income satisfies

$$(1-\gamma_h) Y_t^i = p_t^i Q_{a,t}^i + B^i N_t^i - \mathcal{F}_t^i; \tag{47}$$

(iii) national agricultural revenue equals national crop spending, $\sum_j p_t^j Q_{a,t}^j = \sum_i \bar{\vartheta}_t^i Y_t^i$;

(iv) the numeraire market clears inclusive of operating outlays,

$$\sum_{i\in\mathcal{S}} B^i N_t^i \;=\; \sum_{i\in\mathcal{S}} \Big[\big(1-\gamma_h-\bar{\vartheta}_t^i\big) Y_t^i + \mathcal{F}_t^i \Big]; \tag{48}$$

(v) each region's trade in the traded goods is balanced: the value of its tradable output equals its tradable absorption, $p_t^i Q_{a,t}^i + B^i N_t^i = (1-\gamma_h) Y_t^i + \mathcal{F}_t^i$.

Proof. (i) Housing demand $\gamma_h Y_t^i / r_t^i$ (Lemma 4) equals the fixed endowment $\bar{H}^i$. (ii) Substitute $R_t^i = p_t^i Q_{a,t}^i - w_{A,t}^i L_{a,t}^{D,i} - \mathcal{F}_t^i$ (Proposition 3) and (i) into Y_t^i, and cancel the wage bill using $L_{a,t}^{S,i} = L_{a,t}^{D,i}$. (iii) Sum (25) over j and use $\sum_j \pi_t^{j,i} = 1$. (iv) Sum (ii) over i

and substitute (iii): the residents' numeraire demand is $(1-\gamma_h-\bar{\vartheta}_t^i)Y_t^i$. (v) Regional tradable absorption is $(1-\gamma_h)Y_t^i$ by residents plus $\mathcal{F}_t^i$ by farms; the difference from tradable output is zero by (ii). □

Part (iii) is why, in a closed economy, the level of crop demand is not a free parameter: given the slope γ_y, the calibration sets γ_a so that national crop spending equals national crop revenue (Section 5.2). Part (iv) is the redundant equation: the solver imposes (25) and (26) and uses the residual of (48) as a diagnostic (Appendix E). Residents' consumption alone falls short of non-agricultural output by $\sum_i \mathcal{F}_t^i > 0$.

Remark 1 (The fare and the subsidy in the static accounts). The fare is paid at the origin in consumption goods, so a mover's total goods demand, its utility-relevant consumption plus the resources the move absorbs, equals its income, and regional expenditure does not depend on who moves (Section 4). Where the fare exceeds the share $1-\underline{s}$ of the year's resources, which the spread σ_Y lets happen at any income, the consumption the floor keeps is financed outside the model's accounts. The subsidy of Section 7 and its tax are treated in the same spirit, as a modelling choice: the tax acts on disposable income only, through consumption and the burden of the fare (Proposition 4(d)), and neither the tax revenue nor the payments enter Y_t^i, which is evaluated at market income. The government balances its budget in present value, equation (60), not in every period, so the static accounts are not adjusted for the period-by-period surplus or deficit of the program, nor for its transfers across regions.

D.4. Dynamic choice: Emax, choice probabilities, and the law of motion

Fix (j,k,y,t) and write $\mathcal{V}_i \equiv \beta \overline{V}_{t+1}^{i,k} - \tau^{jk,i}(y)$. Let $\mathrm{EV}(\mu_\xi, \varsigma)$ be the Type-I extreme value law with CDF $\exp[-e^{-(x-\mu_\xi)/\varsigma}]$, and γ_E Euler's constant.

Lemma 6 (Gumbel preliminaries). (a) If $\xi \sim \mathrm{EV}(\mu_\xi, 1)$, then $a+\nu\xi \sim \mathrm{EV}(a+\nu\mu_\xi, \nu)$. (b) The mean of $\mathrm{EV}(\mu_\xi, \varsigma)$ is $\mu_\xi + \varsigma\gamma_E$, so a unit-scale shock has mean zero iff $\mu_\xi = -\gamma_E$. (c) If $X_i \sim \mathrm{EV}(\mu_i, \varsigma)$ are independent, then $\max_i X_i \sim \mathrm{EV}\big(\varsigma \ln \sum_i e^{\mu_i/\varsigma}, \varsigma\big)$.

Proof. (a) $\Pr(a+\nu\xi \le x) = \exp[-e^{-((x-a)/\nu-\mu_\xi)}]$. (b) With $u = e^{-(x-\mu_\xi)/\varsigma}$, $\mathbb{E}[X] = \int_0^\infty (\mu_\xi - \varsigma \ln u) e^{-u} du = \mu_\xi - \varsigma\Gamma'(1) = \mu_\xi + \varsigma\gamma_E$. (c) The product of the CDFs is $\exp[-e^{-x/\varsigma}\sum_i e^{\mu_i/\varsigma}]$. □

Proposition 7 (The Emax formula)**.** *With unit-scale shocks of location μ_ξ entering as $\nu\xi_t^i$, $\mathbb{E}[\max_i\{\mathcal{V}_i + \nu\xi_t^i\}] = \nu \ln \sum_i e^{\mathcal{V}_i/\nu} + \nu(\mu_\xi + \gamma_E)$. Under the mean-zero normalization the second term vanishes and, adding the destination-independent $U(C_t^j(y))$, (8) holds exactly.*

Proof. By Lemma 6(a) and (c) the maximum is $\mathrm{EV}(\nu\mu_\xi + \nu\ln\sum_i e^{\mathcal{V}_i/\nu}, \nu)$; take its mean by (b). □

Proposition 8 (Logit choice probabilities)**.** *The probability that i attains the maximum is* (16)*, whatever the location of the shocks. The probabilities sum to one, lie in* $(0,1)$ *for every* $y>0$ *because every* $\tau^{jk,l}(y)$ *is finite there, and* $\partial[\nu \ln \sum_l e^{\mathcal{V}_l/\nu}]/\partial \mathcal{V}_i = m_t^{jk,i}(y)$*.*

Proof. Conditioning on $\xi_t^i = x$ and using independence, with $D_i \equiv \sum_l e^{(\mathcal{V}_l - \mathcal{V}_i)/\nu}$ and the substitution $u = e^{-(x-\mu_\xi)}$,

$$m_t^{jk,i}(y) = \int_{-\infty}^{\infty} e^{-(x-\mu_\xi)} \exp\left[-e^{-(x-\mu_\xi)} D_i\right] dx = \int_0^{\infty} e^{-uD_i}\, du = \frac{1}{D_i},$$

free of μ_ξ. The derivative follows by differentiating the log-sum-exp. □

Since staying is costless, the stay probability is $m_t^{jk,j}(y) = e^{\beta \overline{V}_{t+1}^{j,k}/\nu} / \sum_l e^{\mathcal{V}_l/\nu}$, and (8) can be written through it,

$$V_t^{j,k}(y) = U_t^j(y) + \beta\, \overline{V}_{t+1}^{j,k} - \nu \ln m_t^{jk,j}(y), \tag{49}$$

the pointwise value from which the value step of Appendix E starts. Because $m_t^{jk,j}(y) \leq 1$, (49) gives the lower bound $V_t^{j,k}(y) \geq U_t^j(y) + \beta \overline{V}_{t+1}^{j,k}$ of Section 4. With $\check{\tau}^{jk,j} = \chi^{j,j} = 0$, the probabilities factor into the income-invariant intercepts and income profiles of equation (53).

Proposition 9 (Choice probabilities and income)**.** *Fix* (j,k,t)*. With* (5)*,* (7) *and* (6)*, and* $\chi^{j,i} > 0$ *for every* $i \neq j$*:*

(a) the log-odds of a move to i relative to staying are

$$\ln \frac{m_t^{jk,i}(y)}{m_t^{jk,j}(y)} = \frac{\beta\left(\overline{V}_{t+1}^{i,k} - \overline{V}_{t+1}^{j,k}\right) - \check{\tau}^{jk,i}}{\nu} - \frac{\mathcal{K}\left(\chi^{j,i} h(y)\right)}{\nu}, \qquad \frac{\partial}{\partial y} \ln \frac{m_t^{jk,i}(y)}{m_t^{jk,j}(y)} = \frac{\mathcal{K}'\left(\chi^{j,i} h(y)\right) \chi^{j,i}}{12\, \nu\, y^2} > 0, \tag{50}$$

so flows identify only cost differences relative to staying, and a lower income lowers the probability of every move relative to staying;

(b) the probability of leaving, $1 - m_t^{jk,j}(y)$, is strictly increasing in y;

(c) conditional on leaving, the expected log distance of the move rises with income wherever $\mathcal{K}'(x) + x\mathcal{K}''(x) \geq 0$ at $x = \chi^{j,i} h(y)$ for the fares from j (at the estimates, fares below about five sixths of the year's income):

$$\frac{\partial}{\partial y} \mathbb{E}\left[\ln \text{dist}^{j,i} \mid i \neq j,\, y\right] = \frac{c}{12\, \nu\, y^2} \operatorname{Cov}\left(\ln \text{dist}^{j,i},\, \mathcal{K}'\left(\chi^{j,i} h(y)\right) \ln \text{dist}^{j,i} \mid i \neq j,\, y\right) \geq 0, \tag{51}$$

where moments are taken over destinations with the probabilities $m_t^{jk,i}(y)/(1 - m_t^{jk,j}(y))$, $i \neq j$.

With $c = 0$ all three derivatives vanish.

Proof. (a) Take logs of (16) for i and j; the denominator cancels, and $\tau^{jk,j} = 0$. Differentiate with $h'(y) = -1/(12y^2)$; $\mathcal{K}' > 0$. (b) $1 - m_t^{jk,j}(y) = 1 - \left[1 + \sum_{i \neq j} e^{o_i(y)}\right]^{-1}$, where the log-odds $o_i(y)$ of (a) are all strictly increasing. (c) Conditional on leaving, the destination probabilities are proportional to $\exp[(\beta \overline{V}_{t+1}^{i,k} - \check{\tau}^{jk,i})/\nu - \mathcal{K}(\chi^{j,i} h(y))/\nu]$. The derivative in y of the log of each normalized probability is $g_i - \mathbb{E}[g \mid i \neq j]$, with $g_i \equiv \mathcal{K}'(\chi^{j,i} h(y))\, c \ln \text{dist}^{j,i}/(12\nu y^2)$, so $\partial\, \mathbb{E}[\ln \text{dist} \mid i \neq j]/\partial y = \text{Cov}(\ln \text{dist}, g \mid i \neq j)$. Where $\mathcal{K}'(x) + x\mathcal{K}''(x) \geq 0$, $\mathcal{K}'(\chi^{j,i} h(y)) \ln \text{dist}^{j,i}$ is nondecreasing in $\ln \text{dist}^{j,i}$, since its derivative is $\mathcal{K}'(x) + x\mathcal{K}''(x)$, and the covariance of a variable with a nondecreasing function of it is non-negative. □

Proposition 9 is the model's statement of Fact 1 and its distance counterpart: within an origin and skill group, poorer workers leave less and, when the fares they face are below about five sixths of a year's income, go less far; for the poorest, whose cost is near its bound $-\ln \underline{s}$, the cost varies little with distance and the sign of the distance gradient is not determined. The intercepts are constant in y because $\check{\tau}^{jk,i}$ does not depend on income and $\overline{V}_{t+1}^{i,k}$ integrates over next period's income distribution, which does not condition on current income.

Aggregation to the estimating equation. Write δ^{ikt} for the value of region i in the location choice of wave t: in the timing of (16) it is $\beta \overline{V}_{t+1}^{i,k}/\nu$, the value of entering i at the end of the wave's window, which Section 5.3 writes $\beta \overline{V}^{ikt}/\nu$. Let $G^{jkit} \equiv \mathbb{E}_{F_t^{jk}}[\mathcal{K}(\chi^{j,i} h(y))]$, the group's mean utility cost of the fare to i of Section 5.3. The log choice probability, $\ln m_t^{jk,i}(y) = \delta^{ikt} - \check{\tau}^{jk,i}/\nu - \mathcal{K}(\chi^{j,i} h(y))/\nu - \ln \sum_l e^{\mathcal{V}_l/\nu}$, and $m_t^{jk,i}$ is a smooth function of the costs $\mathcal{K}(\chi^{j,l} h(y))$ of the destinations l. A second-order expansion around their group means G^{jklt} gives $\mathbb{E}_{F_t^{jk}}[m_t^{jk,i}] = m_t^{jk,i}\big|_{\mathcal{K}=G}\left[1 + O\left(\max_l \text{Var}_{F_t^{jk}} \mathcal{K}(\chi^{j,l} h(y))\right)\right]$, since the first-order term has mean zero. At the group means the log-sum depends on (j,k,t) only and is absorbed, with the cell's population, by the origin effect μ^{jkt}; the cash term is G^{jkit}/ν. This is (28), with an approximation error of second order in the within-cell dispersion of the costs.

Remark 2 (No representative worker). With $c > 0$, $m_t^{jk,i}(y)$ depends on y, and both the flows in (17) and the Emax term in (9) are nonlinear functionals of the whole distribution F_t^{jk}: no single income reproduces both. The model remains tractable because F_t^{jk} is the two-piece distribution of Proposition 4, an atom and a truncated lognormal tail indexed by current wages, so each cell requires one atom evaluation and one one-dimensional quadrature (Appendix E).

D.5. Properties of the migration-cost function

Fix a pair (j,i), $i \neq j$, and a skill group. By (5), (7) and (6),

$$\tau^{jk,i}(y) = \check{\tau}^{jk,i} + \mathcal{K}\Big(\frac{c \ln \mathrm{dist}^{j,i}}{12\,y}\Big). \tag{52}$$

Lemma 7 (Strict concavity and the welfare gap). Let u be strictly concave on $(0,\infty)$ and $\chi > 0$. Then $u(Y) - u(Y-\chi)$ is strictly decreasing in Y on (χ,∞). For affine u it is constant.

Proof. Fix $\chi < Y_1 < Y_2$, let $a \equiv Y_1 - \chi < b \equiv Y_2$ and $\lambda \equiv \chi/(b-a) \in (0,1)$. Then $Y_1 = (1-\lambda)a + \lambda b$ and $Y_2 - \chi = \lambda a + (1-\lambda)b$, so strict concavity gives $u(Y_1) > (1-\lambda)u(a) + \lambda u(b)$ and $u(Y_2-\chi) > \lambda u(a) + (1-\lambda)u(b)$. Adding, $u(Y_1) - u(Y_1-\chi) > u(Y_2) - u(Y_2-\chi)$. For $u(Y) = a_0 + a_1 Y$ the gap is $a_1\chi$. □

Lemma 8 (The exact logarithmic loss). If the fare is paid in full out of annual income $Y = 12y$, without saving or credit, the utility loss at $\mathcal{B}_t^j = 1$ is $-\ln(1-\chi/Y)$ for $Y > \chi$ and $+\infty$ for $Y \leq \chi$. For $x \equiv \chi/Y \in (0,1)$, $-\ln(1-x) = x\,\lambda^*(x)$ with $\lambda^*(x) \equiv \sum_{n\geq 0} x^n/(n+1)$, strictly increasing from $\lambda^*(0^+) = 1$ and diverging as $x \uparrow 1$. The model's cost (6) is this loss, kept finite by the floor $\underline{s}$ and averaged over the spread σ_Y of the year's resources.

Proof. Four of the period's five years are consumed out of Y and one out of $Y-\chi$, so by (14) the period utility falls by $[\ln Y - \ln(Y-\chi)]/\mathcal{B}_t^j = -\ln(1-\chi/Y)/\mathcal{B}_t^j$; the price terms cancel. For $Y \leq \chi$ consumption after the fare is not positive. The series $-\ln(1-x) = \sum_{n\geq 1} x^n/n$ converges for $x \in [0,1)$; dividing by x gives a power series with positive coefficients. □

The floor $\underline{s}$ is what keeps the likelihood of the estimation of Section 5.3 well defined: without it, the loss would be infinite on a set of draws of ε of positive probability, and every observed move would have probability zero.

Lemma 9 (The utility cost in income). For $c > 0$:

(a) $\partial\tau^{jk,i}(y)/\partial y = -\mathcal{K}'(x)\,\chi^{j,i}/(12y^2) < 0$, with $x \equiv \chi^{j,i}h(y)$; $\tau^{jk,i}(y)$ falls to $\check{\tau}^{jk,i} + \mathcal{K}(0)$, with $\mathcal{K}(0) \leq 0$ negligible, as $y \to \infty$, is finite at every $y > 0$, and is bounded above by $\check{\tau}^{jk,i} - \ln \underline{s}$. Where $\mathcal{K}$ is convex on $[0,2x]$, halving income at least doubles the utility cost of the fare, $\mathcal{K}(2x) \geq 2\mathcal{K}(x)$ (at the estimates the doubling holds for $x \leq 0.74$); above, it raises it by less, as the cost approaches $-\ln \underline{s}$. With $c = 0$, $\tau^{jk,i}$ does not depend on income.

(b) The cross-derivative of $\tau^{jk,i}(y)$ in $\ln \mathrm{dist}^{j,i}$ and y is $-c\,[\mathcal{K}'(x) + x\mathcal{K}''(x)]/(12y^2)$: the distance gradient of the cost is steeper at lower income wherever $\mathcal{K}'(x) +$

$x\mathcal{K}''(x) > 0$ (at the estimates, x below about five sixths), in particular where $\mathcal{K}$ is convex.

(c) The cost depends on c and the month base n only through c/n: charging the fare against n months of income is observationally equivalent to a fare larger by $12/n$ charged against twelve months, so the data identify the fare for a given month base. Because $\mathcal{K}$ is not linear, c/n and ν enter (16) separately, not only through $c/(n\nu)$: given ν, the flows identify the fare in reais.

Proof. (a) and (b) follow from (52) and the properties of $\mathcal{K}$; in (a), $2\mathcal{K}(x) \leq \mathcal{K}(0) + \mathcal{K}(2x) \leq \mathcal{K}(2x)$ by convexity on $[0, 2x]$ and $\mathcal{K}(0) \leq 0$. (c) $\mathcal{K}(c \ln \mathrm{dist}/(ny))$ depends on (c, n) through c/n; with $\mathcal{K}$ not linear, $\mathcal{K}(\cdot)/\nu$ is not a function of $c/(n\nu)$ alone. □

Consumption equivalents. Utility is linear in $\ln y$ with slope $5/\mathcal{B}_t^j$, one $1/\mathcal{B}_t^j$ for each of the period's five years. Scaling the income that enters U_t^j by $1 + g$ in every period from t on, with $\mathcal{B}$ held at its value $\mathcal{B}_t^j$ in the worker's region at t, adds $5\ln(1 + g)/\big((1-\beta)\mathcal{B}_t^j\big)$ to lifetime value; equivalently, a change x in lifetime utility is worth the permanent proportional change in income $\exp\big((1-\beta)\mathcal{B}_t^j x/5\big) - 1$, which is the basis of the consumption-equivalent variation (58). It values every utility change at the date-t crop price of the worker's region in the comparison path, and is exact when that price does not change. The scaling applies to the income that enters U_t^j, not to the income against which the fare is charged: the utility cost $\mathcal{K}(\chi^{j,i}h(y))$ of a move is unchanged. Scaling that income too would change $h(y)$ and hence the choice probabilities.

E. Solving the model

The model is solved in time differences, extending the dynamic hat algebra of Caliendo et al. [2019] to migration costs that depend on income. The solution starts from the observed 2010 cross-section and flow matrix of workers and never requires the levels of values, non-monetary costs or productivity fundamentals; the one level it does need is income, because the fare is denominated in money. Write $\dot{u}^{jk}_{t+1} \equiv \exp\big(\overline{V}^{jk}_{t+1} - \overline{V}^{jk}_{t}\big)$ for the change in the value (9) of entering region j with skill k. Period $t = 0$ is 2010, and a period lasts 5 years.

E.1. Dynamic hat algebra with income-dependent costs

The intercept representation. With the total cost (5) and the fare (7)–(6), the choice probabilities (16) factor into an income-invariant intercept and a time-invariant income profile,

$$m^{jk,i}_t(y) = \frac{a^{jk,i}_t\, b^{ji}(y)}{\sum_l a^{jk,l}_t\, b^{jl}(y)}, \qquad a^{jk,i}_t = \exp\Big[\tfrac{\beta \overline{V}^{ik}_{t+1} - \check{\tau}^{jk,i}}{\nu}\Big], \qquad b^{ji}(y) = \exp\Big[-\tfrac{\mathcal{K}(\chi^{j,i} h(y))}{\nu}\Big]. \tag{53}$$

All income dependence sits in b^{ji}, which the fare and the kernel fix once and for all; all time dependence sits in $a^{jk,i}_t$, which does not depend on income. Two consequences follow. First, the intercepts move forward in closed form: since $\check{\tau}^{jk,i}$ is constant, $a^{jk,i}_{t+1} = a^{jk,i}_t \big(\dot{u}^{ik}_{t+2}\big)^{\beta/\nu}$, so

$$\log a^{jk,i}_t = \log a^{jk,i}_0 + \frac{\beta}{\nu} \sum_{s=2}^{t+1} \log \dot{u}^{ik}_s, \tag{54}$$

a cumulative sum over the path of value changes. The update carries no origin index, so the whole path of choice probabilities is stored as the baseline intercepts plus one destination-by-skill vector per period. Second, $m^{jk,i}_t(y)$ can be evaluated at any income, including the agricultural wages a counterfactual produces and the baseline never saw; no income grid is carried and nothing is interpolated.

Baseline intercepts. The period-0 intercepts $a^{jk,i}_0$, normalised by $a^{jk,j}_0 = 1$, are inverted so that the income-averaged probabilities reproduce the observed 2010 origin-conditional flow shares of workers in every region–skill cell (Section 5.2). The inversion iterates $\log a^{jk,i}_0 \leftarrow \log a^{jk,i}_0 + \log M^{jk,i}_{\text{obs}} - \log M^{jk,i}_0$, with $M^{jk,i}_0$ the model's integrated flow share defined below. In the 2010 flow matrix 91.8% of the cells are empty. Under (54) a zero intercept is absorbing: it would remove the destination from the choice set in every period of every counterfactual. The observed shares are therefore mixed, with a small weight, with a prior in which off-diagonal shares decay with distance and the

stay share is the cell's observed one, so the prior leaves each cell's overall migration rate unchanged.

Income distributions and mass flows. Within cell (j,k), disposable income has the distribution F_t^{jk} implied by occupational sorting (12) and the fiscal wedge (13): an atom of mass $1-\pi_{N,t}^{jk}$ at the agricultural wage $w_{A,t}^{j}$, and a lognormal tail truncated below at the occupational cutoff (11), which with $\kappa^{jk}<0$ reaches below the atom. Because the cost depends on income, the flows that enter the law of motion (17),

$$\begin{aligned} M_t^{jk,i} &= \int m_t^{jk,i}(y)\,dF_t^{jk}(y) = \left(1-\pi_{N,t}^{jk}\right) m_t^{jk,i}\left(w_{A,t}^{j}\right) + \int_{\text{tail}} m_t^{jk,i}(y)\,dF_t^{jk}(y), \\ L_{t+1}^{ik} &= \sum_j M_t^{jk,i} L_t^{jk}, \end{aligned} \tag{55}$$

have no closed-form update and are re-integrated every period. The atom is evaluated exactly. The tail is integrated by composite Gauss–Legendre quadrature in the standardised distance above the cutoff, $z=(\eta-\bar{\eta}_t^{jk})/\sigma$, with 24 nodes on each of 4 panels whose breakpoints are equal-probability quantiles of the truncated normal; the upper limit adapts so that it lies several standard deviations above both the cutoff and the mean of η. In z the integrands are smooth and saturate as income rises, because the fare's cost $\mathcal{K}(\chi h(y))$ decays like $1/y$ as income rises and is bounded by $-\ln \underline{s}$ as it falls; Gauss–Legendre in probability space would not converge geometrically, since Φ^{-1} is singular at the upper endpoint. The node count is set by the convergence of the solved path, not of a single integral, because quadrature error compounds over periods and around the fixed point. Expected log income $\mathbb{E}_{F_t^{jk}}[\log y]$ is computed in closed form (atom plus inverse-Mills correction). Probabilities are computed in logs with a max-subtracted softmax, so that $\log m^{jk,j}$ is never formed as the logarithm of a rounded probability where the stay probability is close to one. Since the fare's utility cost is at most $-\ln \underline{s}$, $b^{ji}(y)\geq \underline{s}^{1/\nu}$ at every income. The cost $\mathcal{K}$ is tabulated once on a fine grid in the log of the fare's share of income, with the expectation over ε computed by nine-point Gauss–Hermite quadrature, as in the estimation.

The backward value step. Since staying is costless, (8) evaluated through the stay probability gives the pointwise value $V_t^{jk}(y)=U_t^{j}(y)+\beta\overline{V}_{t+1}^{jk}-\nu\log m_t^{jk,j}(y)$. Integrating over F_t^{jk}, differencing over time, and using (54) to write m_{t+1} in terms of m_t, the non-monetary costs, the fare at any fixed income and all value levels cancel:

$$\log \dot{u}_{t+1}^{jk} = \log \dot{\Omega}_{t+1}^{jk} + \nu\,\mathbb{E}_{F_{t+1}^{jk}}\Big[\log \sum_l m_t^{jk,l}(y)\left(\dot{u}_{t+2}^{lk}\right)^{\beta/\nu}\Big] - \nu\left(\mathbb{E}_{F_{t+1}^{jk}}-\mathbb{E}_{F_t^{jk}}\right)\left[\log m_t^{jk,j}(y)\right], \tag{56}$$

where $\log \dot{\Omega}^{jk}_{t+1} = \mathbb{E}_{F^{jk}_{t+1}}[U^j_{t+1}(y)] - \mathbb{E}_{F^{jk}_t}[U^j_t(y)]$ is the change in the cell's mean period utility. Since U^j_t is linear in $\log y$, it needs only $\mathbb{E}_F[\log y]$ and the price terms, $\log \hat{P}^j_t = \gamma_a \log \hat{P}^j_{a,t} + \gamma_h \log \hat{r}^j_t$ and $\log \hat{\mathcal{B}}^j_t = \gamma_y \log \hat{P}^j_{a,t}$, with hats the period-$t$ solutions of the static block (Appendix F). A term $\beta \log \dot{u}^{jk}_{t+2}$ from the continuation value and its negative from the change in the stay probability cancel exactly and are never formed. The second term is the option value of moving, with period-t probabilities integrated over the period-$(t+1)$ income distribution. The third, the reweighting term, has no counterpart when costs do not depend on income: it values the period-t stay probability under the change in the within-cell income distribution, and it is non-zero exactly to the extent that workers switch sector. It is computed on nodes common to both distributions, with panel breakpoints at both occupational cutoffs, so that the difference is formed node by node rather than as the difference of two quadratures; where the two income maps coincide it reduces to an integral over the band between the cutoffs. Given the probabilities, (56) with the terminal condition is solved in one exact backward sweep. Its derivative with respect to $\log \dot{u}_{t+2}$ sums to β across destinations, so the backward map is a sup-norm contraction of modulus β.

The static block. Given the masses $\{L^{ik}_{t+1}\}$ and the shocks to fundamentals between t and $t+1$, the static equilibrium of Section 4.4 is solved in proportional changes for farmgate prices and agricultural wages (Appendix F). The outer loop is a damped tatonnement in $\log \hat{p}$ on the goods-market conditions (25), wrapped in Anderson acceleration because gross substitutability does not hold here (regional income rises with the local farm price); it stops when the largest log excess demand is below 10^{-10}. For each price conjecture, the agricultural labour markets (26) are cleared region by region by a safeguarded Newton step in $\log \hat{w}_A$, to 10^{-12}. The non-agricultural market (48) is not imposed: it holds by Walras' law and serves as a diagnostic. The static block contains no value object, so each period's equilibrium depends only on that period's masses and fundamentals: within each iteration of the outer fixed point it is solved once per period, warm-started from the previous period's prices, rather than inside the value recursion.

The outer fixed point. The only genuine fixed point is in the path of value changes $x = \{\log \dot{u}^{jk}_{t+1}\}^T_{t=0}$. One evaluation of the map (i) builds the path of intercepts by the cumulative sum (54); (ii) runs forward from the 2010 state, integrating flows and masses by (55) and solving each period's static equilibrium, which regenerates the income distributions F^{jk}_{t+1} at the new wages; and (iii) updates x by one backward sweep of (56). Only the masses carry history: skill is permanent and income is redrawn each period from the cell's distribution, so no individual income history is tracked. The update is damped and accelerated by Anderson mixing, and the iteration stops when the sup

norm, over periods, regions and skill groups, of the change in $\log \dot{u}$ is below 10^{-10}. Every path solve used in the paper meets this tolerance (all of the 10 runs).

E.2. Timing and horizon

Unexpected shocks. Every counterfactual is solved from the same period-0 intercepts $a_0^{jk,i}$ as the baseline. Since $a_0^{jk,i}$ embeds the baseline continuation values, the climate shock, or the policy, is unexpected in 2010 and learned in 2015 (period 1): the period-0 location choice is made believing the baseline path, and choices from period 1 on are made with perfect foresight of the counterfactual path. This is the counterpart, in the intercept representation, of starting a counterfactual from the observed flow matrix in Caliendo et al. [2019]. The subsidy and its tax apply from 2010 on, so the moves of period 0 are already made at the subsidised fare, with baseline continuation values, and the outlays are counted from period 0.

Horizon. The productivity paths run to 2100 (Section 5.4) and are held flat afterwards. The model is solved for $T = 40$ periods, to 2210, with the terminal condition $\dot{u}_{T+1}^{jk} = 1$. Because the backward map contracts at rate β, the terminal condition affects period t by a term of order β^{T-t}. As a check, the baseline and the SSP5-8.5 path are re-solved with $T = 30$: the consumption-equivalent welfare change of climate change (Section 6), for all workers and for the bottom and top quartiles of income, changes by less than 10^{-3} percentage points in 2015, 10^{-3} from 2050 on and 10^{-2} from 2100 on.

E.3. Welfare

Differencing two solved paths. Value levels are not identified, only their changes, so welfare is computed by differencing a counterfactual path C and its comparison path B, which start from the same period-0 state. Write $\log u_t^{jk}$ for $\overline{V}_t^{jk}$. Integrating the pointwise value over F_t^{jk} gives $\log u_t^{jk} = \log \Omega_t^{jk} + \beta \log u_{t+1}^{jk} + \nu \log D_t^{jk}$, with $\log \Omega_t^{jk} = \mathbb{E}_{F_t^{jk}}[U_t^j(y)]$ and $\log D_t^{jk} = -\mathbb{E}_{F_t^{jk}}[\log m_t^{jk,j}(y)]$, both at disposable income. The difference $\Delta_t^{jk} = \log u_t^{jk,\mathrm{C}} - \log u_t^{jk,\mathrm{B}}$ obeys

$$\Delta_t^{jk} = \Delta \log \Omega_t^{jk} + \beta\, \Delta_{t+1}^{jk} + \nu\, \Delta \log D_t^{jk}, \qquad \Delta_{T+1}^{jk} = \frac{\Delta \log \Omega_T^{jk} + \nu\, \Delta \log D_T^{jk}}{1-\beta}, \tag{57}$$

where the terminal value uses that both paths are stationary after T. Every term is computable: each path carries its own cumulated price, rent and wage changes from the common period-0 state, so their difference is known although neither level is. A proportional tax at rate t_{tax} on labour income adds $5 \log(1 - t_{\mathrm{tax}}) / \mathcal{B}_t^j$ to $\log \Omega_t^{jk}$ in every period and lowers the income at which the stay probabilities are evaluated.

Incidence across workers. A worker is described by its microregion, its skill group and its efficiency rank $q \in (0,1)$ within the region–skill cell, with $\eta = \sigma\Phi^{-1}(q)$: below the occupational cutoff it farms and earns $w^j_{A,t}$, above it it earns $w^j_N e^{\mu_k+\eta}$. Neither σ nor κ^{jk} is touched by any shock, so rank q has the same meaning in both worlds; with $\kappa^{jk} < 0$, rank in efficiency is not rank in income. Because the shock is learned in 2015, the population is the same in both worlds then, and workers with the same microregion, skill group and efficiency rank are the same person; at later dates workers have moved differently, and the welfare change at t compares workers with the same three attributes, weighted by the baseline population at t (Section 6). Let $\Delta V^{jk}_t(q)$ be the worker's change in lifetime utility at date t between the two worlds and $\mathcal{B}^{j,\mathrm{B}}_t$ the crop-price term of region j at t in the comparison path.

The welfare measure. Utility is linear in log income with slope $5/\mathcal{B}^j_t$, five years of log consumption per period, so a permanent proportional change g in consumption expenditure (which equals disposable income, since no one saves) from date t on, valued at the crop-price term $\mathcal{B}^{j,\mathrm{B}}_t$ of the worker's region in the comparison path at t, raises lifetime utility by $5\log(1+g)/\left((1-\beta)\mathcal{B}^{j,\mathrm{B}}_t\right)$ (Appendix D.5). Each worker's change in lifetime utility is therefore expressed in log points of income, $\mathcal{B}^{j,\mathrm{B}}_t \Delta V^{jk}_t(q)/5$, and the consumption-equivalent variation of a group $\mathcal{G}$ of workers at date t is

$$\mathrm{CEV}_{\mathcal{G},t} = \exp\Big((1-\beta)\,\mathbb{E}_{\mathcal{G}}\left[\mathcal{B}^{j,\mathrm{B}}_t\,\Delta V^{jk}_t(q)\right]/5\Big) - 1, \tag{58}$$

the permanent proportional change in consumption that gives the group the same average change in lifetime utility, in these units. The average is taken before the exponential, so that the measure is utilitarian, an average of utility changes, and robust to a few large individual gains: averaging the workers' own consumption equivalents instead would let the largest gains dominate the mean through the convexity of the exponential. Groups are cut on the pooled national distribution of baseline market income at date t, weighted by the baseline population of each cell at t, with 200 efficiency ranks per region–skill cell at the midpoints of equal-probability intervals. Each farm atom is one income, so a group boundary that falls inside an atom puts the whole atom on one side, and groups need not be exactly equal-sized.

E.4. The subsidy and its financing

The payment. A worker with disposable income $y < y_{\mathrm{ref}}$ receives, per move from j to i, $\varsigma^{j,i}(y) = \chi^{j,i}\left(1 - y/y_{\mathrm{ref}}\right)$, so that $\left(\chi^{j,i} - \varsigma^{j,i}(y)\right)h(y) = \chi^{j,i}h(y_{\mathrm{ref}})$: the fare net of the payment is the same share of the year's income as the full fare is for an worker at the reference income, so its utility cost $\mathcal{K}(\cdot)$ of every move is the same. Equivalently,

the kernel $h(y)$ is replaced by $\min\{h(y), h(y_{\text{ref}})\}$ in (53). Staying is never paid, and eligibility does not depend on region or skill group. The reference income $y_{\text{ref}} = \text{R\$}\,737$ a month is the 50th percentile of the pooled 2010 distribution of model earnings.

Outlays. The outlay in period t is integrated over the solved flows,

$$O_t = \sum_{j,k} L_t^{jk}\, \mathbb{E}_{F_t^{jk}}\Big[\sum_{i \neq j} m_t^{jk,i}(y)\, \varsigma^{j,i}(y)\Big], \tag{59}$$

with the same quadrature as (55); the fare is paid once per move.

The tax. The subsidy is financed by a proportional tax at rate t_{tax} on the labour income of every worker, constant over time, which balances the budget in present value at the model's discount factor,

$$t_{\text{tax}} \sum_{t=0}^{T} \beta^t\, \mathcal{Y}_t = \sum_{t=0}^{T} \beta^t\, O_t, \qquad \mathcal{Y}_t = \sum_{j,k} L_t^{jk}\, \mathbb{E}_{F_t^{jk}}[\check{y}] \times (\text{months per period}), \tag{60}$$

where $\mathcal{Y}_t$ is market labour income over the period. The tax is a special case of the wedge (13): it acts on disposable income only, leaving market income, occupational sorting and the static block unchanged, but it enters both consumption and liquidity, through $h\big((1 - t_{\text{tax}})\check{y}\big)$ and through eligibility. Outlays depend on flows and flows on t_{tax}, so t_{tax} is iterated to a fixed point: the path is solved at $t_{\text{tax}} = 0$ (the unfunded subsidy), the implied rate is computed from (60), the path is re-solved at that rate, and so on until the implied rate changes by less than a small relative tolerance. The remaining budget gap, relative to the present value of outlays, is below 10^{-5} under SSP5-8.5 and below 10^{-5} without climate change.

E.5. The subsidy's gain by mover type

The split of the subsidy's gain in Section 7.2.3, and of the loss from climate change in Table II, compares each worker in the economies with and without the subsidy at the same idiosyncratic tastes. At the date of measurement, only the option value of moving in an worker's change in lifetime utility depends on the taste draws. Write the best move against staying as one composite option, whose log odds against staying are a without the subsidy and c with it, taken from each economy's choice probabilities at the worker's disposable income, and let the difference of the two taste draws, x, be standard logistic and shared by the two economies. The worker leaves without the subsidy if $x > -a$ and with it if $x > -c$. Workers with $x > \max(-a, -c)$ leave in both economies (inframarginal movers), workers with $-c < x \leq -a$ leave only with the subsidy (induced movers), and the others stay in both. The expected change in the

option value within each group follows from integrals of the logistic distribution, and the rest of the change is the same for every draw. The groups add up exactly to the change in lifetime utility. A simulation with the full choice over all microregions and the same draws in both economies, on a random sample of points, gives group shares and gains within a few percent of the composite.

F. Exact-hat derivations

The solver of Appendix E advances the static equilibrium of Section 4.4 from one period to the next in proportional changes. Fix a period t and drop the time subscript. For any static object x, let x be its value in the period-t equilibrium, x' its value in the equilibrium with the masses $\{L^{ik}_{t+1}\}$ and the fundamentals of period $t+1$, and $\hat{x} \equiv x'/x$. The levers are $\hat{L}^{ik}$ (from the law of motion) and the changes in fundamentals $\hat{A}^i$, $\hat{B}^i$, $\hat{F}^{i0}$, $\hat{X}^i$, $\hat{\bar{H}}^i$ and $\hat{d}^{j,i}$, equal to one unless shocked; climate enters through $\hat{A}^i$. In the counterfactuals of the paper only $\hat{A}^i$ and $\hat{L}^{ik}$ differ from one; the other levers are carried for generality. The parameters α, θ, ϵ, γ_a, γ_y, γ_h, $\bar{a}$, σ, μ_k and the occupational wedges κ^{ik} never take a hat. The unknowns are $\hat{p}^i$ and $\hat{w}^i_A$. No level of productivity, land, operating costs or trade costs is needed: each enters as a hat or through a period-t share. The one level the system needs is income, because the fare is denominated in money and the crop share (23) depends on income relative to $\bar{a}P^i$: the agricultural wage and the location of non-agricultural earnings are carried forward by their own hats, $w'_A = \hat{w}_A w_A$ and $\ell^{ik\prime} = \ell^{ik} + \ln \hat{B}^i$, with $\ell^{ik} = \ln w^i_N + \mu_k$ as in Appendix D.2.

Land and agricultural output. By the definition of v^i in Section 4.4,

$$\hat{v}^i = (\hat{p}^i)^{\frac{1}{1-\alpha}} (\hat{w}^i_A)^{-\frac{\alpha}{1-\alpha}}. \tag{61}$$

By (18) the odds of cultivation are $s/(1-s) = (vA/F^0)^\theta$, so a change multiplies them by $\Theta^i \equiv (\hat{v}^i \hat{A}^i / \hat{F}^{i0})^\theta$,

$$s^{i\prime} = \frac{\Theta^i s^i}{\Theta^i s^i + 1 - s^i}, \qquad \hat{s}^i = \frac{s^{i\prime}}{s^i}. \tag{62}$$

From (20), output, farm revenue and labor demand change as

$$\begin{aligned} \hat{Q}^i_a &= \hat{X}^i \hat{A}^i (\hat{p}^i)^{\frac{\alpha}{1-\alpha}} (\hat{w}^i_A)^{-\frac{\alpha}{1-\alpha}} (\hat{s}^i)^{1-\frac{1}{\theta}}, \qquad \widehat{pQ}^i_a = \hat{X}^i \hat{A}^i (\hat{p}^i)^{\frac{1}{1-\alpha}} (\hat{w}^i_A)^{-\frac{\alpha}{1-\alpha}} (\hat{s}^i)^{1-\frac{1}{\theta}}, \\ \hat{L}^{D,i}_a &= \hat{X}^i \hat{A}^i (\hat{p}^i)^{\frac{1}{1-\alpha}} (\hat{w}^i_A)^{-\frac{1}{1-\alpha}} (\hat{s}^i)^{1-\frac{1}{\theta}}, \end{aligned} \tag{63}$$

three different pairs of price and wage exponents. By Proposition 3(b), operating outlays change as

$$\hat{\mathcal{F}}^i = \hat{X}^i \hat{F}^{i0} \frac{1 - (1-s^{i\prime})^{1-\frac{1}{\theta}}}{1 - (1-s^i)^{1-\frac{1}{\theta}}}, \tag{64}$$

and rents are $R^{i\prime} = \widehat{pQ}^i_a \rho^i - \hat{\mathcal{F}}^i \mathcal{F}^i$, with $\rho^i = (1-\alpha) p^i Q^i_a$. The period-$t$ levels ρ^i and $\mathcal{F}^i$ both follow from farm revenue and s^i by the split ratio (37), which is how the calibration obtains them.

Occupations. The cutoff (11) is a log level, so it changes additively. Since κ^{ik} and μ_k are fixed, $\bar{\eta}^{ik\prime} - \bar{\eta}^{ik} = \ln \hat{w}^i_A - \ln \hat{B}^i$, and the standardized cutoff is revealed by the participation share, $\tilde{\eta}^{ik} = -\Phi^{-1}(\pi^{ik}_N)$ (Lemma 3). The wedge enters only through this period-t value, so

$$\tilde{\eta}^{ik\prime} = \tilde{\eta}^{ik} + \frac{\ln \hat{w}^i_A - \ln \hat{B}^i}{\sigma}, \qquad \pi^{ik\prime}_N = \Phi\big(-\tilde{\eta}^{ik\prime}\big). \tag{65}$$

By (22) and (39), $\pi^{ik}_N \mathbb{E}[e^{\mu_k+\eta} \mid \eta \geq \bar{\eta}^{ik}] = e^{\mu_k+\sigma^2/2}\Phi(\sigma - \tilde{\eta}^{ik})$, so the level factor cancels and

$$\hat{N}^i = \sum_k \varpi^{ik}\, \hat{L}^{ik}\, \frac{\Phi\big(\sigma - \tilde{\eta}^{ik\prime}\big)}{\Phi\big(\sigma - \tilde{\eta}^{ik}\big)}, \tag{66}$$

where ϖ^{ik} is skill group k's period-t share of region i's non-agricultural efficiency units (equivalently, of its non-agricultural wage bill). The income distribution of the next period is the one of Proposition 4 at $(w'_A, \ell^{ik\prime}, \tilde{\eta}^{ik\prime})$, and its expected log income (42) enters the value step of Appendix E.

The agricultural labor market. With period-t clearing, $L^{D,i}_a = L^{S,i}_a = \sum_k L^{ik}\Phi(\tilde{\eta}^{ik})$, condition (26) becomes, one equation per region,

$$\hat{L}^{D,i}_a \sum_k L^{ik}\Phi\big(\tilde{\eta}^{ik}\big) = \sum_k \hat{L}^{ik} L^{ik}\, \Phi\big(\tilde{\eta}^{ik\prime}\big), \tag{67}$$

with $\hat{L}^{D,i}_a$ from (63), $\hat{s}^i$ from (62) and $\tilde{\eta}^{ik\prime}$ from (65).

Proposition 10 (The wage change is unique)**.** *Given $\hat{p}^i > 0$, equation (67) has exactly one solution $\hat{w}^i_A$, and it depends on no other region's price.*

Proof. In $\ln \hat{w}^i_A$, the log of the left-hand side has slope $-[1 + \alpha(\theta-1)(1-s^{i\prime})]/(1-\alpha) < 0$, by the argument of (38), and the log of the right-hand side has slope $\sum_k \hat{L}^{ik} L^{ik} \phi(\tilde{\eta}^{ik\prime})/\sigma$ divided by the right-hand side, which is positive. As $\hat{w}^i_A \to 0$ the left-hand side diverges and the right-hand side vanishes; as $\hat{w}^i_A \to \infty$ the left-hand side vanishes and the right-hand side tends to $\sum_k \hat{L}^{ik} L^{ik} > 0$. Only $\hat{p}^i$ enters. □

Trade and the goods markets. With $\pi^{j,i}$ the period-t share of origin j in region i's agricultural spending (45),

$$\hat{P}^i_a = \Big(\sum_j \pi^{j,i}\big(\hat{p}^j \hat{d}^{j,i}\big)^{1-\epsilon}\Big)^{\frac{1}{1-\epsilon}}, \qquad \hat{\pi}^{j,i} = \big(\hat{p}^j \hat{d}^{j,i}\big)^{1-\epsilon}\big(\hat{P}^i_a\big)^{\epsilon-1}, \tag{68}$$

and neither the level of trade costs nor the weights ψ^j appear. The goods market (25) for origin j, with period-t farm revenue $p^j Q_a^j$ and expenditure Y^i, becomes

$$\widehat{pQ}_a^j \, p^j Q_a^j = \sum_i \pi^{j,i} \hat{\pi}^{j,i} \, \bar{\vartheta}^{i\prime} \, \hat{Y}^i Y^i, \tag{69}$$

where $\bar{\vartheta}^{i\prime}$ is the region's crop share (46) at the primed income distribution, with $w'_A = \hat{w}_A w_A$, $\ell^{ik\prime}$, the primed participation shares and masses, and the primed price terms of (71). The non-agricultural wage needs no equation: $\hat{w}_N^i = \hat{B}^i$.

Income, rents and prices. By (47), $(1-\gamma_h)Y^i = p^i Q_a^i + B^i N^i - \mathcal{F}^i$, and $1-\gamma_h$ is constant, so with the period-t shares v_Q^i, v_N^i and $v_{\mathcal{F}}^i$ of farm revenue, non-agricultural earnings and operating outlays in tradable income ($v_Q^i + v_N^i - v_{\mathcal{F}}^i = 1$),

$$\hat{Y}^i = v_Q^i \, \widehat{pQ}_a^i + v_N^i \, \hat{B}^i \hat{N}^i - v_{\mathcal{F}}^i \, \hat{\mathcal{F}}^i. \tag{70}$$

The income change is explicit given $(\hat{p}, \hat{w}_A)$; housing creates no fixed point. By (24) and (15),

$$\hat{r}^i = \frac{\hat{Y}^i}{\hat{\bar{H}}^i}, \qquad \hat{P}^i = \left(\hat{P}_a^i\right)^{\gamma_a} \left(\hat{r}^i\right)^{\gamma_h}, \qquad \hat{\mathcal{B}}^i = \left(\hat{P}_a^i\right)^{\gamma_y}. \tag{71}$$

The crop share $\bar{\vartheta}^{i\prime}$ in (69) needs the level $\bar{a}P^{i\prime} = \bar{a}P^i \hat{P}^i$, which is carried forward from 2010 like the agricultural wage, and is explicit given $(\hat{p}, \hat{w}_A)$.

Walras' law and the absence of a normalization. Given (67) and (69), the numeraire market (48) clears by Proposition 6, applied to the primed equilibrium; it is the redundant equation and is not part of the system, and its residual serves as a diagnostic. No price is normalized: the numeraire is a traded good with price one everywhere, and $w_N^i = B^i$ anchors the occupational cutoff, so the system is not homogeneous of degree zero in $\{\hat{p}^j\}$, and a normalization would over-determine it. The system is therefore (67), one equation per region in $\hat{w}_A^i$, and (69), one per region in $\hat{p}^j$, with everything else explicit.

The period-t shares. The system uses s^i, π_N^{ik}, ϖ^{ik}, $\pi^{j,i}$, v_Q^i, v_N^i, $v_{\mathcal{F}}^i$, $p^i Q_a^i$, Y^i, L^{ik}, w_A^i, ℓ^{ik}, $\bar{a}P^i$ and $\mathcal{B}^i$, all dated t. Each is advanced to $t+1$ by its own solved hat before the next period is solved: s' and π'_N by (62) and (65), $\pi^{j,i\prime} = \pi^{j,i}\hat{\pi}^{j,i}$, the income shares by their hats divided by $\hat{Y}^i$, which preserves their adding-up by (70), and $\varpi^{ik\prime}$ in proportion to the summands of (66). The algorithm that solves the system is described in Appendix E.

G. Calibration details

This appendix reports details of the calibration of Section 5: the baseline economy, the productivity paths and the fare.

G.1. Baseline economic environment and calibration results

This section presents additional results from the calibration of the baseline economy. By construction, the model matches in each of the 557 microregions: i) the population of each education group, L^{ik}; ii) the agricultural wage, w_A^i; iii) the non-agricultural participation share of each education group, π_N^{ik}, through the occupational cutoff (21); iv) the value of crop production, through goods-market clearing (25); v) total land endowments, X^i; vi) the fraction of land used in agriculture, s^i, through (18), within the bounds imposed where measured crop area exceeds total land; vii) the non-agricultural wage bill of each education group, through the location and the wedge of the occupational cutoff (11); and viii) the origin-conditional migration flow shares of each education group, through the intercepts of the location choice. Table G.II reports these moments (Panel A), the over-identifying moments the calibration does not impose (Panel B) and features of the calibrated baseline (Panel C).

Agriculture. The agricultural labor share α is set externally, at $\alpha = 0.45$, and the level γ_a of the crop share follows from closed-economy adding-up: summing goods-market clearing over origins requires national crop spending to equal national crop revenue (Appendix G.2). The trade block then reproduces each region's crop revenue exactly. The wage bill of farm workers is a further check, which the calibration does not impose: under Cobb–Douglas technology it should equal α times crop revenue, but in the census it is 0.141 times crop revenue, summed over regions that grow crops. Across regions, crop revenue implied by the wage bill, $w_A^i L_A^i / \alpha$, tracks measured crop revenue with $R^2 = 0.57$ (Figure G.I, panel (a)). I read the gap as the return of owner-operators: the census records the earnings of employed crop workers, not the income of farmers who work their own land or of unpaid family workers, and crop revenue is gross of the inputs that large, capital-intensive farms buy. The census wage bill is therefore not a measure of the labor share of crop revenue, and α is kept at its external value.

Non-agricultural earnings and the occupational wedge. Without the wedge κ^{ik}, the occupational block has one free object per region and education group, and the cutoff sits at the agricultural wage. Matching the participation share then fixes the earnings of non-agricultural workers by selection alone, and with participation shares above nine in ten it puts their mean far above the agricultural wage, well above the

TABLE G.I
Calibration of the model economy, 2010

Object	Interpretation	Target / source	Value	Provenance
Panel A. Scalars				
β	Five-year discount factor	0.97^5	0.859	external
ν	Gumbel scale of tastes	Set externally	1.75	external
α	Agricultural labour share	Literature, 0.4–0.5	0.45	external
θ	Fréchet shape, land quality and operating cost	Midpoint of Costinot et al. (2016) and Sotelo (2020)	2.0	external
ϵ	Armington elasticity across regional varieties	Literature, 4–8	5.0	external
δ_d	Trade-cost distance elasticity	$\ln d^{j,i} = \delta_d \ln \text{dist}^{j,i}$	0.15	external
K	Skill groups	Census educational attainment	4	external
c	Fare, 2010 R$ per log km ($\chi^{j,i} = c \ln \text{dist}^{j,i}$)	Income changes 2000–2010 within origins	1,000	data
$\underline{s}$	Floor: share of the year's income a mover keeps at least	Estimated jointly with c	0.035	data
σ_Y	Spread of the year's resources around measured earnings (log)	Estimated jointly with c	0.25	data
—	Months of income against which the fare is charged	Assumption, $Y = 12\,y$	12	external
γ_h	Housing expenditure share	2010 Census rent over household income (renters)	0.139	data
γ_a	Crop expenditure share at income $\bar{a}$	Closed-economy adding-up, crop spending $= \sum_j R^j$	0.169	inverted
γ_y	Slope of the crop share in log income	POF 2008–09 Engel slope, scaled to farm prices	−0.0493	data
$\bar{a}$	Income normalisation of the crop share, 2010 R$ a month	2010 geometric-mean earnings	783	data
σ	Dispersion of idiosyncratic log efficiency	Truncation-corrected sd of non-farm log earnings	0.739	inverted
T^*	Damage function optimum, °C	$\ln A$ on annual mean temperature ($\theta = 2$)	19.79	data
—	Damage function curvature	$-\hat{\beta}_2$ of the same regression	0.0179	data
Panel B. Varying by region: mean (SD) across microregions or region×skill cells				
$\ln B^i + \mu_k$	Non-farm location, by region×skill	Participation π_N^{ik} and census non-farm wage bill, both exactly	6.54 (0.52)	inverted
κ^{ik}	Occupational wedge (log points), by region×skill	Same two moments: non-farm iff $\ln y_N \geq \ln w_A^i + \kappa^{ik}$	−0.77 (0.47)	inverted
w_A^i	Agricultural (reservation) wage, R$/month	Census earnings, manual crop workers	446 (185)	data
s^i	Cultivated share of land endowment	PAM crop area / total land	0.166 (0.201)	data
X^i	Land endowment, km^2	Microregion shapefiles, geodesic area	15,324 (29,632)	data
$\pi^{i,i}$	Own-region crop expenditure share	Goods-market clearing (inverted $\psi^j (p^j)^{1-\epsilon}$)	0.083 (0.092)	inverted
λ_F^i	Farm operating outlays / tradable income	Function of s^i and θ only	0.068 (0.057)	inverted

Notes. Provenance: *external*, set from outside the data; *data*, measured or estimated on Brazilian data; *inverted*, chosen so that the baseline reproduces the data exactly. The fare c, the floor $\underline{s}$ and the spread σ_Y are estimated jointly, conditional on ν. Panel B reports the unweighted mean and standard deviation across the 557 microregions, or across region–skill cells. The cultivated share s^i is bounded to $[10^{-4}, 0.95]$: above, because crop area sums over crops and seasons, so double cropping can put it above total land, and a share of one would switch off the extensive margin; below, to keep $\ln s^i$ finite. Amounts in R$ are 2010 reais per month, except the fare.

TABLE G.II
Calibration fit: matched and over-identifying moments

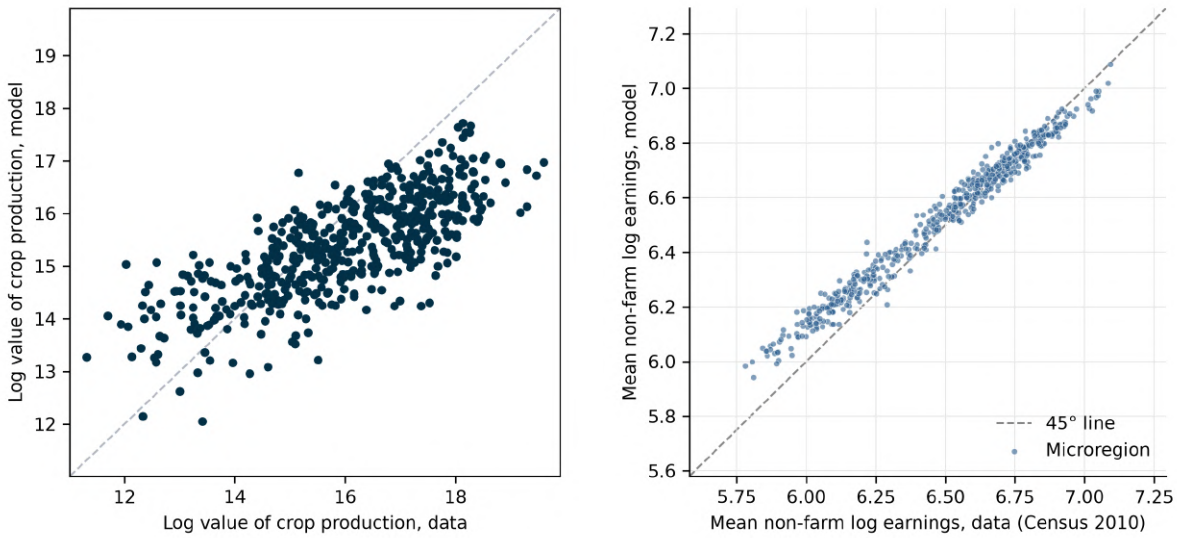
	Moment	Mean	Min	Max
Panel A. Matched exactly, region by region				
L^i	Workers	120,766	3,541	5,251,787
L^i_A / L^i	Agricultural employment share[a]	0.132	0.000	0.474
w^i_A	Agricultural wage, R$/month	446	110	1,185
X^i	Land endowment, km^2	15,324	280	333,727
s^i	Cultivated share of land[b]	0.166	0.000	0.950
R^i	Crop revenue, R$ m/month[c]	23.28	0.08	325.09
w^i_N	Mean non-farm earnings, R$/month[d]	929	506	1,782
$\bar{w}^i$	Mean labour earnings, R$/month[d]	869	432	1,773
Panel B. Over-identifying, reported not imposed				
	Mean non-farm log earnings, model minus data: mean			−0.015
	root-mean-squared / maximum absolute			0.068 / 0.349
	SD of non-farm log earnings, data / model			0.673 / 0.685
	Non-farm workers below w^i_A, data[e] / model			25.1% / 27.2%
	Geometric-mean income, census over model			1.007
	Mean inverse earnings, model over census[f]			0.944
	by education, less than primary to university			0.865 / 1.009 / 1.042 / 1.026
	Non-additivity of $\ln B^i + \mu_k$ in region and skill (RMSE)			0.073
Panel C. Features of the calibrated baseline				
	Household earnings over regional income (labour share)			0.778
	Crop agriculture, share of tradable income			0.271
	Farm operating outlays, share of tradable income			0.068
	Mean own-region expenditure share			0.083
	Bilateral flow cells with zero migrants			91.8%
	Cells with π^{ik}_N capped at $1 - 10^{-4}$, number / pop. share			97 / 4.3%

Notes. 2010. Panel A: unweighted mean, minimum and maximum across the 557 microregions; Panels B and C: one value each. Panel B is reported, not imposed; Figure G.I shows its fit region by region. [a] Farm workers over all workers, summed over education groups. [b] Crop area over total land, bounded to $[10^{-4}, 0.95]$: 9 microregions with 10.8% of the population are at a bound, and 5 microregions with 10.4% of the population grow no crops. [c] Annual value of crop production from the municipal crop survey, divided by twelve. [d] Matched through the non-agricultural wage bill of each region and education group, and with it labor earnings. [e] From each cell's lognormal at its census mean and standard deviation of non-agricultural log earnings. [f] Population-weighted mean of $1/y$ in the model's 2010 income distributions over the same mean on the census earnings nodes of the fare's estimation.

census. The wedge adds the second free object, pinned by the non-agricultural wage bill, so that both participation and the wage bill are matched exactly. It averages -0.77 across cells (standard deviation 0.47) and -1.24 across non-agricultural workers. Mean non-agricultural log earnings, which the calibration does not target, then fit with a root-mean-squared gap of 0.07 log points across cells and a maximum of 0.35 (Table G.II), and the model reproduces their variation across regions with $R^2 = 0.98$ (Figure G.I, panel (b)). The standard deviation of non-agricultural log earnings is 0.68 in the model and 0.67 in the data, and 27% of non-agricultural workers earn less than the agricultural wage in the model, against about 25% in the census. The farm atom holds 6.4% of workers, as in the data.

FIGURE G.I
Calibration fit by microregion, 2010: not targeted

(a) Crop revenue implied by the farm wage bill *(b) Mean non-agricultural log earnings*



Notes. 2010, by microregion; dashed lines are 45-degree lines. Panel (a): log value of crop production, the municipal crop survey (data) against the value implied by the census farm wage bill, $w_A^i L_A^i / \alpha$ (model); not imposed, since the calibration matches crop revenue through goods-market clearing (25). Panel (b): mean non-agricultural log earnings (the education groups of each microregion weighted by their non-agricultural workers), model against census; not targeted, since the calibration matches the participation share and the non-agricultural wage bill.

The scale of earnings. The model's earnings are on the scale of the census: the census geometric-mean income is 1.007 times the model's, so the fare is charged against earnings on the same scale in the model as in its estimation. The scale is matched in geometric means, while the fare's utility cost depends on low earnings, to first order through mean inverse earnings. The farm wage, a single value in each region, puts mean inverse earnings below the census in the least educated cells: mean inverse earnings are 0.944 times the census overall and 0.865 times among the least educated, because the farm earnings of a region have no spread below them. The model therefore understates how hard the fare bites on the poorest workers, by up to about 14% among the least educated to first order, and by less where the fare is a large share of the year's income, since its cost is bounded.

G.2. Food spending across incomes

In the model the crop share of spending falls with income (23), as food does in the data, where it weighs much more in the budgets of the poor. Figure G.II shows food spending by family income class from Brazil's household budget survey, the Pesquisa de Orçamentos Familiares (POF) 2008–2009 [IBGE, 2010], the survey closest to the model's base year. The POF records the monetary and non-monetary spending of a

national sample of families over twelve months; food includes food eaten at home and away from home, at retail prices. Per person, families in the lowest income class spend 29.6 percent of their consumption on food and families in the highest class 12.7 percent; across all families the share is 19.8 percent.

The food share falls close to linearly in the logarithm of consumption. A share linear in log consumption, the price-independent generalized logarithmic (PIGLOG) form of Deaton and Muellbauer [1980], fits the seven classes with a root-mean-squared error of 0.52 percentage points and a slope of -0.0734 per log point. A subsistence requirement (Stone–Geary) fits worse, 2.6 points, because it implies too steep a share at the bottom: 34.8 percent in the lowest class against 29.6 in the data. The more flexible PIGL form [Boppart, 2014] fits best, 0.23 points, but it changes the curvature of utility in income, on which the estimate of the fare rests. PIGLOG keeps the marginal utility of income proportional to the inverse of income, so the fare's utility cost keeps the form under which it is estimated. The factor of proportionality, $1/\mathcal{B}_t^i$, moves with crop prices; the fare's utility cost (6) is evaluated at $\mathcal{B}_t^i = 1$, its value at the 2010 national-average crop price.

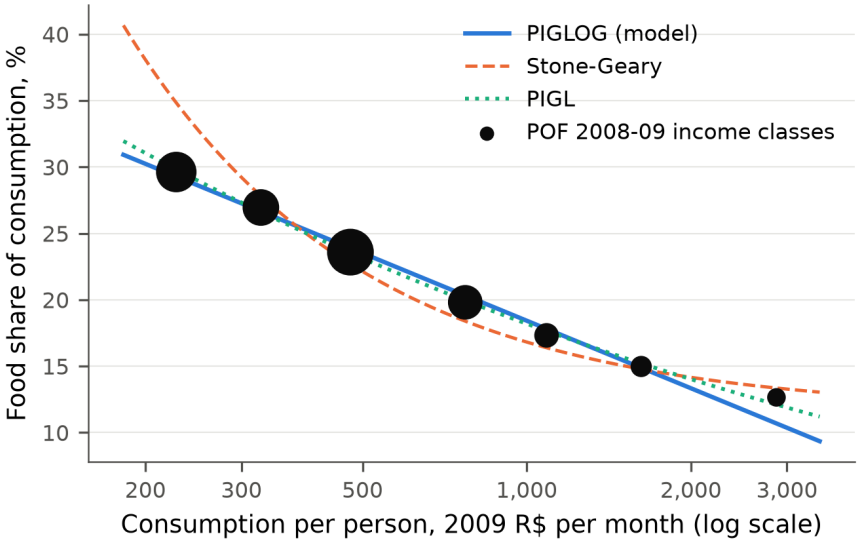


FIGURE G.II
The food share of consumption across incomes

Notes. Dots: the food share of consumption of the seven income family income classes of the POF 2008–2009 [IBGE, 2010], against consumption per person (log scale); dot area proportional to the number of families. Consumption: consumption expenditure (*despesas de consumo*), monetary and non-monetary, per family and month divided by the mean family size of the class; food: food at home and away from home (*alimentação*), at retail prices. Lines: population-weighted fits of a share linear in log consumption (PIGLOG), a subsistence requirement (Stone–Geary) and the PIGL form.

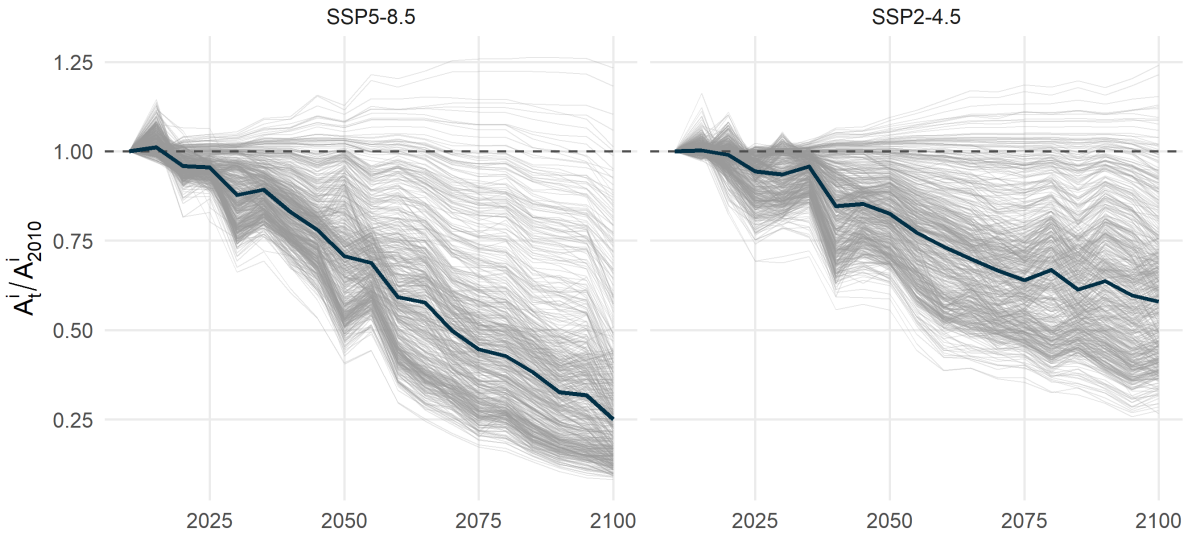
The model takes the slope of the POF profile and the level of crop spending from the calibrated economy. Its good is crops at farm prices, while the POF records food at retail prices, which include processing, distribution and meals away from home: crops

take 13.3 percent of national income in the model, against a POF food share of 19.8 percent. I therefore scale the POF slope by the ratio of the two shares, 0.67, so that $\gamma_y = -0.0493$. This assumes that farm produce is the same fraction of food spending at every income: every worker's crop share is then the same fraction of its POF food share, and the ratio of the shares of low and high incomes is that of the POF. Given γ_y, the level $\gamma_a = 0.169$ is set so that national crop spending equals national crop revenue, iterated with the inversion of relative crop prices; the landowners of each region spend at the region's worker share (46). Earnings, wage bills and crop revenue are matched as in Appendix G.1. In 2010 workers in the bottom decile of national earnings spend 23.0 percent of their income on crops, and those in the top decile 8.5 percent. At very high incomes the share (23) would turn negative; in demand it is bounded below by a small positive value, which binds for 0.0055 percent of workers in 2010.

G.3. Counterfactual productivity paths

The productivity paths of Section 5.4 combine the quadratic temperature–productivity relationship of Figure III, estimated at the model's $\theta = 2$ with an optimum of 19.8°C in annual mean temperature, with temperature projections from the CMCC-ESM2 model of CMIP6 ScenarioMIP. The model's anomaly in monthly near-surface mean temperature, relative to the model's own 2006–2015 mean, is added to each microregion's observed 2010 annual mean temperature from Xavier et al. [2022] (the delta method) and averaged over five-year periods from 2010 to 2100; productivity is held at its 2100 level thereafter. The baseline is the single year 2010, the year of every other calibration moment, rather than a decade mean. Only the shape of the damage function enters, since each period's productivity change is a ratio of damage factors. Each region is anchored to its own baseline temperature, so cooler and hotter regions move along different segments of the quadratic, and a common warming has heterogeneous effects on productivity; the projected warming itself also differs across microregions. The main scenario is SSP5-8.5, with SSP2-4.5 as the alternative. The crop-level estimates of Appendix B.6 do not enter the paths. Figure G.III shows the path of every microregion under both scenarios. The paths are not smooth: a single climate model carries its own year-to-year variability into the five-year averages, and the last period averages only the years 2098–2100. Under SSP5-8.5, by 2100 some regions lose up to 91.8 percent of their fundamental productivity, while a few cool ones gain up to 23.3 percent. Under SSP2-4.5 the median loss by 2100 is 42.1 percent.

FIGURE G.III
Counterfactual agricultural productivity paths



Notes. Fundamental agricultural productivity relative to 2010, A_t^i / A_{2010}^i, for each of the 557 microregions (grey) and the median microregion (dark line), under SSP5-8.5 and SSP2-4.5. The paths combine the quadratic temperature–productivity relationship of Figure III (at $\theta = 2$) with CMCC-ESM2 (CMIP6) projections of annual mean temperature, added to each microregion's 2010 observed value by the delta method and averaged over five-year periods.

TABLE G.III
Projected change in fundamental productivity

	SSP5-8.5			SSP2-4.5		
Year	Median	Min	Max	Median	Min	Max
2030	−12.2%	−33.8%	+5.5%	−6.5%	−31.2%	+5.5%
2050	−29.3%	−59.5%	+12.8%	−17.5%	−44.4%	+9.5%
2100	−75.0%	−91.8%	+23.3%	−42.1%	−73.3%	+24.0%

Notes. Median, minimum and maximum change across the 557 microregions in fundamental agricultural productivity relative to 2010, under SSP5-8.5 and SSP2-4.5. The changes apply the quadratic temperature–productivity relationship (at $\theta = 2$) to projected annual mean temperatures from CMCC-ESM2 (CMIP6, ScenarioMIP, near-surface mean temperature), added to each microregion's 2010 observed value from Xavier et al. [2022] and averaged over five-year periods.

G.4. Migration costs

This appendix reports the sample on which the fare of Section 5.3 is estimated and the moments of the estimated location choice against the two facts.

Sample. The fare is estimated on earners, the model's population, whose earnings are the income of the model's earners at the baseline. They are 59% of Fact 1's adults and 36% of its bottom quintile: 41% of Fact 1's adults do not work or report no earnings, and they are concentrated in its bottom quintile. The fare says nothing about how the cash cost binds for those without earnings.

The estimated location choice against the facts. Table G.IV reports the untargeted moments of Section 5.3 for the calibrated economy and for the estimated location choice of Figure V, fitted to the census earnings and flows of 2000 and 2010. Unlike the calibrated economy, it has both waves, so it can be compared with Fact 2. Because the fare is estimated from the same kind of variation, the comparison on Fact 2 is a check of consistency rather than an independent validation.

TABLE G.IV
The two facts: data, estimated model and calibrated economy

	Data	Estimated model	Calibrated economy
Fact 2: emigration from lagging places			
Post × shift × return, ρ	+6.11 (3.49)	+4.52 (2.77)	
Fact 1: emigration by income, Q5 − Q1 (pp), quintiles of own earnings			
Within origin × wave	+4.5 (0.8)	+9.4	+7.5
Within origin × education × wave	+3.9 (0.8)	+10.5	+7.9
Distance by income: distance elasticity, quintiles of own earnings			
Lowest quintile	−1.77	−2.04	−1.99
Highest quintile	−1.59	−1.48	−1.54

Notes. Workers, 2000 and 2010. Estimated model: the location choice of Figure V. Calibrated economy: the 2010 baseline, with its workers ranked into national quintiles of own earnings; emigration and the flows of movers by quintile from its location choice. It has one wave, so Fact 2 cannot be computed on it, and the data it is compared with pool 2000 and 2010 (2010 alone, within origin: +4.4). Fact 2: the triple-difference regression of Fact 2 on the origin × education cells of the estimation, with the observed and with the predicted emigration rate on the left, weighted by population, standard errors clustered by origin; a group is an education level, with one income shift each. The shift is the national growth of mean real total income of each education level and the return is that of Fact 2, both computed on all adults; on all adults the education-level estimate is 7.67. Fact 1: quintiles of monthly earnings formed by wave among workers, with the fixed effects of the row, without age effects; in the model, the choice probabilities are evaluated at every earnings node of every cell, whose aggregate emigration differs from the cell's fitted rate by at most 1.00 percentage points, and nodes are grouped into quintiles of their own earnings. Distance: the gravity equation of Fact 1's distance counterpart on flows by quintile, origin, destination and wave.

The time units of the fare. The model's period is five years, and the period's utility (14) counts the log consumption of the five years. The fare is paid out of the income of the year of the move, and its loss is a loss of log consumption in that year. A real paid through the fare and a real of consumption are therefore valued in the same units, out of the same year's income. At the 2010 national crop price a real of consumption is worth $1/(12\,y)$ of utility, and a real of fare relief $\mathcal{K}'(\chi h(y))/(12\,y)$ (Appendix D.5): about the same when the fare is a small share of the year's income, more when it is a larger share, close to $1/(12\,y-\chi)$, and less again as the mover approaches the floor. Comparisons in reais, such as the outlays of a subsidy, a cash transfer of the same cost and the ratio of output gained to outlays, are therefore made directly.

H. Additional quantitative results

H.1. Climate losses by group and sector

Losses by group. Table H.I reports the welfare change from climate change under SSP5-8.5 by group of workers. The bottom quartile loses more than the top at every date, nationally, in the North and Northeast and in the hottest regions. In the South, Southeast and Centre-West the gap is smaller, and in the coolest regions the bottom quartile loses less than the top in 2015 and 2050. The loss falls with education: from 2050 on the least educated lose 6.63 percent and university graduates 4.70 percent.

Losses by sector. Farmers, who are 6 percent of workers in 2015, lose 16 percent more than other workers from 2050 on (Table H.II). The gap between low and high-income workers is largest within agriculture: from 2050 on, the bottom quartile of farmers loses 46 percent more than the top quartile of farmers, and outside agriculture the bottom quartile loses 18 percent more than the top.

Migration responses. In 2050, 0.63 percent of workers live in a different microregion than they would in the baseline, and 2.55 percent in 2100 (Table H.III). By 2100 the number of workers in the hottest regions is 4.26 percent lower than in the baseline, and in the North and Northeast 4.58 percent lower. The number of workers in the hottest regions falls in every skill group but university graduates, whose number rises slightly; it falls most for the least educated. Under SSP2-4.5, 1.31 percent of workers live elsewhere by 2100.

TABLE H.I
Welfare change from climate change, by group of workers

Workers	2015	2050	2100
All workers	−2.56	−6.08	−10.82
By national earnings quartile			
Bottom quartile	−2.82	−6.72	−11.69
Second quartile	−2.57	−6.13	−11.13
Third quartile	−2.52	−5.98	−10.71
Top quartile	−2.34	−5.50	−9.73
Bottom vs. top quartile (% difference)	+20.3	+22.2	+20.2
By region			
North and Northeast	−3.01	−7.15	−11.92
bottom quartile	−3.20	−7.79	−12.83
top quartile	−2.57	−5.90	−10.02
difference (%)	+24.5	+32.1	+28.0
South, Southeast, Centre-West	−2.38	−5.68	−10.42
bottom quartile	−2.48	−5.90	−10.89
top quartile	−2.29	−5.41	−9.66
difference (%)	+8.2	+9.0	+12.7
Hottest regions	−3.00	−7.09	−11.93
bottom quartile	−3.21	−7.78	−12.89
top quartile	−2.57	−5.91	−10.09
difference (%)	+25.1	+31.7	+27.8
Coolest regions	−2.11	−5.09	−9.51
bottom quartile	−2.12	−5.04	−9.39
top quartile	−2.17	−5.16	−9.34
difference (%)	−2.0	−2.3	+0.5
By skill group			
Less than primary	−2.79	−6.63	−11.64
Primary	−2.63	−6.25	−11.17
Secondary	−2.50	−5.94	−10.64
University	−1.97	−4.70	−8.50

Notes. Consumption-equivalent variation (CEV, percent) of climate change under SSP5-8.5 relative to the baseline without climate change: the permanent proportional change in consumption, from the date on, that gives the group the same average change in lifetime utility (Appendix E.3). *Difference (%)*: the percent by which the bottom quartile's loss exceeds the top quartile's, $100\,(\mathrm{CEV}_{Q1}/\mathrm{CEV}_{Q4} - 1)$; negative where the bottom quartile loses less. Workers are compared at the same microregion, skill group and efficiency rank, weighted by the baseline population at the date (in 2015, the same workers in both worlds) and grouped by their earnings in the baseline economy at the date; quartiles are national, pooled across regions and skill groups, also within the regional rows. Hottest (coolest) regions: the microregions with a 2010 annual mean temperature above 24.6°C (below 21.8°C), which hold a third of workers. Skill groups are education levels. Negative values are losses.

TABLE H.II
Welfare change from climate change, by sector

	Climate change, SSP5-8.5 (%)		
Workers	2015	2050	2100
Agricultural workers			
All	−2.74	−7.02	−12.22
bottom quartile	−3.30	−8.77	−13.48
top quartile	−2.41	−5.99	−12.65
Non-agricultural workers			
All	−2.55	−6.02	−10.73
bottom quartile	−2.76	−6.48	−11.48
top quartile	−2.33	−5.48	−9.69
Both sectors			
All workers	−2.56	−6.08	−10.82

Notes. CEV (percent) of climate change under SSP5-8.5 relative to the baseline, as in Table H.I, for agricultural and non-agricultural workers at the date (a worker is agricultural if its efficiency rank puts it below its cell's occupational cutoff in the baseline). Quartiles are cut within each sector on baseline earnings at the date. Negative values are losses.

TABLE H.III
Migration responses to climate change: workers relocated and the change in the number of workers by region, against the baseline

				Hottest regions				
Scenario	Year	Relocated	N/NE pop.	All	<Primary	Primary	Secondary	University
SSP5-8.5	2050	0.63	−1.34	−1.17	−2.01	−1.29	−0.55	+0.18
SSP5-8.5	2100	2.55	−4.58	−4.26	−7.18	−4.97	−2.24	+0.92
SSP2-4.5	2050	0.40	−0.83	−0.73	−1.26	−0.80	−0.33	+0.11
SSP2-4.5	2100	1.31	−2.57	−2.36	−4.19	−2.66	−1.06	+0.64

Notes. Percent, climate path against the baseline, in 2050 and 2100. *Relocated*: half the sum over microregions and skill groups of the absolute difference in workers between the two paths, over national population: the share of workers who live elsewhere than in the baseline. *N/NE pop.*, *Hottest regions*: change in the number of workers living in the North and Northeast, and in the hottest regions (2010 temperature above 24.6°C, a third of workers), all workers and by skill group (less than primary, primary, secondary, university).

H.2. The subsidy averaged over all workers

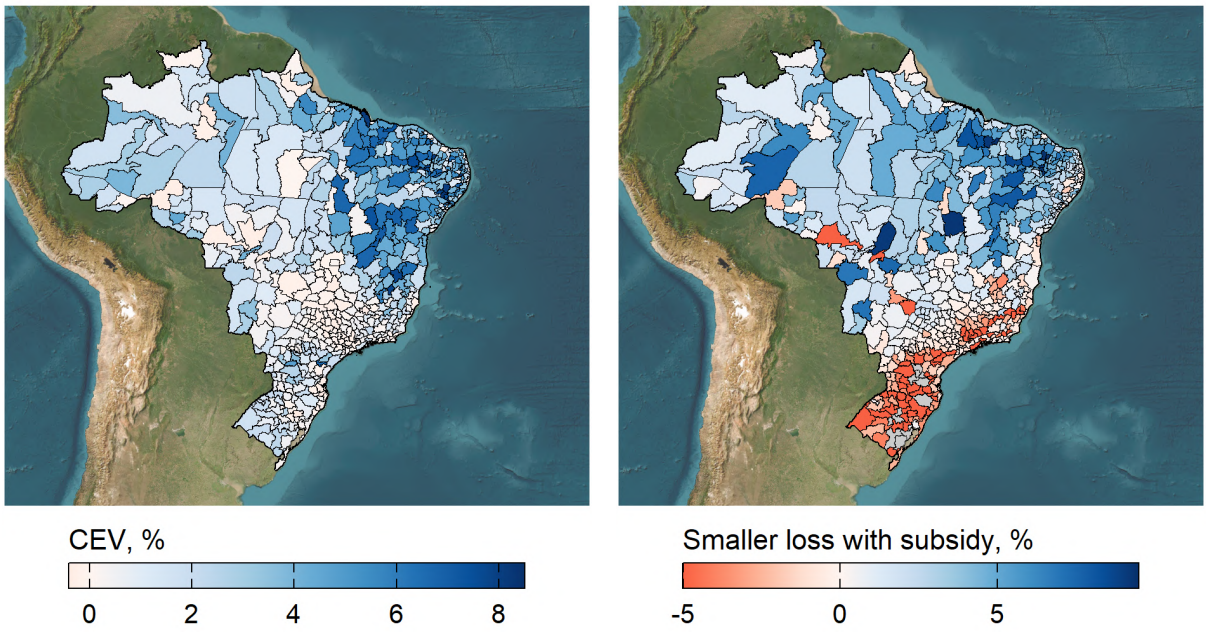
Section 7 measures the subsidy's effect on welfare on the workers it induces to move. This appendix reports the same effects averaged over all workers of each group, including those whose choice the subsidy does not change, most of whom would stay in their microregion even without the fare.

Under SSP5-8.5, the gain of the bottom quartile in 2015 is the equivalent of 1.53 percent of its consumption, permanently, and that of the lowest decile 2.09 percent (Figure VII, Panel A(c)). The gains are largest in the North and Northeast and in the

hottest regions: 2.71 percent for the bottom quartile in the North and Northeast and 2.36 percent in the hottest regions, against 0.49 percent in the rest of the country (Table II, Gain; Figure H.I, map (a)). Without climate change the gains are almost as large: 1.50 percent for the bottom quartile and 2.28 percent in the hottest regions. Workers above the reference income gain as well, up to the ninth decile, because earnings are redrawn every period and they may be eligible later; their gain falls with earnings because high-income workers belong to cells that rarely draw low incomes, and the top decile, which pays the tax, loses slightly.

FIGURE H.I
The subsidy and the bottom quartile, by microregion, 2015: all workers

(a) Welfare gain from the subsidy *(b) How the subsidy changes the climate loss*



Notes. As Figure VIII, for all workers of the bottom national quartile of baseline earnings; in map (a) the few microregions with a negative value are drawn as zero.

The bottom quartile. For the bottom quartile as a whole, the loss from climate change in 2015 is 1.0 percent smaller with the subsidy, and 2.5 percent smaller in the North and Northeast and 2.4 percent in the hottest regions. In the South, Southeast and Centre-West the loss is slightly larger with the subsidy. The loss is smaller in 93 percent of the microregions of the North and Northeast (Figure H.I, map (b)).

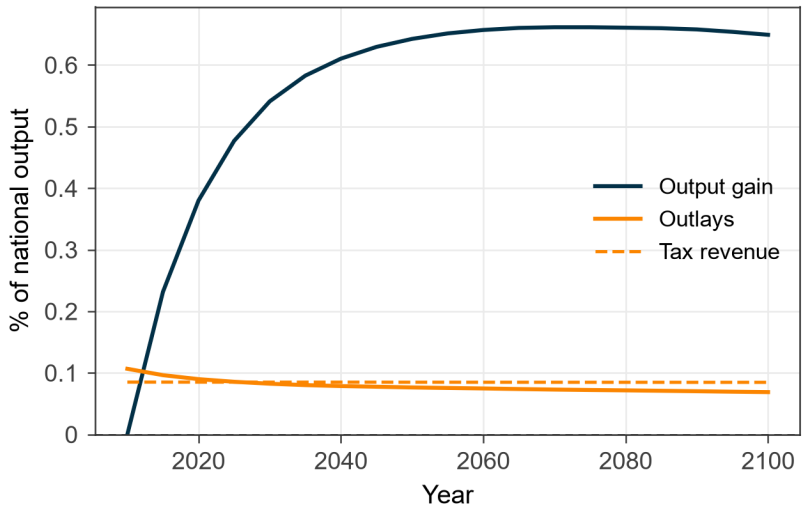
Compensation. Most of the subsidy's gain to the bottom quartile goes to workers whose choice in 2015 it does not change (Figure VII, Panel B(b)). The induced movers receive 14 percent of the quartile's gain, those who leave anyway 15 percent, and those who stay 71 percent: their incomes are redrawn every period, so they can draw a low

income and move with the subsidy later (Appendix E.5). Set against the loss from climate change, both measured in lifetime utility, the subsidy's gain offsets 53 percent of the bottom quartile's loss in 2015, and 71 percent in the hottest regions (Appendix Table H.IV). These shares are ratios of changes in lifetime utility, not of the consumption-equivalent percentages, which are measured against different consumption paths. Because the losses from climate change grow over the century while the gains of the subsidy do not, the share falls for later years: for the bottom quartile it is 20 percent in 2050 (Appendix Table H.IV).

H.3. The subsidy over time

This appendix extends the results of Section 7, reported there for 2015, to later years. Table H.IV reports the welfare gain of the income-targeted subsidy, and the share of the climate loss it offsets, in 2015, 2050 and 2100. Figure H.II shows the output gain, the outlays and the tax revenue in each period.

FIGURE H.II
The migration subsidy over time: output gain, outlays and tax revenue in each period



Notes. The subsidy financed by a proportional tax, 2010–2100, under SSP5-8.5: the change in national output (the sum of regional incomes) against the path without the subsidy, the outlays of the subsidy, and the revenue of the tax, in percent of national output of the period. The tax balances revenue and outlays in present value over the horizon of the model, 2210, not in each period; Figure IX shows the discounted cumulated values.

H.4. Alternative policies

The subsidy of Section 7 is one way to lower the burden of the fare on low-income workers. This subsection compares it with two alternatives, each financed by a proportional tax on earnings: the same payment paid only on moves out of the hottest

TABLE H.IV
Welfare gains from the migration subsidy, by group of workers, 2015–2100

	Gain under SSP5-8.5 (%)			Share of the climate loss offset (%)		
Workers	2015	2050	2100	2015	2050	2100
All workers	+0.60	+0.50	+0.45	23	8	4
By national earnings quartile						
Bottom quartile	+1.53	+1.43	+1.32	53	20	11
Second quartile	+0.62	+0.50	+0.44	24	8	4
Third quartile	+0.23	+0.15	+0.13	9	2	1
Top quartile	+0.01	−0.05	−0.06	1	−1	−1
By region						
North and Northeast	+1.70	+1.74	+1.68	55	23	13
bottom quartile	+2.71	+2.83	+2.77	82	34	20
top quartile	+0.42	+0.43	+0.44	16	7	4
South, Southeast, Centre-West	+0.14	+0.04	+0.02	6	1	0
bottom quartile	+0.49	+0.36	+0.32	20	6	3
top quartile	−0.08	−0.16	−0.17	−3	−3	−2
Hottest regions	+1.32	+1.30	+1.26	43	18	10
bottom quartile	+2.36	+2.41	+2.33	71	29	17
top quartile	+0.25	+0.24	+0.25	10	4	2
Coolest regions	+0.14	+0.03	−0.04	7	1	0
bottom quartile	+0.42	+0.28	+0.18	19	5	2
top quartile	−0.06	−0.15	−0.18	−3	−3	−2
By skill group						
Less than primary	+1.18	+1.09	+1.01	42	16	8
Primary	+0.56	+0.41	+0.35	21	6	3
Secondary	+0.25	+0.15	+0.13	10	3	1
University	−0.11	−0.12	−0.11	−6	−3	−1

Notes. Gain under SSP5-8.5: CEV (percent) of the financed subsidy against the same economy without it, net of the tax that finances it. *Share of the climate loss offset*: the gain of the subsidy under climate change as a share of the group's loss from climate change, both in lifetime utility, in percent; above 100 when the group is better off with climate change and the subsidy than in the baseline without either. Workers compared at the same microregion, skill group and efficiency rank, weighted by the baseline population in that year and grouped by national quartile of baseline earnings in that year; columns give the year in which the CEV is measured (Section 6). Hottest (coolest) regions: the microregions with a 2010 mean temperature above 24.6°C (below 21.8°C); each group holds a third of 2010 workers. Skill groups are education levels.

regions, and a cash transfer of the same present value to the same eligible workers. The comparison turns on whom each policy induces to move.

Targeting by place instead of by income. The subsidy is targeted by income, not by place: every eligible worker is paid, wherever it lives. Its gains are nonetheless concentrated in the hottest regions (Section 7.2.3), because that is where low-income

workers and the cells from which they leave are concentrated. The place-targeted subsidy has the same payment and the same eligibility by income, but is paid only on moves out of the hottest regions to destinations outside them (Tables H.V and H.VI). It pays for fewer moves and costs less: the tax that finances it is 0.028 percent of earnings, against 0.109 percent, and the present value of its outlays is 0.022 percent of output, against 0.085 percent. It raises output by 0.310 percent of the present value of output, against 0.450 percent. The moves it pays for are those from the low-wage farm regions to the rest of the country, where the income gain of a move is largest, while the income-targeted subsidy also pays for moves between regions with similar wages; so its output gain is 1,425 percent of its outlays, against 527 percent. In the hottest regions the bottom quartile gains 1.37 percent, against 2.36 percent under the income-targeted subsidy. In the South, Southeast and Centre-West it does not: the bottom quartile there gains 0.49 percent under the income-targeted subsidy and loses 0.16 percent under the place-targeted one, which it pays for through the tax and is rarely paid. Nationally, the bottom quartile gains 0.54 percent, against 1.53 percent. Targeting by place yields more output per real of outlays; targeting by income reaches the low-income workers who bear the fare wherever they live.

A cash transfer instead of a fare subsidy. A cash transfer to low-income workers would also lower the burden of the fare, since the burden falls with income, and it would raise their consumption whether or not they move. I compare the subsidy with a transfer to the same eligible workers, of the same present value, paid every month whether or not they move, financed in the same way (Tables H.V and H.VI). Spread over every eligible worker in every month, the same budget pays a transfer of at most R$ 6.37 a month, and R$ 2.54 for the average recipient, because the subsidy's outlays are concentrated on the few workers who move. The transfer raises the welfare of the bottom quartile by 0.63 percent, against 1.53 percent for the subsidy, and 0.84 against 2.36 percent in the hottest regions. It does not change who moves: the bottom quartile's leaving rate in 2015 is 3.1 percent with the transfer and 3.1 percent without a policy, against 7.8 percent with the subsidy. Its output gain is 1 percent of the outlays, against 527 percent for the subsidy. Of the bottom quartile's loss from climate change, the transfer offsets 22 percent and the subsidy 53 percent. The subsidy lowers the cost of a move and so changes who moves; the transfer does not. The fare and consumption are valued in the same units, out of the year's income (Appendix G.4): a real of the transfer and a real of fare relief are worth the same to a worker when the fare is a small share of the year's income, and a real of relief is worth more when it is a larger share, and less where the floor binds. In a model of seasonal migration in Bangladesh, Lagakos et al. [2023] find that a migration subsidy and an unconditional transfer have similar average welfare gains, with the subsidy raising the welfare of the lowest workers more.

TABLE H.V
Three policies compared: welfare gains under SSP5-8.5, by group of workers, 2015 and 2050

	Income-targeted subsidy		Place-targeted subsidy		Cash transfer	
Workers	2015	2050	2015	2050	2015	2050
All workers	+0.60	+0.50	+0.12	+0.10	+0.30	+0.29
By national earnings quartile						
Bottom quartile	+1.53	+1.43	+0.54	+0.53	+0.63	+0.60
Second quartile	+0.62	+0.50	+0.07	+0.03	+0.30	+0.28
Third quartile	+0.23	+0.15	−0.02	−0.05	+0.19	+0.19
Top quartile	+0.01	−0.05	−0.09	−0.12	+0.09	+0.09
By region						
North and Northeast	+1.70	+1.74	+0.86	+1.04	+0.58	+0.56
bottom quartile	+2.71	+2.83	+1.33	+1.56	+0.89	+0.87
top quartile	+0.42	+0.43	+0.35	+0.49	+0.19	+0.18
South, Southeast, Centre-West	+0.14	+0.04	−0.18	−0.25	+0.19	+0.18
bottom quartile	+0.49	+0.36	−0.16	−0.25	+0.39	+0.38
top quartile	−0.08	−0.16	−0.19	−0.25	+0.07	+0.07
Hottest regions	+1.32	+1.30	+0.81	+0.95	+0.50	+0.48
bottom quartile	+2.36	+2.41	+1.37	+1.57	+0.84	+0.81
top quartile	+0.25	+0.24	+0.30	+0.41	+0.16	+0.15
Coolest regions	+0.14	+0.03	−0.19	−0.27	+0.16	+0.15
bottom quartile	+0.42	+0.28	−0.18	−0.27	+0.36	+0.34
top quartile	−0.06	−0.15	−0.19	−0.27	+0.06	+0.06
By skill group						
Less than primary	+1.18	+1.09	+0.33	+0.31	+0.52	+0.50
Primary	+0.56	+0.41	+0.08	+0.03	+0.34	+0.32
Secondary	+0.25	+0.15	0.00	−0.02	+0.19	+0.18
University	−0.11	−0.12	−0.07	−0.08	−0.06	−0.06

Notes. CEV (percent) of each policy, financed by a proportional tax on earnings that balances its budget in present value, against the calibrated economy under SSP5-8.5 without the policy, net of the tax, for workers in 2015 and in 2050. *Income-targeted subsidy*: the subsidy of Section 7.2.1. *Place-targeted subsidy*: the same payment and eligibility, paid only on moves out of the hottest regions (the microregions with a 2010 mean temperature above 24.6°C, 33 percent of 2010 workers) to destinations outside them. *Cash transfer*: a monthly transfer to the same eligible workers, falling linearly from R$ 6.37 at zero earnings to zero at the reference income, paid whether or not they move, with the same present value of outlays as the income-targeted subsidy; in 2015 a recipient receives R$ 2.54 a month on average. Groups by national quartile of baseline earnings in that year; regions as in Section 6.

H.5. The welfare effect of a marginal change in the subsidy

The subsidy of Section 7 is a benchmark rule, not an optimum. This subsection derives the first-order welfare effect of changing it over a range of incomes, the building block of an optimal schedule, and the statistics that govern it.

Set-up. Write the payment as a share of the fare, $\varsigma^{j,i}(y) = \iota(y)\chi^{j,i}$ with $0 \leq \iota(y) \leq 1$, so that the payment never exceeds the fare and the utility cost of the fare becomes $\mathcal{K}\big((1-\iota(y))\chi^{j,i}h(y)\big)$, with $h(y) = 1/(12y)$, $\mathcal{K}$ the loss of Appendix D.5 and y disposable

TABLE H.VI
Three policies compared: cost, output and mobility

	SSP5-8.5				No climate change			
	Income-targeted	Place-targeted	Cash transfer	No policy	Income-targeted	Place-targeted	Cash transfer	No policy
Tax (% of earnings)	0.109	0.028	0.110		0.109	0.026	0.109	
Transfer at zero earnings (R$ a month)			6.37				6.31	
PV cost (% of output)	0.085	0.022	0.085		0.085	0.021	0.085	
PV output gain (% of output)	+0.450	+0.310	+0.001		+0.448	+0.307	+0.002	
Output gain (% of outlays)	527	1,425	1		526	1,489	3	
Leaving rate, bottom quartile, 2015 (%)	7.8	4.2	3.1	3.1	7.8	4.1	3.1	3.1
Leaving rate, top quartile, 2015 (%)	8.3	8.3	8.3	8.3	8.3	8.3	8.3	8.3
Climate loss offset, bottom quartile, 2015 (%)	53	19	22					

Notes. Each policy financed by a constant proportional tax on earnings that balances its budget in present value over the horizon of the model. PV: present value, discounted at 3 percent a year, in percent of the present value of national output. Output gain: the change in national output against the economy without the policy under the same climate. Leaving rate: the share of workers who leave their microregion within five years. Climate loss offset: the gain of the policy under SSP5-8.5 as a share of the bottom quartile's loss from climate change, both in lifetime utility.

income; the benchmark is $\iota(y) = (1 - y/y_{\text{ref}})^{+}$. A utilitarian planner maximises

$$W = \sum_{j,k} L_0^{jk} \int V_0^{j,k}(y)\, dF_0^{jk}(y), \tag{72}$$

the sum of the lifetime utilities of the workers of the initial period, weighted equally. W is the objective of the marginal analysis, not the consumption-equivalent aggregate in which the paper reports its results. The schedule is financed by the proportional tax of Appendix E.4, which balances the budget in present value. Three assumptions keep the derivation first-order. (i) Prices, rents and wages, and hence, at a given tax rate, the distributions of disposable income F_t^{jk}, are held at their values along the path; general equilibrium is discussed below. (ii) The schedule is known before the choices of the initial period are made, unlike the unexpected subsidy of Section 7, so that every period's choices respond to it. (iii) The sums run over the model's horizon, $t = 0, \ldots, T$, with the terminal continuation value held fixed.

Objects. Consider raising ι by $d\iota$ on an interval $\mathcal{I}$ of disposable incomes, in every period, region and skill group. Let

$$\mathcal{X}(\mathcal{I}) = \sum_{t=0}^{T} \beta^t \sum_{j,k} L_t^{jk} \int_{\mathcal{I}} \sum_{i \neq j} m_t^{jk,i}(y)\, \chi^{j,i}\, dF_t^{jk}(y) \tag{73}$$

be the present value of the fares paid on the moves made at incomes in $\mathcal{I}$, and $\mathcal{H}(\mathcal{I})$ the same sum with each fare weighted by $\eta^{j,i}(y) \equiv h(y)\, \mathcal{K}'\big((1 - \iota(y))\chi^{j,i} h(y)\big)$, the utility of a real of fare relief; integrating against dF_t^{jk} handles the farm atom. Let $e(\mathcal{I}) \equiv \partial \ln \mathcal{X}(\mathcal{I}) / \partial \iota(\mathcal{I})$, at a given tax rate and including the change in the masses L_t^{jk},

be the semi-elasticity of these fares, the response of moves in $\mathcal{I}$ to the subsidy. Let $\mathcal{D}$ be the present value of the budget surplus, revenue less outlays, and

$$\lambda_G \equiv -\frac{\partial W/\partial t_{\text{tax}}}{\partial \mathcal{D}/\partial t_{\text{tax}}} > 0$$

the utility cost of a real of budget raised by the tax, every response included: through consumption, through the fare's utility cost $\mathcal{K}\big((1-\iota)\chi h((1-t_{\text{tax}})\breve{y})\big)$, through eligibility and through the flows. Finally, let $\mathcal{R}(\mathcal{I})$ be the change in the present value of the budget per unit of $d\iota$, at a given tax rate, other than the change in outlays on moves at incomes in $\mathcal{I}$: the change in tax revenue as workers locate elsewhere, less the change in outlays on moves at incomes outside $\mathcal{I}$, among them later moves of the workers the subsidy induces to move.

Proposition 11 (Marginal welfare effect of the subsidy)**.** *Under assumptions (i)–(iii), with the budget balanced by the tax,*

$$\frac{dW}{d\iota(\mathcal{I})} = \underbrace{\mathcal{H}(\mathcal{I})}_{\textit{mechanical gain}} - \lambda_G \Big[\underbrace{\mathcal{X}(\mathcal{I})\big(1+\iota\, e(\mathcal{I})\big)}_{\textit{outlays}} - \underbrace{\mathcal{R}(\mathcal{I})}_{\textit{fiscal externality}} \Big], \tag{74}$$

with ι the share on $\mathcal{I}$, taken constant on a narrow interval. The moves the subsidy induces have no first-order effect on the welfare of the workers who make them; they enter only through the budget.

Proof. By (8), $V_t^{j,k}(y)$ depends on ι through the migration costs $\tau^{jk,i}(y) = \breve{\tau}^{jk,i} + \mathcal{K}\big((1-\iota(y))\chi^{j,i}h(y)\big)$ and through the continuation values $\overline{V}_{t+1}^{i,k}$. The derivative of the log-sum in (8) with respect to $-\tau^{jk,i}(y)$ is the choice probability $m_t^{jk,i}(y)$ of (16), so, for $y \in \mathcal{I}$,

$$\frac{\partial V_t^{j,k}(y)}{\partial \iota} = \sum_{i \neq j} m_t^{jk,i}(y)\, \eta^{j,i}(y)\, \chi^{j,i} + \beta \sum_i m_t^{jk,i}(y)\, \frac{\partial \overline{V}_{t+1}^{i,k}}{\partial \iota},$$

and only the second term outside $\mathcal{I}$. The change in the choice probabilities does not appear, because each probability maximises the expected value of the worker's choice (the envelope theorem for the log-sum). Integrate over incomes with (9), at the distributions F_{t+1}^{ik} held fixed by (i), and sum over origins with (17): the continuation term of period t becomes the masses L_{t+1}^{ik} times $\partial \overline{V}_{t+1}^{i,k}/\partial \iota$. Iterating forward to T gives $\partial W/\partial \iota = \mathcal{H}(\mathcal{I})$ at a given tax rate. The outlay on moves in $\mathcal{I}$ is ι times the fares in (73); its derivative is $\mathcal{X} + \iota\, \partial \mathcal{X}/\partial \iota = \mathcal{X}(1+\iota e)$. Every other effect on the budget at a given tax rate is $\mathcal{R}$ by definition, so the surplus falls by $\mathcal{X}(1+\iota e) - \mathcal{R}$. Restoring it with the tax costs λ_G per real, by the definition of λ_G. □

The optimal share. Setting (74) to zero on a narrow interval around y, where $\mathcal{H} = \bar{\eta}(y)\mathcal{X}$ with $\bar{\eta}(y)$ the fare-weighted mean of $\eta^{j,i}(y)$, gives the condition for an interior optimum,

$$\iota^*(y)\, e(y) \;=\; \bar{g}(y) - 1 + \frac{\mathcal{R}(y)}{\mathcal{X}(y)}, \qquad \bar{g}(y) \equiv \frac{\bar{\eta}(y)}{\lambda_G}. \tag{75}$$

The condition is necessary, not sufficient, and implicit, because e, $\mathcal{R}$ and λ_G depend on the whole schedule and $\bar{g}(y)$ depends on $\iota(y)$ through $\mathcal{K}'$. With $0 \leq \iota \leq 1$, the optimum is at $\iota^*(y) = 0$ where the right-hand side is negative at $\iota = 0$, and at $\iota^*(y) = 1$ where the right-hand side exceeds e at $\iota = 1$.

The three statistics. The first is $\bar{g}(y)$, the value of a real paid to a mover at income y relative to the cost of raising it. Because the fare is paid in money, $\eta^{j,i}(y) = \mathcal{K}'(x)/(12y)$, with x the share of the year's income that the fare net of the payment takes. Away from the floor both factors fall with income, so $\bar{g}(y)$ falls with y although the planner weighs every worker equally; where the floor binds, $\mathcal{K}'(x)$ rises with income. To a first approximation, ignoring the tax's effect on the fare's utility cost and on the flows, raising the rate by dt_{tax} costs every worker about $5\,dt_{\text{tax}}$ in utility per period at the 2010 national crop price, five years of log consumption, and raises $60\,\breve{y}\,dt_{\text{tax}}$ reais from each. Hence $\lambda_G \approx 1/(12\,\bar{y})$, with $\bar{y}$ the discounted mean market income of workers, and $\bar{g}(y) \approx \bar{\mathcal{K}}'(y)\,\bar{y}/y$, with $\bar{\mathcal{K}}'(y)$ the fare-weighted mean of $\mathcal{K}'(x)$. The factor $\bar{y}/y$ is redistributive. A real of fare relief and a real of consumption are valued out of the same year's income (Appendix G.4). The factor $\bar{\mathcal{K}}'$ is about one where the fare net of the payment is a small share of the year's income, exceeds one where it is a larger share, as $\mathcal{K}'(x) \approx 1/(1-x)$, and falls towards zero where the floor binds: a marginal payment to a worker whose fare is far above a year's income lowers its cost little, while a payment large enough to bring the fare net of the payment below the level at which the floor binds lowers it by much. W need not then be concave in $\iota(y)$, and (75), a local condition, can miss the optimum there.

The second is $e(y)$, the response of moves to the subsidy. Payments to the workers it induces to move cost the budget but, at the margin, buy them nothing, since they were indifferent; workers who move anyway receive a pure transfer. A larger response therefore lowers the optimal share. In the results, 39 percent of the subsidised moves of the bottom quartile in 2015, all below the reference income, would have been made without the subsidy (Section 7.2.3). With a constant semi-elasticity that share would be $\exp(-\iota e)$, which puts ιe at about one. This is a rough guide only: the share counts leavers rather than fare-weighted moves, refers to one period, averages over the quartile's shares $\iota(y)$, and comes from a subsidy that is unexpected and not marginal.

The third is $\mathcal{R}(y)$, the fiscal externality. Induced movers who earn more at their destinations pay more tax, but they are also paid on their later moves, and moves

driven by tastes can lead to lower incomes, so the sign of $\mathcal{R}$ is not given. Its tax part is small, because the tax that finances the subsidy is small; its payment part is not bounded by the tax. With a pre-existing income tax, induced moves to higher earnings would raise the optimal share.

What the formula leaves out. At given prices a move has no external effect other than through the budget, so the case for the subsidy rests on redistribution and on the value of fare relief relative to consumption, both in $\bar{g}$, and on the fiscal externality. In general equilibrium, the induced moves also change crop prices, land rents and farm wages at origins and destinations. These add pecuniary terms that have redistributive value; changes in rents also shift income between workers and the landowners, who are outside W. Climate change shifts all three statistics over time and space, through the incomes, flows and fares it changes, so the optimal schedule itself depends on the climate path.

H.6. The induced movers

TABLE H.VII
Whom the subsidy induces to move: the bottom quartile, 2015

	Composition			All workers of the row			Workers induced to move		
	Share of quartile (%)	Share of induced (%)	Induced, % of the row	Climate change (%)	Loss smaller by (%)	Gain (%)	Climate change (%)	Loss smaller by (%)	Gain (%)
Bottom quartile	100	100	4.7	−2.82	1.0	+1.53	−2.97	2.9	+4.51
Sector									
Farm	11	18	7.8	−3.23	3.7	+4.45	−3.27	5.8	+6.81
Non-farm	89	82	4.4	−2.76	0.6	+1.17	−2.91	2.2	+4.01
Schooling									
Less than primary	55	60	5.2	−3.00	1.5	+2.09	−3.20	3.8	+5.14
Primary	19	18	4.7	−2.71	0.4	+1.11	−2.78	1.7	+4.04
Secondary	25	21	4.0	−2.54	0.2	+0.68	−2.56	0.8	+3.27
University	2	1	3.4	−1.99	0.1	+0.27	−2.00	0.4	+2.33
Region									
North and Northeast	47	61	6.2	−3.20	2.5	+2.71	−3.27	4.6	+5.47
South, Southeast, Centre-West	53	39	3.5	−2.48	−0.7	+0.49	−2.51	−0.6	+3.03
Hottest third	48	59	5.9	−3.21	2.4	+2.36	−3.31	4.7	+5.19
Middle third	30	24	3.9	−2.71	0.3	+1.06	−2.77	0.9	+4.09
Coolest third	23	17	3.5	−2.12	−2.2	+0.42	−2.09	−3.4	+2.77
National earnings decile									
First	40	46	5.5	−2.93	1.6	+2.09	−3.08	3.8	+5.28
Second	40	38	4.5	−2.77	0.7	+1.25	−2.91	2.2	+4.08
Third (lower half)	20	16	3.7	−2.68	0.4	+0.97	−2.82	1.6	+3.35

Notes. SSP5-8.5, 2015. Composition: share of the bottom quartile and of its induced movers; induced rate: share of the row induced to leave in 2015. Remaining columns as Table II. Farm: workers at the farm atom of the baseline.

Additional references

Timo Boppart. Structural change and the Kaldor facts in a growth model with relative price effects and non-Gorman preferences. *Econometrica*, 82(6):2167–2196, 2014.

Jonah B. Gelbach. When do covariates matter? And which ones, and how much? *Journal of Labor Economics*, 34(2):509–543, 2016.

IBGE. *Pesquisa de Orçamentos Familiares 2008–2009: Despesas, Rendimentos e Condições de Vida*. Instituto Brasileiro de Geografia e Estatística, Rio de Janeiro, 2010.